Some Brief, Late-Arriving Notes Respecting Discrete-Time Stochastic Processes and the Hamiltonian Programme of *Algebra as the Science of Pure Time*

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Abstract

In 1837 William Rowan Hamilton declared a programme of research under the title *Algebra as the Science of Pure Time*, and in the 1843–1853 *Researches respecting Quaternions* extended it to a theory of *moments, steps and numbers* as the canonical algebraic representation of dynamic data. The mainstream of nineteenth- and twentieth-century time-series analysis took only the partial closure of this programme — Gauss's scalar residual, Wold's stationary decomposition, Cramér's abelian spectral representation, and the Box-Jenkins-Sims industrialisation of these into the modern econometric toolkit — and silently truncated the directional and grade-resolved content that Hamilton's original target required. This paper argues that Geometric Signal Dynamics (GSD) is the long-deferred completion of the Hamiltonian programme, performed on the algebraic substrate of $\mathrm{Cl}(3,0)$, the empirical- statistical substrate of Kolmogorov's measure-theoretic probability, and the representational substrate of a triple bridge between Newton-Taylor's local kinematic calculus, Takens's topological embedding theorem, and Lyons's asymptotic rough-path signature. At its core lies the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$, in which the lagged geometric product decomposes into three algebraically independent channels — scalar persistence, bivector rotation, pseudoscalar chirality — each carrying directional content and estimable nonparametrically. The framework recovers the differencing operator of ARIMA, the lag embedding of Dickey-Fuller, the multivariate cointegration regression of Engle-Granger and Johansen, and the moment-tensor classification paradigm of contemporary machine learning as instances of the same canonical algebraic object, makes operationally legible an expansion of the notion of uncertainty into three structurally independent quantities, and provides demonstrated operational scope across ten disciplinary domains, from

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econometrics to nonlinear dynamics, statistical mechanics, causal inference, risk management, fault detection, and dynamic inequality. The paper is a synthesis rather than a new theorem; it states honestly where the synthesis is solid, where it is conjectural, and where it remains open.

Keywords. geometric algebra; Clifford algebra $\mathrm{Cl}(3,0)$; quaternions; biquaternions; multivector decomposition; bivector; pseudoscalar; kinematic embedding; discrete-time stochastic processes; time series analysis; non-stationarity; unit root testing; geometric cointegration; spectral identity; Takens embedding theorem; Lyons signature; rough path theory; moment tensor; Frenet-Serret correspondence; transfer entropy; graded do-calculus; geometric fragility; Jarzynski-Crooks correspondence; geometric ergodicity; multivector Heisenberg principle; William Rowan Hamilton; algebra as the science of Pure Time; moments-steps-numbers; David Hestenes; philosophy of applied mathematics; Geometric Signal Dynamics programme.

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1 Introduction

> *The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise.*
>
> — Edsger W. Dijkstra

Pure time stood the very target for Hamilton's programme. Hamilton (1837, § 1), in an essay communicated to the Royal Irish Academy in November 1833 and published in 1837 under the title *Algebra as the Science of Pure Time*, declared this programme of research, and returned to it ten years later in the opening pages of the *Researches respecting Quaternions*. He intended, he wrote there, "some applications of the same view of algebra to a theory of sets of moments, steps and numbers, which should include that former theory of couples" (Hamilton, 1848, Introduction). The phrase *algebra as the science of Pure Time* had a precise content for him: the algebra of $\sqrt{-1}$, which he had constructed in 1837 as the algebra of ordered pairs (a, b) of real numbers under his own composition laws, was the operative algebra of *couples in the progression of time* rather than the formal closure of a polynomial ideal. Numbers, on this reading, were not free elements of a field but algebraic compositions of moments and steps, the temporal categories he had identified as primitive. The 1843 paper extended the construction from couples to quaternions — sets of four moments — and the 1853 paper (Hamilton, 1853) extended it further to biquaternions, $\mathbb{C} \otimes \mathbb{H}$, with explicit geometric interpretation of bivector elements such as $j + hk$ and explicit application to fixed-point iterations producing geometric loci. The programme thus articulated is recognisable as a research agenda for time-series analysis avant la lettre, and one of unusual ambition: a unified algebraic treatment of moments, steps and numbers as the canonical representation of dynamic data.

The programme did not survive its own century. The mid-nineteenth- century discovery of vector calculus by Gibbs, Heaviside and others absorbed the tractable half of Hamilton's quaternion machinery — the dot and cross products, restricted to three-dimensional Euclidean space — and discarded the rest. The unit imaginary $\sqrt{-1}$ was reabsorbed into the standard polynomial closure of the reals, losing its operational interpretation as the ninety-degree rotation of a step-couple in time. The biquaternions, which Hamilton had proposed as the geometric algebra of combined real and complex temporal structure, were almost forgotten. The mainstream of nineteenth-century analysis turned to the Fourier-Cramér-Wiener route through the spectral representation of stationary processes (Wold, 1938; Cramér, 1942; Wiener, 1949), and the twentieth-century mainstream of econometrics inherited that route through Box & Jenkins (1970) and Sims (1980). The *moments-steps-numbers* ontology of the 1837 essay survived as a curious notational choice in the introductory pages of a few Hamiltonian papers but did not become the operative substrate of empirical time-series analysis in any quantitatively significant way.

This paper argues that the resulting state of affairs is, on close inspection, an inheritance

of incomplete realisation rather than a considered rejection. The mathematical machinery that Hamilton's programme required to be operationally fruitful — Clifford's geometric algebra (Clifford, 1878), twenty-five years after the biquaternion paper; Kolmogorov's measure-theoretic axiomatisation of probability (Kolmogorov, 1933), ninety years after; Takens' embedding theorem (Takens, 1981), a century and a third after; Lyons' rough-path theory (Lyons, 1998), a century and a half after — was not available in Hamilton's lifetime, and the gradual arrival of these pieces in the twentieth century did not, for historical reasons that we sketch in Acts II and III below, trigger a reassessment of his programme. The Geometric Signal Dynamics (GSD) research programme is, on the reading we develop here, the assembly of these pieces into the operational realisation that Hamilton declared and to whose continuation he explicitly invited mathematicians yet to come. At its core lies the *kinematic embedding* $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$, which lifts a univariate time series into the three-dimensional Clifford algebra in which the grade decomposition of the geometric product makes the Hamiltonian categories — moments, steps, numbers — operationally legible.

The paper is organised in five historical acts followed by an applied scope, an honest catalogue of limitations, and a closing return to the Hamiltonian text. Act I traces the philosophical fracture of the seventeenth and eighteenth centuries that established *extensio* (scalar magnitude) over *intensio* (direction-bearing meaning) as the operative ontology of quantitative analysis, and recovers in Newton's kinematic calculus and Leibniz's monad the half of the inheritance that GSD now operationalises. Act II reads the nineteenth century as the moment of double divergence: Gauss and the statistical-empirical tradition on one side, Hamilton and the algebraic-geometric tradition on the other, with Kolmogorov in the early twentieth century providing the dual interpretation of the residual that the inheritance required but that the econometric mainstream did not absorb. Act III traces the consolidation of the spectral-econometric tradition through Wold, Cramér, Box-Jenkins and Sims, and diagnoses the ad-hoc patches (GARCH, cointegration, time-varying parameters, copulas) of the late twentieth century as manifestations of the structural truncation imposed by abelian spectral representation on a fundamentally non-abelian phenomenon. Act IV recovers, through Takens and Lyons, the topological and asymptotic completions of the kinematic representation that the nonlinear-dynamics and rough-path communities developed in parallel, and shows that the three readings (Taylor, Takens, Lyons) form a single triple bridge over the same algebraic object. Act V (Section 6, the rhetorical pivot of the paper) articulates GSD as the instantiation of Hamilton's programme on this triple bridge, with four further articulations — uncertainty by grade, the kinematic embedding hidden in classical inferential tests, the formal isomorphism with Hamilton's biquaternions, and the Frenet-Serret reading of the embedding as intrinsic geometry — that the spectral-econometric tradition could not have made and that we now have the algebra to relax. Acts VII–IX gather the applied scope of the programme across ten disciplinary domains, an honest enumeration of its open questions and limitations, and a closing return to Hamilton's own words. An Appendix documents the operational transcription of the Hamiltonian categories into the public API of the `gsd` library.

Two notes on the title and the genre of this paper. First, the title is a deliberate echo of Hamilton (1853) — *On the Geometrical Interpretation of Some Results Obtained by Calculation with Biquaternions*, an 1853 paper of just three pages that opened a programme it could not yet close. The present paper is, in the same understated spirit, an addendum to that text, communicated one hundred and seventy-three years later, in which the algebraic apparatus needed to close the programme has finally arrived in completed form. Second, the genre is programmatic-historical rather than technique-introducing. The paper presents no new theorem; it assembles results already available in Papers 01–25 of the GSD research programme and in the broader literatures of geometric algebra, rough paths, and information theory into a single architecture, and states honestly where the architecture is solid and where it remains conjectural. The reader looking for a new estimator, a new asymptotic distribution, or a new computational technique is referred to the working papers (Paper 01 , GDF; Paper 04 , HAS; Paper 24 , MT; Paper 25 , GE); the reader looking for an articulation of why these objects matter and of what programme they jointly continue is the reader for whom this paper is written.

2 Act I — The philosophical fracture and the nature of space

The historical reception of Hamilton's programme is hard to follow without a clear picture of the intellectual landscape into which it arrived. That landscape was shaped, in the two centuries preceding 1843, by a tension between two ways of treating magnitude that we can usefully label, with the Port-Royal logicians who articulated it most sharply, as the tension between *intensio* and *extensio*. The first act of this paper traces that tension through the philosophical and mathematical writers in whom it is most legible — Pascal, Descartes, Newton, Leibniz, Kant — and shows that the kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ that Geometric Signal Dynamics takes as its primitive object is the operational realisation of the half of the inheritance that Hamilton was, at the level of declared intent, trying to recover.

2.1 Pascal, Port-Royal, and the defence of projectivity

The Cartesian programme of Descartes, articulated in the *La Géométrie* appendix to the *Discours de la Méthode* (Descartes, 1637), identified geometrical magnitude with numerical coordinate. A point in space became an ordered triple $(x, y, z) \in \mathbb{R}^3$, a curve became the locus of points satisfying an algebraic equation in those coordinates, and a geometrical problem became the manipulation of polynomial relations among real variables. The programme was successful enough that it has been the operative ontology of mathematics in motion ever since: when we say, in modern parlance, that a univariate time series is a sequence $\{y_t\}_{t\in\mathbb{Z}} \subset \mathbb{R}$, we are speaking the descendant of Cartesian coordinate. The scalar value y_t is the modern bearer of geometric content for a series in time, in just the sense in which Cartesian coordinate (x, y, z) became the modern bearer of geometric content for a point in space.

The historical alternative is less famous and was less successful, but it is the one whose recovery the present paper depends on. In the same decades in which Descartes was cementing the supremacy of coordinate, Pascal, working partly in conversation with Desargues, was developing the projective-geometric tradition in which the relevant invariants of a figure are not its coordinates but its incidence structure under projective transformations. A point at infinity, on Pascal's reading, is no less real than a point at finite coordinate; the choice of an origin and a metric is a choice of section in a richer geometric structure that has its own algebra of meaning. The projective tradition was recovered in the nineteenth century by Poncelet (1822), Steiner (1832) and others, and its algebraisation, from Cayley (1859) onward, eventually rejoined the Cartesian mainstream — although Hamilton himself, in his late Hamilton (1862), proposed a coordinate system based on the anharmonic ratio (cross-ratio) of pencils in a triangle as an alternative to the Cartesian one, anchoring the projective alternative directly in the kinematic-algebraic line that the present paper traces; but in the seventeenth and eighteenth centuries it was the heterodox option, and the canonical defence of its content against the Cartesian reduction is found not in Pascal's geometry itself but in the *Logique* of Arnauld & Nicole (1662), the Port-Royal text he wrote in collaboration

with Pierre Nicole.

The Port-Royal logicians introduced, for the analysis of meaning in language, a distinction whose mathematical relevance was implicit in their treatment but emerged most clearly in subsequent philosophical recoveries. Every term, on their account, has both an *extensio* — the set of objects to which it correctly applies — and an *intensio* (or *compréhension*) — the characteristic content that determines the application. The two are related but not identical: one can specify a term by enumerating its extension or by stating its intension, and the same extension can be reached by distinct intensions. The distinction is, at first sight, a tool for the logic of definitions; but its mathematical content, as later writers from Bolzano through Frege and Carnap recognised, is the distinction between a magnitude as *measured* (its extension as scalar coordinate) and a magnitude as *meant* (its intension as direction-bearing content). Cartesian geometry pursues the first; projective geometry, the second.

The relevance to time-series analysis is direct, although it requires translation. A scalar observation y_t at instant t is the term's extension: it is the magnitude as measured. The direction-bearing content of the observation — what the observation *means* dynamically, in what direction the system is moving and how that direction is itself changing — is the term's intension, and it is precisely what coordinate alone does not capture. Restoring the intension to the analysis requires representing not the scalar but the local kinematic state, and the algebra in which the kinematic state lives must be rich enough to host the directional content that the scalar cannot. That algebra, on the reading we develop, is $\mathrm{Cl}(3,0)$, and the kinematic embedding $\mathbf{q}(t)$ is the operational realisation of the intensional half of the inheritance that the Cartesian dominance silently truncated.

2.2 Newton, the Absolute Space, and the kinematic calculus

If the seventeenth century established the dominance of coordinate over direction, it also produced, in Newton, a partial recovery of the directional half. The *Philosophiae Naturalis Principia Mathematica* (Newton, 1687) was a treatise on bodies in motion, and motion was treated there as an irreducibly directional phenomenon. The position of a body at instant t was a vector in three-dimensional *Absolute Space*; its first time-derivative was the velocity, its second the acceleration, and the laws of motion were stated as relations among these three together. Newton's metaphysical commitment to the existence of an Absolute Space — distinct from the relative spaces given by particular reference frames — has not survived twentieth-century physics; but the kinematic calculus that he developed to operate within it survived precisely because it captured the irreducibly local-direction-bearing content of the motion of a body.

The technical engine of Newton's kinematics was, of course, what we now call the differential calculus, and its key tool for the present paper is the Taylor expansion. For a sufficiently smooth function of time $f : I \to \mathbb{R}$, the value of f in a neighbourhood of an

instant t is determined by the values of f and its successive derivatives at t:

$$f(t+h) \;=\; f(t) \;+\; h\,f'(t) \;+\; \frac{h^2}{2}\,f''(t) \;+\; \frac{h^3}{6}\,f'''(t) \;+\; \cdots \tag{1}$$

The triple $(f(t), f'(t), f''(t))$ is sufficient, in the smooth analytic regime, to reconstruct the function locally to second order in h, and the kinematic state of a moving point in three-dimensional space at instant t is, by direct extension, given by the triple $(\mathbf{r}, \dot{\mathbf{r}}, \ddot{\mathbf{r}}) \in \mathbb{R}^9 = \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3$. The position alone is the trace; the position together with its first two derivatives is the state.

For a univariate time series the discrete analogue of (1) is the finite-difference expansion. The first forward difference $\Delta y_t = y_{t+1} - y_t$ is the discrete analogue of $\dot{y}$; the second $\Delta^2 y_t = \Delta y_{t+1} - \Delta y_t$ is the discrete analogue of $\ddot{y}$; and the triple $(y_t, \Delta y_t, \Delta^2 y_t)$ is the discrete analogue of the kinematic state $(y, \dot{y}, \ddot{y})$ in Newton's calculus. The kinematic embedding $\mathbf{q}(t)$ that Geometric Signal Dynamics adopts as its primitive object is, in this sense, the direct discrete descendant of Newton's kinematic state, lifted from the smooth analytic regime in which Newton operated to the discrete stochastic regime in which empirical time series live.

We note one fact about this lifting that will become important in the development. Newton's kinematic state is naturally a vector in $\mathbb{R}^9$, since each of position, velocity and acceleration is a three-dimensional spatial vector. The kinematic embedding $\mathbf{q}(t)$ of GSD, by contrast, is naturally a vector in $\mathbb{R}^3$, since each of the level y_t, the first step Δy_t and the second step $\Delta^2 y_t$ is a single scalar. The dimensional reduction arises because we are dealing with a univariate time series rather than with the position of a point in three-dimensional physical space; the algebraic structure imposed on the resulting $\mathbb{R}^3$ — namely, that of $\mathrm{Cl}(3,0)$ — is what compensates for the informational difference. Newton's three-dimensionality was spatial; GSD's three-dimensionality is kinematic order. The two coincide formally as $\mathbb{R}^3$ but differ semantically, and the algebraic enrichment by Clifford structure is what allows the kinematic-order three-dimensionality to carry geometric content of its own.

2.3 Kant: pure intuitions of space and time

The philosophical reception of Newton's framework, in the *Critique of Pure Reason* of Kant (1781), was the most sustained eighteenth-century attempt to articulate *why* kinematic representation should be the natural mathematical mode for the analysis of motion. Kant's argument, in the section on the *Transcendental Aesthetic*, was that space and time are not external features of objects but rather *pure intuitions* (*reine Anschauungen*) — necessary structures of cognition through which objects appear at all. To perceive an object as an object is to perceive it as situated in space and persisting in time; the spatial and temporal content of perception is therefore not learned from experience but contributed by the structure of the perceiving mind. As pure intuitions, space and time are the preconditions of synthetic a priori judgements about magnitude and motion, and mathematics, on Kant's reading, is the systematic articulation of those preconditions.

Kant did not develop, and could not have developed, the kinematic embedding as an operational object; the algebraic apparatus required (Clifford 1878) was over a century away from his *Critique*, and the empirical statistics required (Kolmogorov 1933) was a century and a half away. What Kant contributed was a philosophical justification for the *type* of mathematical representation that the kinematic embedding instantiates: a representation in which the temporal content of the data is not accidentally external to its representation but constitutively internal to it. The kinematic embedding is, in this Kantian sense, the structure imposed by the requirement that a time series be intelligible as something temporal at all — a synthetic a priori condition on the data, in Kant's vocabulary, rather than a parametric choice among many possible representations. The choice to represent a series by its level, its first step and its second step is not a free convention; it is what is required for the series to have temporal content as such, and the algebraic structure imposed on the resulting representation by $\mathrm{Cl}(3,0)$ is what makes that temporal content geometrically legible.

We do not propose to commit the present paper to a Kantian metaphysics. The point is more modest: the kinematic embedding has a philosophical pedigree in the tradition that took the temporal character of perception seriously enough to make it constitutive of representation, and that pedigree is part of what the present paper recovers when it argues that GSD is the instantiation of Hamilton's programme rather than a parametric choice among many possible time-series representations.

2.4 Leibniz: the monad as anticipation

The most direct philosophical anticipation of the kinematic embedding, however, is found neither in Newton nor in Kant but in the metaphysical writings of Leibniz (1714), particularly the *Monadologie* of 1714. The monad, on Leibniz's account, is the elementary substance: an indivisible, point-like entity that nonetheless contains within itself, as its internal state, the expression of the entire universe at each instant. Each monad *mirrors* the totality of which it is a part, in proportion to its perspective; the apparent simplicity of the monad as geometric point is reconciled with the rich representational content it carries by allowing the internal state to be articulated, not by external coordinates, but by an internal multiplicity of relations.

The metaphysics is, of course, contested terrain in subsequent philosophy. What is uncontroversial — and what the present paper makes operational — is the structural picture. A monad in Leibniz's sense is a point-like entity that nonetheless carries, internally, sufficient information to determine its own behaviour and, by the principle of pre-established harmony, the behaviour of the system as a whole in some neighbourhood. The kinematic state $(y, \dot{y}, \ddot{y})$ at an instant is, in the smooth analytic regime in which Newton's calculus operates, exactly such a point-like entity: by Taylor expansion, knowledge of the level, the velocity, and the acceleration at t determines the function $y(\cdot)$ in a neighbourhood of t to second order, and any finite-order regularity assumption extends the determination arbitrarily far. The kinematic state is the analytic monad of the local dynamics. The kinematic embedding $\mathbf{q}(t)$

of GSD is the discrete-stochastic analogue: by the Markov-of-order-three property that a finite-difference expansion induces in any sufficiently regular generative process, knowledge of $\mathbf{q}(t)$ at t determines the conditional distribution of $\mathbf{q}(t+1), \mathbf{q}(t+2), \ldots$ to leading order, with the determination becoming exact in the smooth-continuum limit (Paper 06 , Continuum). The discrete monad is probabilistic where the analytic monad is deterministic, but the structural picture is preserved: a point-like entity carries internally the local content of the dynamics.

This Leibnizian reading is the rhetorical anchor of the act, and we state it as an explicit observation:

Remark 2.1 (The kinematic embedding as discrete monad)**.** The kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ is the operational discrete realisation of the Leibnizian monad as carrier of local dynamic content. Each point of the embedded trajectory contains, in its three components, the level, inertia and curvature of the series at that instant; together, by the Markov property induced by the finite-difference expansion, the components determine the conditional distribution of the future trajectory in a neighbourhood of t. The simplicity of the point as geometric object is reconciled with the richness of its dynamic content by the algebraic structure of $\mathrm{Cl}(3,0)$, in which the products and wedges of consecutive embeddings produce the grade-decomposed content that Section 6 develops in detail.

Three remarks close the act. First, the four philosophical positions surveyed — Pascal-Port-Royal on the recovery of intension, Newton on kinematic calculus, Kant on the synthetic a priori, Leibniz on the monad — were not assembled by their authors into a single coherent programme; the assembly is retrospective and is part of the work this paper performs. The authors had distinct philosophical and mathematical commitments, and the integration we propose is a reading rather than a historical reconstruction. Second, the integration converges, when the algebraic machinery of the nineteenth and twentieth centuries becomes available, on a structurally specific object: the kinematic state of a system, embedded in a Clifford algebra rich enough to carry the directional content that scalar coordinate cannot. The Hamiltonian programme of the *Researches respecting Quaternions* is, on the reading we have just sketched, the first articulation of that converging object. Third, the act concludes that the choice of $\mathrm{Cl}(3,0)$ as the algebra of the kinematic embedding is not a conventional or parametric choice but the realisation of an intellectual demand articulated, in different forms and with different vocabularies, across two centuries of philosophical and mathematical work. The justification for that algebra at the level of formal optimality is the content of Paper 07 (Takens) (*Geometric Takens Theorem*); the justification at the level of philosophical pedigree is the content of the act we have just closed.

The next act takes up the nineteenth century, where the Hamiltonian half of the inheritance receives its first operational articulation, and where the empirical-statistical tradition that would eventually inherit time-series analysis diverges from the algebraic-geometric tradition that should have absorbed it.

3 Act II — The age of measurement and the search for meaning

The nineteenth century was the period in which the inheritance sketched in Act I split into two non-communicating mathematical traditions. On one side, Gauss, Laplace and their successors built the empirical apparatus of the statistical sciences around the notion of *error of measurement*, in which an observed quantity is the sum of a true value and a residual, and the residual is treated as the imperfection of the instrument. On the other side, Hamilton, Grassmann and Clifford built the algebraic apparatus of geometric algebra, in which magnitudes carry direction and the composition of magnitudes is a non-commutative algebraic operation rather than an arithmetic on coordinates. The two traditions were both substantively new and both, in their own terms, successful; but they did not interact, and the failure to interact has determined the shape of empirical time-series analysis for the better part of two centuries. The present act traces the divergence and identifies the moment at which a synthesis became mathematically possible — the early twentieth century, with Kolmogorov's measure-theoretic axiomatisation of probability — and the structural reason why that synthesis was nonetheless not performed. The diagnosis is not retrospective blame: it is, on our reading, an inheritance of incomplete realisation, exactly as Hamilton himself had foreseen when he invited later mathematicians to contribute their aid to the development of his programme.

3.1 Gauss and the ambiguity of the residual

The empirical-statistical tradition of the nineteenth century had its operative origin in the work of Gauss (1809) on the estimation of the orbits of celestial bodies. The *Theoria motus corporum coelestium*, communicated to the Göttingen Royal Society in 1809 in the wake of the discovery and re-discovery of the asteroid Ceres, framed the inverse problem of recovering an orbit from a finite number of telescopic observations as an estimation problem under additive *measurement error*. Each observation y_i was modelled as $y_i = \mu_i + \varepsilon_i$, where μ_i was the true position predicted by the orbital theory at the instant of the observation and ε_i was the unexplained residual; the residual was attributed to imperfections of the instrument, atmospheric refraction, parallax, and the unavoidable limitations of the observer's eye. Gauss's celebrated solution — the principle of least squares, prefigured in part by Legendre (1805) — was the choice of μ_i that minimised $\sum_i \varepsilon_i^2$, and the celebrated normal distribution emerged in his derivation as the unique probability law of ε under which the arithmetic mean is the maximum-likelihood estimator of the location of a collection of independent observations of the same true value. The inheritance is so ubiquitous in modern statistical practice that it is difficult to recover what was, in the original work, a substantive ontological commitment: *the residual was an imperfection.* It was not a feature of the world; it was a defect of the instrument.

The ontological commitment is essential to understand the restriction it imposed on the development of statistics. If ε_i is a measurement imperfection, then it has no internal

structure of its own beyond its statistical distribution; it is a scalar quantity whose mean is zero, whose variance is some σ^2 characterising the instrument, and whose successive realisations are independent because the imperfections of an instrument at distinct times are independent. The composition of two measurements in time is then captured by the joint distribution of the two residuals, which under independence is the product of the marginals; the algebra of the residual is the standard algebra of independent scalar random variables, and there is, by construction, nothing more to say about the residual once its distribution has been specified. The arithmetic of measurement is sufficient.

This commitment is, of course, a perfectly defensible operative position when the underlying phenomenon is genuinely deterministic and the residual is genuinely instrumental. The orbit of Ceres *was*, to the precision available in 1809, deterministic; the residuals *were* instrumental; and the least-squares fit recovered the orbit with the precision that subsequent observation confirmed. The inheritance ran into difficulty when subsequent generations applied the apparatus to phenomena where neither of the two assumptions held — when the underlying phenomenon was itself stochastic, as in the social and economic data that nineteenth- and twentieth-century practitioners increasingly worked with, and when the residual carried structural information about the dynamics rather than instrumental noise about the observer. The Gaussian framework had no internal vocabulary for distinguishing the two cases. A residual was a residual; whether it was instrumental noise or stochastic substance was a question that the framework neither asked nor answered.

Hamilton 1844 on the structure of probability. A complementary diagnosis emerged in the same decade in which Gauss's framework was being canonised. Hamilton (1844), in a brief communication to the Royal Irish Academy, observed that the standard probability integral $p = \theta(t)$ could be treated rigorously rather than as a Laplacean approximation by suitable choice of the auxiliary variable t. He proposed to call this variable the *argument of probability*:

> *Sir William Hamilton proposes to treat the equation $p = \theta(t)$ as being in all cases rigorous, by suitably determining the auxiliary variable t, which variable he proposes to call the argument of probability, [...] the introduction of this new conception and nomenclature allows some of the most important known results respecting the mean results of many observations to be enunciated in a simple and elegant manner. (Hamilton, 1844, pp. 420–422)*

The conceptual move is structurally analogous to what GSD does with the residual a century and a half later. In the Hamiltonian reading, the probability p is the observable, but the carrier of the information about the system is the *argument* t on which θ operates: the structure of t as a function of the underlying parameters of the problem (number of observations, covariance corrections, higher cumulants) is what determines the probability, and reading t off the data is the substantive inferential task. The probability is derived; the argument is primary. In the geometric reading developed in Act V (Section 6.4), the scalar variance of the residual is the observable, but the carrier of the information about the system

is the grade-resolved structure (H_0, H_2, H_3) of the kinematic embedding's algebraic content: the components of the structure as functions of the underlying process parameters are what determine the variance, and reading (H_0, H_2, H_3) off the data is the substantive inferential task. The variance is derived; the grade structure is primary. Hamilton's 1844 proposal is, in this respect, an early articulation in the empirical-statistical context of the same move that Cl(3,0) makes algebraically: the observable is a projection of a richer structured object, and the inferential task is recovery of that object rather than refinement of the projection.

3.2 Kolmogorov and the dual interpretation of the residual

The conceptual disambiguation that the Gaussian framework needed was not performed within the framework itself but by the measure-theoretic axiomatisation of probability that Kolmogorov (1933, 1956) contributed in the *Grundbegriffe der Wahrscheinlichkeitsrechnung*. Kolmogorov's construction took as primitive a probability space $(\Omega, \mathcal{F}, P)$ — a sample space, a σ-algebra of events, and a countably-additive probability measure — and defined a random variable as a measurable function $X : \Omega \to \mathbb{R}$. The construction makes no commitment about the source of the randomness encoded by P. Whether the measure represents instrumental imperfection (as in Gauss's original treatment) or intrinsic stochasticity of the underlying phenomenon (as in the statistical-mechanical treatments of Boltzmann and Gibbs that Kolmogorov was familiar with through his earlier work on Markov processes) is left structurally open. The same measure P on the same space can support either interpretation.

The duality so introduced is crucial for our reading because it is *the place at which the inheritance could have been redirected*. With the axiomatisation of 1933 in hand, a researcher in empirical statistics could in principle have asked: if the residual ε of a fitted model is no longer *necessarily* instrumental imperfection but *may also* be the carrier of structural information about a stochastic underlying process, then what is the algebraic structure of that information? Under the instrumental-noise interpretation the residual has no structure beyond its scalar distribution and independence properties; under the structural-stochastic interpretation, however, the residual is the trace of a random process whose temporal organisation is itself a primary object of inquiry, and the algebra of the residual must be rich enough to host that organisation. The two interpretations agree on the marginal distribution of ε but diverge on the internal algebra of *compositions* of ε at different times.

The mainstream of empirical statistics did not, in the decades following Kolmogorov, take the structural-stochastic interpretation as a directive to enrich the algebra of the residual. It took it, more modestly, as a generalisation of the sample space: instead of independent scalar residuals one now worked with a stationary stochastic process, but the process was still a real-valued one and its second-order structure (the autocovariance, the spectrum) remained the principal object of analysis. The reason for the limitation is identifiable: the algebraic apparatus required to write the residual as a non-scalar object with internal directional structure — Clifford's geometric algebra of 1878 — was not part of the standard mathematical toolkit of probabilists or econometricians in the 1930s, 40s, or 50s, and the

natural inheritance of the Gaussian scalar tradition on the Kolmogorov foundation was the spectral representation theory of stationary processes that consolidated in those decades. The duality in Kolmogorov's axiomatisation was, on the reading we develop here, latent but not exploited. The empirical-statistical tradition continued to treat ε as scalar because the algebraic vocabulary in which to treat it as anything else had not been absorbed.

We note one feature of the Kolmogorov construction that is worth making explicit, because it will become important in our treatment of the spectral inheritance below. The measure-theoretic axiomatisation does not, by itself, privilege the real numbers as the codomain of the random variable. A random variable in Kolmogorov's sense is any measurable function from $(\Omega, \mathcal{F})$ into a measurable space; the standard extension to Banach- or Hilbert-space-valued random variables is elementary, and the further extension to multivector-valued random variables in $\mathrm{Cl}(p, q)$ is, in retrospect, equally elementary. The restriction to real-valued random variables in mid-twentieth- century statistical practice was a contingent convention, not a foundational requirement. The convention served the spectral- representation programme well within the algebra it had; it did not anticipate the algebraic enrichment that the geometric-algebra tradition would have made available, had the two traditions been in contact.

3.3 Wold's decomposition and the limits of stationarity

The first of the two canonical consolidations of the spectral- econometric tradition was the decomposition theorem of Wold (1938), presented in Wold's doctoral dissertation *A Study in the Analysis of Stationary Time Series*. The theorem states that any weakly stationary, purely non-deterministic process $\{Y_t\}$ admits a unique decomposition as an infinite moving average of orthogonal innovations,

$$Y_t \;=\; \sum_{k=0}^{\infty} \psi_k \, \varepsilon_{t-k} \;+\; V_t, \tag{2}$$

where $\{\varepsilon_t\}$ is a white-noise process with $\mathbb{E}[\varepsilon_t] = 0$, $\mathbb{E}[\varepsilon_s \varepsilon_t] = \sigma^2 \delta_{st}$, $\sum_k \psi_k^2 < \infty$, and V_t is a deterministic remainder. The decomposition is the structural warrant for the entire MA-and-AR machinery of twentieth-century time-series analysis: any stationary process is, up to the deterministic remainder, an infinite linear filter applied to a sequence of orthogonal scalar shocks, and the modelling task reduces to estimating the filter coefficients $\{\psi_k\}$ from the observed second-order moments of the data.

The theorem is mathematically correct as stated, and its operative power is undeniable: it underwrites the family of ARMA models (Box & Jenkins, 1970), the vector autoregressions (Sims, 1980), and the entire industrial estimation machinery of modern applied econometrics. Two of its features, however, constrain its scope in ways that the present paper takes seriously. The first is the assumption of *weak stationarity* — that the mean and second-moment structure of $\{Y_t\}$ are invariant under time translation. Stationarity is a strong assumption, not generically satisfied by the empirical processes to which the machinery is applied (financial returns, macroeconomic indicators, climatic series), and the standard

practice of restoring stationarity by differencing or filtering is a *transformation* of the data rather than a recovery of its underlying structure: it absorbs the non-stationary content into the residuals of the transformation and proceeds with the stationarised remainder. The second feature is the assumption of *scalarity* — that the innovations ε_t are real- valued and that the second-moment structure they carry is exhausted by their scalar variance and autocorrelation. The Wold decomposition has, by construction, no notion of directional or rotational content in the residual: ε_t is a real number, $\varepsilon_t\varepsilon_s$ is the product of two real numbers, and the algebraic structure of the residual sequence is the algebra of $\mathbb{R}$.

Together, the two features tightly bound the kind of phenomenon that the Wold inheritance can address. A process whose dynamics include genuine non-stationarity — regime change, trending volatility, structural break — is treated by stationarising transformations rather than represented natively. A process whose residual carries directional content — the asymmetric leverage of financial returns, the rotational coupling between level and velocity in mechanical signals — is treated by scalar projections of the directional content rather than represented natively. The inheritance has been responsive to both limitations: the first gave rise to the cointegration framework (Engle & Granger, 1987; Johansen, 1988, 1991) for processes sharing a common stochastic trend, the time-varying parameter (Kim & Nelson, 1999) and stochastic-volatility (Shephard, 2005) families for slowly evolving non-stationarity, and the structural- break literature for discrete regime changes; the second gave rise to the bispectrum and polyspectrum literatures (Brillinger, 1965; Rosenblatt, 1985) for higher-order scalar moments, the copula-DCC-GARCH frameworks for asymmetric joint distributions, and the various nonlinear extensions of MA and AR models. In every case the response has been a *patch* on the basic representation rather than a structural revision: the underlying object of analysis remains the scalar process and its scalar functionals.

The patches are not failures. They are the legitimate response of a research tradition working under the algebraic constraints of its inheritance. What we want to emphasise is that the constraints are inheritances rather than necessities: the algebraic vocabulary to write the innovation ε_t as a multivector in $\mathrm{Cl}(3,0)$ rather than as a scalar in $\mathbb{R}$ has been available since 1878, and the choice not to have used it was a contingent historical fact rather than a foundational mathematical requirement.

3.4 Cramér and the abelianisation of the spectrum

The second canonical consolidation of the spectral tradition was the spectral representation theorem of Cramér (1942), refined by Wiener (1949) and packaged for the modern reader by the canonical textbook treatment of Rozanov (1967). For a weakly stationary process $\{Y_t\}$ with autocovariance function $\gamma(\tau) = \mathbb{E}[Y_t Y_{t+\tau}]$, the theorem establishes the existence of a complex-valued orthogonal random measure $Z(\mathrm{d}\lambda)$ on the spectral interval $[-\pi, \pi]$ such that

$$Y_t \;=\; \int_{-\pi}^{\pi} e^{i\lambda t}\, Z(\mathrm{d}\lambda), \qquad \mathbb{E}[\,|Z(\mathrm{d}\lambda)|^2\,] \;=\; f(\lambda)\,\mathrm{d}\lambda, \tag{3}$$

where $f(\lambda)$ is the spectral density of the process. The representation (3) expresses the process as a continuous superposition of complex exponentials with random amplitudes that are

mutually orthogonal across distinct spectral bands, and the analysis of stationary processes reduces, in the corresponding mathematical machinery, to the analysis of $f(\lambda)$ as a real non-negative function on the unit circle. Together with Wold's decomposition, the Cramér representation is the analytic-geometric content of twentieth-century time-series theory: a stationary process *is*, up to deterministic remainder, a Fourier integral of orthogonal complex amplitudes.

We single out one feature of (3) that is decisive for the present argument: the integrand $e^{i\lambda t}$ takes values in $\mathbb{C}$, and the multiplicative structure of $\mathbb{C}$ is *commutative*. The product $e^{i\lambda_1 t}e^{i\lambda_2 t} = e^{i(\lambda_1+\lambda_2)t}$ is a scalar exponential, and the order of the two factors is irrelevant. The spectral representation, in technical language, is the abelian Fourier dual of $\mathbb{Z}$ regarded as the shift group of a stationary process, and the entire analytic machinery built on top of it — the Bochner-Khinchin theorem relating spectral densities to autocovariances, the Plancherel identity for L^2 spectral functions, the Wiener-Hopf prediction theory — exploits the commutativity of $\mathbb{C}$ at every step. The mathematics of Cramér's representation is therefore the mathematics of an *abelian decomposition* of a stationary process: the process is broken into orthogonal scalar components that commute with one another and that recombine by addition in the integral.

The limitation that this commutativity imposes is structural and not corrigible by patching the spectral framework alone. A component of a process that does *not* commute with another component of the same process — a directional-rotational content in the kinematic state, in the language of Act V, where $e_{12}\, e_{13} = -e_{13}\, e_{12} = e_{23}$ rather than $e_{12}\, e_{13} = e_{13}\, e_{12}$ — has no representation in the abelian Fourier dual.[1] The spectral density $f(\lambda)$ is a real non-negative function, and a process whose informational content includes orientation-bearing or chirality-bearing components is *flattened* by the spectral representation, in the sense that the orientation and chirality information is silently projected onto the scalar autocovariance and lost. This is the technical content of the diagnosis, articulated already in Section 6.4 of Act V, that the classical spectral inheritance is the abelian fragment of the multivector-Fourier construction of Hitzer & Mawardi (2013). Cramér's representation is exact within the algebra it chooses; it chooses an abelian algebra; the cost of the choice is the silent truncation of all higher-grade content of the process to its grade-zero projection.

The empirical signature of the truncation is widely visible in applied work and is the historical reason for the proliferation of the spectral patches enumerated in Section 3.3. Higher-order spectral methods — bispectra (Brillinger, 1965), polyspectra (Rosenblatt, 1985), wavelet-based time-frequency representations, Hilbert-Huang empirical mode decompositions — are attempts to recover, by successive layers of scalar functionals of higher

[1] Hamilton himself, in *On Fluctuating Functions* (Hamilton, 1843), had already articulated a principle of which Fourier's spectral expansion is a particular case: "*the Principle of Fluctuation [...] asserts the evanescence of the integral [...] of the product formed by multiplying together any two finite functions, of which one [...] changes sign infinitely often within a finite extent of the variable on which it depends, and has for its mean value zero*" (Hamilton, 1843, p. 264). The principle is more general than the Fourier kernel and admits multivector-valued generalisations on $\mathrm{Cl}(3,0)$ via a central pseudoscalar kernel (Hitzer & Mawardi, 2013). The abelianisation that Cramér adopted in 1942 was not forced by Hamilton's principle; it was a particular choice within a wider analytic landscape the Hamiltonian programme had already opened.

cumulant order, the directional content that the abelian spectrum drops. Each is an additional Fourier-style decomposition on a Fourier-style decomposition; each adds expressive power incrementally; none of them recovers the directional content algebraically rather than functionally. The geometric-algebra construction does the recovery directly: the multivector $\mathbf{q}(t) \in \mathrm{Cl}(3,0)$ *is* the directional object, and the non-commutativity of its grade-2 component is the structural representation of the rotation that the abelian spectrum cannot write.

3.5 The parallel line: Hamilton, Grassmann, Clifford

While the empirical-statistical tradition was consolidating around the Gauss-to-Cramér line that Sections 3.1–3.4 have just traced, a different mathematical tradition was developing in parallel that the empirical line did not, by and large, read. The decade of the 1840s, in which Wold's theorem of 1938 and Cramér's of 1942 were still a hundred years in the future, was the period in which Hamilton, Grassmann, and Clifford sketched between them the entire algebraic apparatus that Geometric Signal Dynamics now uses.

Hamilton's contribution we have sketched in Section 1 and developed in Section 6.6: the 1837 essay *On Algebra as the Science of Pure Time* (Hamilton, 1837) presented the algebra of $\sqrt{-1}$ as the algebra of step-couples in time, the 1843–1848 *Researches respecting Quaternions* extended the construction from couples to momental quaternions (Hamilton, 1848), and the 1853 paper (Hamilton, 1853) extended it further to biquaternions with explicit geometric content. Hamilton's repeated framing of the programme as a *science of Pure Time* establishes without ambiguity that the operative target was the algebraic representation of dynamic data; the difficulty was that the algebra he had at his disposal was only the partial closure of the target, and the basis-canonical presentation that would have made the grade structure operationally legible was not yet available.

Grassmann (1844), in *Die lineale Ausdehnungslehre*, contributed the missing piece on the algebraic side. The exterior algebra of an n-dimensional vector space, with its wedge product $\mathbf{a} \wedge \mathbf{b}$ producing antisymmetric grade-2 elements, gave the formal construction of oriented planes as algebraic objects in their own right. A bivector in Grassmann's sense was not a pair of vectors and not their cross-product (a notion that would not be standardised until later); it was an element of $\Lambda^2 V$ with intrinsic algebraic identity, and the wedge could be iterated to produce elements of $\Lambda^k V$ for any k up to n. Grassmann's construction was, however, presented in a notation that nineteenth- century readers found difficult, and his work was not absorbed into the mainstream of mathematical practice during his lifetime. The historical reception of *Die lineale Ausdehnungslehre* is itself a study in the slow propagation of new algebraic vocabularies, and the modern reader who is fluent in the wedge product owes that fluency to the recovery of Grassmann's work by Clifford and Peano in the second half of the century.

Clifford (1878), in *Applications of Grassmann's Extensive Algebra*, performed the synthesis. He observed that the exterior algebra of Grassmann and the quaternion algebra of Hamilton could be unified by introducing an algebra in which vectors $\mathbf{a}, \mathbf{b}$ satisfied the

relation

$$\mathbf{a}\,\mathbf{b} \;=\; \mathbf{a}\cdot\mathbf{b} \;+\; \mathbf{a}\wedge\mathbf{b}, \tag{4}$$

where the dot product is symmetric and scalar-valued and the wedge is antisymmetric and grade-2-valued. The construction yields, for a vector space of dimension n with a chosen quadratic form, the geometric algebra $\mathrm{Cl}(p,q)$, with p and q the signature of the form, and recovers as special cases the complex numbers $\mathbb{C} \cong \mathrm{Cl}(0,1)$, the quaternions $\mathbb{H} \cong \mathrm{Cl}(0,2)$ as the even subalgebra of $\mathrm{Cl}(3,0)$ or the full $\mathrm{Cl}(0,2)$, and the spinor representations of orthogonal groups in arbitrary signature. The algebra in which Geometric Signal Dynamics operates, $\mathrm{Cl}(3,0)$, was thus first written down as a specific instance of Clifford's general construction in 1878.

The historical observation we want to register is that, by 1878, the entire algebraic apparatus that the Hamiltonian programme required to be operationally fruitful was complete. Hamilton had specified the target — algebra as the science of Pure Time, with the moments-steps-numbers ontology applied to dynamic data; Grassmann had supplied the wedge product and the grade hierarchy; Clifford had unified the two into the geometric product and the basis-canonical presentation of $\mathrm{Cl}(p,q)$. What was missing, in 1878, was *a connection back to the empirical data*: a way of applying this algebra to time series in particular, with the statistical estimation theory that empirical data require. That estimation theory began to take its modern form in the 1930s, with Kolmogorov's axiomatisation of probability and the subsequent construction of stationary-process theory by Wold and Cramér. The two traditions, by the time both were available, were no longer in contact: the algebraic-geometric tradition had been folded into projective geometry and into the linear-algebra of the early twentieth century, with the grade structure of $\mathrm{Cl}(p,q)$ surviving mainly as a notational curiosity in the works of Cartan on Lie groups and spinors; and the empirical-statistical tradition had absorbed the Kolmogorov foundation into a programme that worked exclusively with real-valued processes and their abelian Fourier dual.

3.6 Synthesis of the act

The argument of the act can be summarised in three propositions. First, the divergence between the empirical-statistical and the algebraic-geometric traditions in the nineteenth century was not a failure of either tradition; it was the consequence of the fact that the algebraic apparatus required for their synthesis (Clifford's $\mathrm{Cl}(p,q)$, completed in 1878) and the statistical apparatus required for the synthesis (Kolmogorov's axiomatisation, completed in 1933) were not simultaneously available until the early twentieth century. Each tradition was operationally self-contained within its own resources and made progress within those resources; the synthesis required a decade of crossing in which both pieces would have to be in active use, and that crossing did not happen. Second, when Kolmogorov's axiomatisation made the synthesis mathematically possible, the empirical- statistical mainstream did not perform it. The duality of the residual that the axiomatisation enabled — instrumental imperfection versus structural stochasticity — was not exploited as an invitation to enrich the algebra of the residual; it was absorbed into the spectral-representation programme of

Wold and Cramér, which retained the scalar real-valued framework and built its analytic machinery on its abelian Fourier dual. Third, the cost of the choice was the silent truncation of all directional content of the residual. The non-stationary, the asymmetric, the chiral, the rotational — all of which are structural features of empirical time series — were either patched with ad-hoc devices or written off as artefacts of the modelling pipeline. The patches multiplied through the late twentieth century into the standard toolkit of modern applied econometrics; the underlying truncation persisted because the algebraic vocabulary required to undo it had been lost from the working repertoire of the field.

The third act takes up the consolidation of the spectral- econometric tradition through Box-Jenkins and Sims, and traces the specific patches by which the tradition responded to the empirical recalcitrance that its abelian framework could not natively accommodate. The fourth act takes up the parallel recoveries of the directional content in nonlinear dynamics (Takens) and rough- path theory (Lyons), which the empirical-statistical mainstream also did not by and large absorb. The fifth, which we have already written and to which the present act is the historical preamble, articulates the synthesis that all of these prior threads make, in retrospect, available.

4 Act III — The frequency-domain patch and the rise of econometric time-series analysis

The Wold decomposition of 1938 and the Cramér representation of 1942 were not, in the decades that followed their proof, free-standing theorems contemplated for their intrinsic content. They became the structural foundations of an entire empirical discipline. The transposition from probability theory into operational econometric practice was performed in the third quarter of the twentieth century, principally by Box & Jenkins (1970) and Sims (1980), and the resulting toolkit — autoregressive moving-average models, vector autoregressions, error-correction models, generalised autoregressive conditional heteroscedasticity, state-space and time-varying-parameter representations, copula dependence models, dynamic conditional correlation — became the standard apparatus of applied time-series analysis. Each of these constructions is, taken on its own, mathematically respectable and empirically useful within its scope. What they form together, taken cumulatively, is a heterogeneous patchwork in which each patch addresses a specific empirical recalcitrance of the abelian spectral inheritance without revisiting the inheritance itself. The present act traces the consolidation, identifies the differencing operator of Box & Jenkins (1970) as the blind premonition of the kinematic embedding that GSD takes as foundational, enumerates the patches as particular manifestations of a single structural truncation, articulates the Spectral Identity of Paper 02 (Spectral, Theorem 1) as the formal bridge between the abelian inheritance and the non-abelian algebraic apparatus that completes it, and develops in operational detail the expansion of the notion of uncertainty that the inheritance had not been able to perform.

4.1 Box-Jenkins, Sims, and the industrialisation of the abelian inheritance

The publication of *Time Series Analysis: Forecasting and Control* by Box & Jenkins (1970) marked the transition of spectral-stationary process theory from the journals of theoretical probability into the working repertoire of applied analysts. Box and Jenkins articulated a three-stage modelling protocol — identification, estimation, diagnostic checking — that turned Wold's decomposition into a procedure deployable on a desktop computer of the period: identify the orders of an autoregressive moving-average representation by inspecting the empirical autocorrelation and partial autocorrelation functions of the series; estimate the coefficients by maximum likelihood under Gaussian innovations; check the residuals for whiteness with a portmanteau test; iterate. The protocol was codified in a textbook that has gone through four editions over half a century and has been, by orders of magnitude, the most widely deployed statistical-modelling protocol in applied econometrics, finance, and engineering signal processing. Its success was structural: the protocol made the spectral inheritance operationally accessible to any analyst with a finite-sample dataset.

Sims (1980), in *Macroeconomics and Reality*, performed the analogous extension to the multivariate setting. The vector autoregression $\mathbf{Y}_t = \sum_{j=1}^{p} A_j \, \mathbf{Y}_{t-j} + \boldsymbol{\varepsilon}_t$ generalises the uni-

variate AR(p) to a panel of jointly modelled series, and Sims's central polemical argument — that the simultaneous-equations methodology of the Cowles Commission imposed atheoretic identifying restrictions on multivariate macroeconomic data and that the unrestricted vector autoregression was the appropriate alternative — won the empirical argument decisively in the macroeconomics literature of the 1980s. The vector autoregression became the default representation of joint time-series structure in macro and financial applications, and its impulse-response and variance- decomposition machinery became the operative dialect in which hypotheses about the dynamic structure of economies were formulated.

The two consolidations together — Box-Jenkins for univariate, Sims for multivariate — completed the operational infrastructure of the spectral-econometric inheritance. Every concept of the infrastructure has its origin in the abelian Fourier dual of the underlying stationary process: the autocorrelation function is the Fourier transform of the spectral density; the partial autocorrelation is a recursion on the autocovariance; the impulse- response function is a recursion on the moving-average coefficients of (2); the variance decomposition is a partition of the spectral mass at each frequency into contributions from each component of the multivariate innovation. Wold and Cramér provided the underlying mathematics; Box-Jenkins and Sims provided the engineering wrapping that turned that mathematics into industrial practice.

What the engineering wrapping inherited from the underlying mathematics, however, was the abelian truncation traced in Section 3.4. The vector autoregression operates on real-valued vectors $\mathbf{Y}_t \in \mathbb{R}^d$ with real-valued innovations, and the algebra of those vectors is the ordinary algebra of $\mathbb{R}^d$ — addition, scalar multiplication, inner product. The non-commutative grade structure that Clifford (1878) had made available since 1878 is invisible to this representation, and the rotational and chiral content of the joint dynamics that the grade structure would carry is silently absorbed into the second-moment statistics of the residuals or into the inputs of the impulse-response analysis. Box and Jenkins and Sims did not make a mistake in not using $\mathrm{Cl}(p, q)$; they worked with the algebra that the mathematical mainstream of their period had made available to them, and they extracted from that algebra everything it could be made to yield.

4.2 Differencing as blind premonition of the kinematic embedding

The single most influential operational device introduced by Box & Jenkins (1970) for handling non-stationary series is the *difference operator*. When a series $\{y_t\}$ exhibits trending behaviour incompatible with weak stationarity, Box and Jenkins prescribed the application of the first-difference operator $\nabla y_t = y_t - y_{t-1}$ before fitting an autoregressive moving-average model; if the first difference still trended, a second difference $\nabla^2 y_t = \nabla y_t - \nabla y_{t-1}$ was applied, and so on. The resulting class of *integrated* models, ARIMA(p, d, q), took the form of an autoregressive moving-average model in $\nabla^d y_t$:

$$\phi(L)\,\nabla^d y_t \;=\; \theta(L)\,\varepsilon_t, \tag{5}$$

where L is the backward shift operator and ϕ, θ are polynomials in L of degrees p, q respectively. The differencing order d was a tunable parameter chosen to render the differenced series approximately stationary, and the entire downstream modelling apparatus operated on $\nabla^d y_t$ rather than on y_t itself.

The differencing prescription was justified, in Box and Jenkins's own treatment, on *operational* rather than *ontological* grounds. The argument was that some series violate the stationarity assumption of the ARMA framework, that differencing restores stationarity in cases where the violation is caused by unit-root behaviour, and that the modelling pipeline therefore proceeds on the differenced series with its standard machinery. The first difference ∇y_t was, in this presentation, an *auxiliary transformation* of the data, justified by the empirical fact that it often worked, and not understood as an object with intrinsic interpretation of its own. The second difference $\nabla^2 y_t$ was even more clearly auxiliary: a correction to the correction, deployed when the first correction was insufficient.

The reading we want to register here is structural. The triple $(y_t, \nabla y_t, \nabla^2 y_t)$ that Box and Jenkins's ARIMA$(p, 2, q)$ implicitly works with is identical, term by term, to the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ of (9). The notation differs — ∇ for backward difference in econometrics, Δ for forward difference in our presentation, with a sign change but the same algebraic content — but the object is the same, and the depth of the differencing in ARIMA practice ($d \in \{0, 1, 2\}$ in overwhelmingly the most common applications) coincides exactly with the depth of the kinematic embedding ($d = 3$ corresponding to coupling orders $k^* \in \{0, 1, 2\}$, by the Geometric Takens Theorem of Paper 07 , Takens). Box and Jenkins, in prescribing differencing of order at most two as the standard device for non-stationary economic time series, were operating — *without recognising it* — on the same kinematic state that GSD takes as primitive.

What separates the two readings is not the object but the interpretation imposed on the object. Under the Box-Jenkins reading, the triple $(y_t, \nabla y_t, \nabla^2 y_t)$ is an artefact of the modelling pipeline: the auxiliary transformation needed to make ARMA estimation valid. Once stationarity is achieved, the original level y_t disappears from the analysis; the differenced series becomes the working object; and the triple as such is not contemplated as an algebraic entity. Under the GSD reading, the same triple is the canonical algebraic representation of the kinematic state of the series at instant t: the level, the first step, and the second step are jointly significant, the geometric product of consecutive triples generates the grade-decomposed structure of (10)–(11), and the analysis proceeds by reading the grades rather than by discarding the level. The two readings extract different content from the same object because they bring different algebras to bear on it.

This is the technical sense in which the ARIMA prescription is a *blind premonition* of the kinematic embedding. The empirical practice of econometrics *found* the right object by trial and error in the 1960s, justified its use on local technical grounds (asymptotic distribution theory of Dickey-Fuller, residual stationarity for ARMA estimation), and then promptly threw away the level component to proceed on the differenced series. A quarter-century before that, Hamilton had already declared the algebraic programme that would have given the triple its proper interpretation. The two trajectories did not meet; the

technical practice of econometrics produced the kinematic embedding as a modelling device without recognising what it had produced, and the algebraic tradition that would have made the recognition available was, by the 1960s, not part of the working repertoire of the field.

The historical observation has a corollary that we want to make explicit. The success of the ARIMA modelling protocol — the fact that it works empirically on a wide range of economic and financial time series — is not, on the reading we develop here, a coincidence. It is the silent operational ratification of the fact that the kinematic embedding is the right object. Box and Jenkins's protocol succeeds because it tacitly respects the kinematic structure of the data, even though it does not name what it is respecting; the ARIMA orders that practitioners empirically find to be appropriate ($d \leq 2$ in the overwhelming majority of applications) coincide with the coupling orders $k^* \in \{0, 1, 2\}$ that the Geometric Takens Theorem identifies as the operationally relevant family. What ARIMA cannot do — and what GSD does — is read the level and the steps as a single algebraic object whose structure carries the dynamics, rather than as an auxiliary transformation pipeline whose intermediate stages are discarded.

Extension to unit-root and cointegration tests. The same diagnostic applies, with no qualification, to the inferential machinery that grew on top of the ARIMA framework. The Dickey-Fuller test of Dickey & Fuller (1979) reparametrises the unit-root regression as a regression of Δy_t on y_{t-1}, which is the two-dimensional kinematic embedding $(y_{t-1}, \Delta y_t)$ in the lag-and-difference basis: the choice that makes the test statistic asymptotically non-degenerate is the choice of the kinematic embedding, although it is not named as such. The Engle-Granger cointegration procedure (Engle & Granger, 1987) runs a Dickey-Fuller test on the residual of a static regression between two integrated series, which is therefore a unit-root test on the kinematic embedding of the cointegrating residual. The Johansen vector error-correction machinery (Johansen, 1988, 1991) regresses $\Delta \mathbf{y}_t$ on $\mathbf{y}_{t-1}$ and on lagged $\Delta \mathbf{y}_{t-j}$, which is by construction a kinematic- embedding regression in the multivariate setting; the cointegrating coefficient matrix β is recovered as a parameter of the regression where the GSD reading would recover it as a cross-level bivector $\mathbb{E}[\mathbf{q}^{(X_i)}(t) \wedge \mathbf{q}^{(X_j)}(t)]$ read directly off the multivariate kinematic field (Paper 03 , GC). The pattern is uniform across the three procedures: the kinematic embedding is present operationally and absent conceptually. We develop the algebraic re-reading of these specific tests in detail in Section 6.5 of Act V and refer the reader interested in the full re-articulation to that treatment.

4.3 The proliferation of patches

The decades following the consolidation of the spectral- econometric inheritance in Box-Jenkins and Sims were marked by the emergence, one after another, of a family of methodological patches addressing specific empirical features that the basic framework could not natively accommodate. We enumerate the most influential of these here, with the aim of identifying which algebraic feature of the data each patch is recovering and which grade of

the kinematic embedding the recovery would correspond to under the geometric reading.

Conditional heteroscedasticity (ARCH, GARCH). Engle (1982), in the autoregressive-conditional- heteroscedasticity model, observed that the variance of the residuals of a stationary mean model could itself follow an autoregressive structure, and proposed estimating the variance as $\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \, \varepsilon_{t-i}^2$. Bollerslev (1986) extended this to the generalised model GARCH(p, q) with autoregressive variance dynamics. The patch recovers *second-order temporal structure of the residual*, which under the abelian inheritance is invisible because the spectral density assumes second-moment stationarity. Under the geometric reading, the time-variation of σ_t^2 is captured non-parametrically by the bivector $b_{12}(t, \tau)$, which encodes the rotation of the level-velocity plane of the embedding; ARCH-GARCH is, in this reading, the scalar projection of grade-2 content of the kinematic embedding back onto a parametric scalar variance representation.

Cointegration (Engle-Granger, Johansen, VECM). The recognition that two unit-root series might share a common stochastic trend, with stationary residuals around a linear combination, gave rise to the cointegration literature of the 1980s and 1990s (Engle & Granger, 1987; Johansen, 1988, 1991). The patch recovers *cross-series directional structure*, expressed as a linear restriction $\beta' \mathbf{Y}_t = u_t$ on the joint process. Under the geometric reading, the cointegrating direction is the cross-level bivector $\mathbb{E}[\mathbf{q}^{(X)}(t) \wedge \mathbf{q}^{(Y)}(t)]$ in $\mathrm{Cl}(2d, 0)$, with the cointegrating coefficient β recovered as the ratio of two bivector components and no separate estimation step required (Paper 03 , GC). The Johansen procedure's reduced-rank machinery is, when re-read through the lens of Paper 08 (MPRT), a precursor of the random-matrix-theoretic rank estimator on the multivariate bivector field.

Time-varying parameter and state-space models. Kim & Nelson (1999) and predecessors articulated the use of state- space representations with time-varying parameters as a way of accommodating slow non-stationarity in econometric models. The patch recovers *drift in the autoregressive coefficients themselves*, which under the abelian inheritance is incompatible with the stationarity assumption of Wold's decomposition. Under the geometric reading, drift in the coefficients is drift in the bivector signature $(\Phi^{(0)}(t), \Phi^{(2)}(t))$ over windows (Paper 05 , GTE), and the windowed-Wasserstein-distance machinery proposed for Paper 10 of the GSD programme makes the drift quantitatively measurable.

Stochastic volatility and dynamic conditional correlation. Shephard (2005) catalogued the stochastic-volatility family in which the variance is itself a latent stochastic process; the multivariate extensions through dynamic conditional correlation (Engle, 2002) couple the volatility processes of multiple series via time-varying correlation matrices. The patches recover *joint asymmetry of distributional structure*, including asymmetric leverage effects in financial returns. Under the geometric reading, joint asymmetry is the trivector content $\mathbb{E}[\mathcal{T}^Y]$ for univariate fragility (Paper 16 , GF) and the cross-trivector content $\Phi^{(3)}_{X \to Y}$ for bivariate spillover (Paper 17 , FC), and the entire copula-DCC machinery is, in

retrospect, an attempt to reach grade-3 content via successive scalar projections of higher cumulants.

Common factor. The enumeration is not exhaustive — we omit the long-memory ARFIMA, the threshold and Markov-switching models, the regression- based structural-break tests, the wavelet-based time-frequency methods, and several others — but the structural pattern is the same across all of them. Each patch identifies an empirical feature of the data that the basic spectral framework cannot natively accommodate; each adds a parametric or semiparametric device to recover a scalar projection of that feature; and each leaves the underlying scalar real-valued representation of the process intact. The cumulative effect of half a century of patches is the heterogeneous methodological landscape that the applied analyst now navigates, in which the choice between ARIMA, ARFIMA, GARCH, EGARCH, GJR-GARCH, ARCH-M, IGARCH, FIGARCH, VECM, TVP-VAR, MS-VAR, copula-DCC, dynamic-factor models and a dozen further variants is made by appeal to a combination of empirical fit, computational tractability, and tradition. The landscape is not a coherent algebra; it is a accumulation of fixes that together cover the empirical territory the basic framework left exposed.

The geometric reading does not propose to discard any of these patches. Each is a legitimate response to a specific empirical recalcitrance, and each contains content that the geometric framework also has to recover. What the geometric reading proposes is that the underlying object the patches are trying to recover is, in every case, a specific grade or cross-grade component of the kinematic multivector, and that the natural articulation of the underlying object is in the algebra in which its grade structure is intrinsic rather than reconstructed parametrically.

4.4 The Spectral Identity of Paper 02

A central technical bridge between the abelian-spectral inheritance of Acts II and III and the non-abelian geometric apparatus of Act V is the *Spectral Identity* of Paper 02 (Spectral, Theorem 1), which we record here because it makes the connection structurally explicit.

For a weakly stationary univariate process $\{y_t\}$ with autocovariance $\gamma(\tau) = \mathbb{E}[y_t\, y_{t+\tau}]$ and spectral density $f(\lambda)$ on $[-\pi, \pi]$, the mean of the level-velocity component of the lagged geometric product, $b_{12}(t,\tau) = y_t\, \Delta y_{t+\tau} - \Delta y_t\, y_{t+\tau}$, satisfies the identity

$$\mathbb{E}\big[\, b_{12}(t,\tau)\,\big] \;\propto\; \gamma(1) \;-\; \gamma(0) \;=\; \int_{-\pi}^{\pi} \big(\cos\lambda \;-\; 1\,\big)\, f(\lambda)\, \mathrm{d}\lambda \;\leq\; 0, \tag{6}$$

with equality if and only if f has a unit-root atom at $\lambda = 0$. The identity (6) expresses the bivector mean — a grade-2 algebraic functional of the kinematic embedding — as a weighted integral of the abelian spectral density, with the weight $\cos\lambda - 1$ vanishing at zero frequency and strictly negative elsewhere. The bivector mean is, in this sense, *the spectral integral of an antisymmetric weight against the ordinary spectral density of the process*, and the connection between the abelian inheritance and the non-abelian apparatus is not metaphorical but algebraic.

Three observations on the identity are worth recording for the present argument.

First, the identity provides a *translation key* between the grade-2 mean of GSD and the spectral integrals familiar to practitioners trained in the spectral-econometric tradition. A researcher accustomed to thinking about a stationary process in terms of its autocovariance or its spectrum can compute the bivector mean from (6) without leaving the spectral framework; the geometric content emerges as a specific weighted integral, and the strict negativity of the weight $\cos\lambda - 1$ on $\lambda \neq 0$ is the spectral fingerprint of the rotational content the bivector recovers. The identity is the operative point of contact between two ways of reading the same data.

Second, the identity makes precise the sense in which the classical Dickey-Fuller test of Dickey & Fuller (1979) is a spectral measurement. A unit root corresponds to spectral mass at $\lambda = 0$, where the weight $\cos\lambda - 1$ vanishes; the bivector mean accordingly vanishes for unit-root processes and is strictly negative for stationary ones. The Geometric Dickey-Fuller test of Paper 01 (GDF) reads the unit-root null off the bivector mean directly, without the intermediate step of fitting a regression and computing the asymptotic distribution of a $\hat{\rho}$ statistic. The two tests measure the same underlying spectral object; the geometric form measures it more directly.

Third, the identity (6) is the analogue, for the grade-2 channel, of the relationship between the autocovariance and the spectral density itself. Just as $\gamma(\tau) = \int e^{i\lambda\tau} f(\lambda)\mathrm{d}\lambda$ is the abelian Fourier inversion that recovers the autocovariance from the spectrum, $\mathbb{E}[b_{12}(\tau)]$ is recovered from the spectrum by the corresponding antisymmetric weight. Higher-grade analogues exist for $\mathbb{E}[\mathcal{T}(\tau_1, \tau_2)]$ and for the multivariate cross-grade objects, expressing each grade-wise mean as a spectral integral against a grade-specific weight. The abelian Fourier representation is, in this reading, not the *end* of the analysis but the *starting point* from which the higher-grade structure is reachable; the bridge is (6) and its grade-3 and cross-grade generalisations.

The identity reinforced from several angles in the GSD programme. The Spectral Identity does not stand alone in the GSD programme; it is reinforced by a family of complementary results that articulate different aspects of the same spectral- geometric correspondence. We catalogue four of them here because each adds an independent line of support to the diagnostic (6) and because their joint content is the operational reason to take the bridge as more than a single-paper phenomenon.

(i) Beveridge-Nelson decomposition. For an $I(1)$ process $\Delta y_t = \psi(L)\,\varepsilon_t$ admitting the Beveridge- Nelson decomposition $y_t = \tau_t + c_t$ with τ_t the permanent random-walk component and c_t the transitory stationary component, Paper 02 (Spectral) establishes a *permanent–transitory split between grades*: the scalar mean $\bar{s}$ scales with the long-run multiplier $\psi(1)^2 \propto f(0)$ and therefore reads the permanent component, while the bivector mean $\bar{b}_{12}$ scales with the integral $\int(\cos\lambda - 1)\, f_c(\lambda)\,\mathrm{d}\lambda$ over the spectral density f_c of the transitory component (note $f_c(0) = 0$ by construction) and therefore reads the transitory component. Grade zero captures permanence; grade two captures reversion. The split is not a heuristic interpretation; it is the algebraic shape of the lagged geometric product as projected onto

the Beveridge-Nelson representation.

(ii) Distance from the unit circle. For an AR(p) with characteristic polynomial $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$, Paper 02 (Spectral, Proposition B(iii)) establishes that $|\mathbb{E}[\bar{b}_{12}]|$ is monotone increasing in the distance of the dominant root of $\phi(z) = 0$ from the unit circle. The bivector mean is therefore not merely a binary unit-root indicator; it is a continuous proximity measure whose magnitude encodes the spectral gap. The geometric reading turns the spectral density into a quantitatively legible directional field, with the strength of the reversion encoded in the bivector magnitude.

(iii) Multivariate cross-level extension. Paper 03 (GC) extends the identity to the multivariate setting in $\mathrm{Cl}(2d, 0)$ for cointegrated systems: the cross-level bivector $\mathbb{E}[\mathbf{q}^{(X)}(t) \wedge \mathbf{q}^{(Y)}(t)]$ between two integrated series is the geometric analogue of the cointegrating direction in the Engle-Granger and Johansen frameworks (Engle & Granger, 1987; Johansen, 1988, 1991), with the cointegrating coefficient β recovered as a ratio of bivector components rather than estimated by a separate regression step. The spectral identity (6) generalises to a *cross-spectral* identity in which the weight is the imaginary part of the cross-spectrum, and the spectral conjecture of Paper 05 (GTE, Eq. 5) — $\Phi^{(2)}_{XY} \geq c \cdot |\mathrm{Im}\, f_{XY}(\lambda^*)|$ — is the multivariate-causal analogue at the level of grade-resolved transfer entropy.

(iv) Grade-3 spectral extension. The Jarzynski-Crooks correspondence of Paper 18 (JC, Theorem 2) extends the spectral reading to grade three: under AR(1) skew-normal innovations, the discrete entropy production rate $\langle \Sigma \rangle$ is $\kappa(\phi)\, \mathbb{E}[\mathcal{T}(1,2)]^2 + O(\gamma_3^4)$, with the proportionality constant $\kappa(\phi)$ derived as a closed-form spectral functional of the AR coefficient. The grade-3 mean is therefore the spectral signature of irreversibility, in exactly the sense in which the grade-0 and grade-2 means are the spectral signatures of permanence and reversion respectively. The three- grade ladder $(\bar{s}, \bar{b}_{12}, \bar{\mathcal{T}})$ is the grade-resolved spectral signature of the kinematic state, with each grade reading off the spectral density a structurally distinct component of the dynamics.

(v) Empirical validation. The spectral identity is not purely theoretical content. Paper 02 (Spectral, Section 4) reports a FRED-MD application on 805 monthly observations covering 1959–2026, in which the bivector mean cleanly separates 9 $I(1)$ series ($|\bar{b}_{12}| < 0.10$) from 5 stationary series ($|\bar{b}_{12}| > 0.17$, with VIX at -0.528 as the strongest mean-reverting case), and identifies 4 interest-rate spreads as cointegrated stationary combinations through their intermediate $\bar{b}_{12} \in [-0.44, -0.32]$. The empirical fingerprint of the spectral identity in real macroeconomic data is therefore concretely available, and the geometric reading we develop is not asymptotic-only or simulation-only but operationally validated on the same datasets that the spectral-econometric mainstream uses for its own diagnostics.

The five angles above — Beveridge-Nelson grade split, root- proximity monotonicity, cross-level extension, grade-3 spectral signature, FRED-MD validation — are not independent theorems but coherent facets of the single spectral-geometric correspondence that (6) initiates. They collectively support the reading that the abelian Fourier dual is not the bottom layer of the analysis but the projection onto grade zero of a richer multigrade spectral representation that the geometric algebra makes operationally legible.

4.5 Expanding the notion of uncertainty

The cumulative effect of the spectral-econometric consolidation is that, in the working vocabulary of late twentieth- and early twenty-first-century applied analysis, the word *uncertainty* acquired a specific and structurally narrow meaning. The uncertainty of a stationary time series is, by default, the variance of its residual after the conditional mean has been fit; the heteroscedasticity literature extends this to a time-varying variance under the same scalar interpretation; the copula and DCC literatures extend the joint-distributional asymmetry of the multivariate residual but again under successive scalar functionals of higher cumulant order. The structurally common element is that uncertainty, on this reading, is a *scalar magnitude*: a single non-negative number per instant of time, possibly varying with time, possibly carrying joint structure across components, but in every case ultimately reducible to scalar functionals of a real-valued residual.

The geometric reading proposes — and the GSD programme operationally demonstrates — that this scalar conception is the grade-zero projection of a structurally richer object. Under the kinematic embedding into $\mathrm{Cl}(3,0)$, the uncertainty content of a stationary process decomposes algebraically into three independent quantities, each carrying its own information and its own operational interpretation. The decomposition is not a refinement of the scalar variance into finer scalar functionals; it is a *partition* of the uncertainty into structurally distinct algebraic objects, made precise by the antisymmetric-entropy decomposition $H_{AS} = H_0 + H_2$ of Paper 04 (HAS) and its three-grade extension $H_{\mathrm{geom}} = H_0 + H_2 + H_3 - \mathrm{TC}$ of Paper 18 (JC), with the total-correlation remainder empirically negligible ($\mathrm{TC}/H_{\mathrm{geom}} < 0.5\%$ on canonical generative processes; see Paper 18 , JC, Theorem 1).

The structural content of the partition is worth making explicit and concrete. The grade-zero entropy H_0 measures the *persistence uncertainty* of the process: how unpredictable is the level y_{t+1} given the history up to t, in the information-theoretic sense of the differential entropy of the conditional distribution of the level alone. This is the classical conception of uncertainty as carried by the spectral- econometric inheritance, and it remains a meaningful and estimable quantity under GSD. The grade-2 entropy H_2 measures the *rotational uncertainty* of the process: how unpredictable is the change in the bivector encoding the level-velocity- acceleration plane of the kinematic state. A process with high H_0 but low H_2 has unpredictable levels but a stable direction of kinematic evolution; a process with low H_0 but high H_2 has predictable levels but a fluctuating direction of evolution. The two are independent in the algebraic sense (Lemma 1 of Paper 04 , HAS) and empirically nearly independent in the statistical sense, and a process can score arbitrarily on one without constraining its score on the other. The grade-3 entropy H_3 measures the *volumetric or chiral uncertainty*: how unpredictable is the orientation-asymmetry of the trivector across three lagged states. Under time-reversibility, $\mathbb{E}[\mathcal{T}] = 0$ exactly and H_3 encodes pure dispersion of the chiral content; under irreversibility, $\mathbb{E}[\mathcal{T}] \neq 0$ and H_3 encodes both dispersion and a directional bias whose square is proportional, in the Gaussian-Markov regime, to the Jarzynski-Crooks entropy production rate of the underlying process (Paper 18 , JC,

Theorem 2).

The empirical signature of the expansion is visible in cases where the three components carry strikingly different information about the same data. The Jarzynski-Crooks proxy $|\mathbb{E}[\mathcal{T}]|^2/\mathrm{Var}(\mathcal{T})$ applied to the SP500 around the 1929 crash returns a z-score of 4.30 in the year of the crash, while the scalar-volatility content for the same period is unremarkable (Paper 18 , JC, Section 6). Seven distinct foreign-exchange crises between 1971 and 2026 are picked up by the same proxy as top-five peaks aligned with documented events, and the alignment is structural rather than coincidental: the scalar variance was, in each case, a conditioned response to the event, while the chiral asymmetry $\mathbb{E}[\mathcal{T}]$ was a leading indicator of the breakdown of detailed balance in the system. The empirical work of Paper 17 (FC) on macro-financial transmission and of Paper 25 (GE) on regime classification provides further demonstrations of the same point: the grade-resolved decomposition of uncertainty captures structural content that the scalar inheritance silently flattens.

The expansion has a final aspect that we want to register explicitly because it bears on the philosophical reading of Act I. Under the scalar inheritance, uncertainty is a single quantity that admits more or less of itself; a process is more or less variable, more or less predictable. Under the geometric expansion, uncertainty is three structurally distinct quantities that admit independent variation; a process can be more variable in one channel and more predictable in another, and the two movements are not conflated. This is, in the language of the Port-Royal logicians, a recovery of the *intension* of the notion of uncertainty: where the scalar inheritance gave only the extension (the magnitude under the chosen measure), the geometric expansion gives the directional content of that magnitude (its distribution among the three structurally distinct algebraic channels). The recovery is, in this respect, the operational realisation of the philosophical movement that Section 2.1 of Act I traced from Pascal's defence of projectivity through Port-Royal's distinction between intension and extension.

Energy by grade as the dual of entropy by grade. The entropy decomposition $H_{\mathrm{geom}} = H_0 + H_2 + H_3 - \mathrm{TC}$ is one face of the grade-resolved uncertainty content; the other is the grade-resolved *energy*. For each grade $g \in \{0, 2, 3\}$ the quadratic functional

$$E_g \;:=\; \left\|\mathbb{E}[\mathbf{X}_g\,\mathbf{X}_g^\top]\right\|_F^2 \;=\; \mathrm{tr}\big(\mathbb{E}[\mathbf{X}_g\,\mathbf{X}_g^\top]^2\big) \tag{7}$$

of the grade-g random variable $\mathbf{X}_g$ measures the average squared magnitude of the channel — the rotational energy of the bivector field for $g = 2$, the persistence energy of the scalar field for $g = 0$, the volumetric energy of the pseudoscalar for $g = 3$. The Frobenius energy of the bivector field E_2 was already introduced in Paper 04 (HAS, Proposition 2), where it is shown that E_2 and H_2 are quantitatively independent across canonical generative processes ($\rho(H_2, E_2) \approx 0.2$–$0.4$): a process can be energetic and predictable, energetic and chaotic, weak and predictable, or weak and chaotic, with the four cases distinguishable from one another by the joint reading of the two functionals. The same independence holds, by direct extension, at grades zero and three.

The pair (E_g, H_g) for each grade is the discrete analogue of the dispersion-magnitude pair that Hitzer and Mawardi's (Hitzer & Mawardi, 2013) multivector Heisenberg principle binds together at the level of multivector-valued distributions on $\mathbb{R}^3$. Where the Heisenberg principle in Section 6.4 of Act V states that the dispersion of a multivector-valued function is bounded below by a function of the dispersion of its Clifford Fourier transform, the energy-entropy pair on the kinematic embedding states that the average squared magnitude and the differential entropy of each grade-resolved channel are independently free quantities of the process, with no compositional rule reducing one to the other. The two structures are dual and they live on the same algebra; the spectral-frequency Heisenberg of Hitzer-Mawardi and the energy-entropy partition of the grades are two faces of the same fact about uncertainty in $\mathrm{Cl}(3,0)$.

The MSDS as the maximally granular reading. The energies and entropies of (7) and (13) are summary functionals of the grade-wise distributions; they reduce each grade to one or two scalars. The *Multigrade Signal Dynamics Signature* (MSDS), formalised in Paper 24 (MT, Section 2), is the maximally granular reading of the same algebraic content. For each of the sixteen Clifford channels of the kinematic path $\mathbf{q} : [0, T] \to \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ — three vector components q_1, q_2, q_3, four scalar/bivector products of $\mathbf{q}(t)\,\mathbf{q}(t+\tau_i)$ for each of three lags τ_i, and one trivector $\mathcal{T}(t, \tau_1, \tau_2)$ — the MSDS records the first four Pearson moments (mean, variance, standardised skewness, standardised excess kurtosis), giving a total of $16 \times 4 = 64$ coordinates. Where H_g and E_g each compress a grade into a single number, the MSDS preserves the moment-by-moment structure of every channel and lag. The compression matters when summary suffices; the granularity matters when the structural content of the joint moments is the substantive object — for classification of generative processes (Paper 24 , MT, Section 8), for empirical regime detection (Paper 25 , GE, Section 6), for fragility-causality spillover analyses (Paper 16 , GF; Paper 17 , FC), and in general for any application in which the joint behaviour of the grades is more informative than either grade alone.

The MSDS is, in this sense, the operational realisation of the *Leibnizian expressiveness* that the kinematic monad of Section 2.4 announced philosophically. A Leibnizian monad contains, internally, the entire dynamic content of the system at its instant; the MSDS is the empirical-statistical realisation of that internality, in which 64 coordinates articulate the local kinematic state along sixteen direction- bearing channels and four orders of statistical content. The expressiveness comes not from any single coordinate but from the joint articulation: it is the MSDS as a whole, not its individual entries, that captures the dynamic monad of a series; and the information-theoretic upper bound on what can be extracted from the path is, by the Hambly-Lyons uniqueness theorem (Hambly & Lyons, 2010), the truncation order of the Lyons signature on the same kinematic path (Paper 24 , MT, Theorem 5). Granularity is the operational price of expressiveness, and the MSDS pays it.

Energy-entropy ratios and the diagnostic algebra of relations. The dual structure of E_g and H_g at each grade admits a natural family of *ratios* that operate as diagnostic indices for the dynamic regime of the process. The ratio E_g/H_g — energy over entropy at grade g — admits a four-fold typology: high E_g with low H_g corresponds to a process whose grade-g content is energetic and predictable (strong rotation in regular pattern, for $g = 2$); low E_g with high H_g to a process whose grade-g content is weak and chaotic; high-high to energetic and chaotic; low-low to a near- stationary regime in that grade. The four-fold typology is available independently at each grade, giving a $4^3 = 64$-cell joint regime classification that no scalar functional of the process can reach. The diagnostic content of the ratio is the *relational structure* between the two faces of the grade-wise uncertainty, and the joint reading across grades is the *relational structure* between the three grades themselves.

This is the place at which the philosophical content of the act becomes operationally concrete. The expressiveness of the kinematic monad is not only granular (in the MSDS sense) but *relational*: the dynamic content of the process is not exhausted by any one channel or any one grade or any one moment, and it is not even exhausted by their joint listing — it lives in the relations among them, in the ratios and joint regimes that the algebraic structure of $\mathrm{Cl}(3,0)$ makes legible. The information-theoretic decomposition gives the floor (independent quantities at each grade); the energy decomposition gives the ceiling (joint magnitudes); the MSDS gives the granular reading (four moments per channel); and the relational diagnostics (E_g/H_g ratios, intra-grade and cross-grade comparisons, windowed signatures, ergodicity exponents) give the interpretative layer that turns the granularity into intelligible content. The whole structure is a recovery, on empirical- statistical data, of the kind of articulated representation that Hamilton declared in 1837 as the operative target of *algebra as the science of Pure Time* and that the scalar inheritance, for the algebraic reasons traced in Acts II and III, could not perform.

4.6 Synthesis of the act

Three propositions consolidate the act and prepare the transition to Act IV.

First, the spectral-econometric inheritance was industrialised in the third quarter of the twentieth century by Box-Jenkins and Sims into the operational toolkit that has dominated applied time-series analysis for the half-century since. The toolkit was not a free-standing intellectual construction but the engineering wrapping of the underlying probabilistic apparatus — Wold, Cramér, Kolmogorov — into a deployable protocol; its mathematical content is the abelian Fourier dual of stationary processes, and its successes and limitations are the successes and limitations of that dual.

Second, the operational practice of the toolkit produced, at several points, blind premonitions of the algebraic apparatus that GSD now articulates. The differencing operator of ARIMA (5) produces, term by term, the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$; the ARIMA depth $d \leq 2$ that empirically suffices for economic and financial data coincides with the coupling-order family $k^* \in \{0, 1, 2\}$ that the Geometric Takens Theorem identifies

as operationally relevant; the Dickey-Fuller test reads the spectral mass at zero frequency that the Spectral Identity (6) identifies with the unit-root signature of the bivector mean. What the toolkit lacks is not the right object but the right language for the object.

Third, the proliferation of patches that has characterised the last fifty years of methodological development in applied time series — GARCH, cointegration, TVP-VAR, copula-DCC, state-space — is the systematic accumulation of scalar projections of higher- grade content. Each patch is a legitimate response to a specific empirical recalcitrance of the abelian framework, and each recovers a piece of structure that the abelian framework cannot natively accommodate; the patches do not, in their cumulative effect, constitute a coherent algebra, but they do constitute the empirical demonstration that there is a coherent algebra waiting to be articulated. The articulation is the content of the algebraic apparatus that GSD adopts as foundational, and the expansion of the notion of uncertainty into the three-grade decomposition $H_{\text{geom}} = H_0 + H_2 + H_3 - \text{TC}$ is the partial consolidation of the patches under that articulation.

The fourth act takes up the parallel mathematical traditions — Takens' embedding theorem in nonlinear dynamics, Lyons' rough-path theory in stochastic analysis — that recovered, independently of the spectral-econometric mainstream and from outside the empirical-statistical context, the same algebraic object that ARIMA was implicitly working with. The triple bridge between Taylor's calculus, Takens' topological reconstruction, and Lyons' asymptotic functional representation is what completes the algebraic-empirical synthesis on the side of representation, and prepares the ground on which the fifth act has already articulated the synthesis as instantiation of the Hamiltonian programme.

5 Act IV — Takens, Lyons, and the triple bridge

The previous two acts traced a divergence: the Hamiltonian algebraic-geometric tradition completed by Clifford in 1878 became mathematically available, the Kolmogorov measure-theoretic foundation completed in 1933 made the synthesis with empirical data structurally possible, but the spectral-econometric mainstream consolidated by Wold and Cramér and industrialised by Box-Jenkins and Sims did not perform the synthesis. The fourth act records that, during the same decades in which the empirical- statistical mainstream was developing in the abelian register, two parallel mathematical traditions outside that mainstream were recovering the directional-kinematic content of dynamic data through different routes. *Nonlinear dynamics* recovered it in 1981, when Takens established that a single observable of a dynamical system carries the topological information of the full attractor through the lag embedding. *Rough-path analysis* recovered it from 1998 onwards, when Lyons established that the asymptotic functional content of a continuous path is captured by its iterated-integral signature. The two recoveries proceeded in almost complete isolation from the spectral-econometric tradition and, until recently, from each other; and yet, when read together with the discrete kinematic calculus of Newton, they form a single triple bridge whose three sides — local-analytic, topological, asymptotic-functional — converge on the same algebraic object that the Hamiltonian programme had identified as its target. The present act articulates the bridge as architecture, and prepares the ground on which Act V has already articulated the synthesis as instantiation of Hamilton's *Researches*.

5.1 Takens 1981: topology from a single observable

The embedding theorem of Takens (1981) was published as a short paper in the Springer *Lecture Notes in Mathematics* volume *Dynamical Systems and Turbulence*, edited by Rand and Young in the wake of the conference at the University of Warwick of the same name in 1980. The conference was a meeting point of the early nonlinear-dynamics community, with contributions from Ruelle, Eckmann, Yorke, Mandelbrot and others, and Takens' contribution was structurally distinct from the bulk of the volume in being purely mathematical: a theorem on generic properties of smooth maps under a particular embedding construction, with no specific application or empirical illustration. The theorem is short and worth stating in full.

Theorem 5.1 (Takens 1981, informal statement)**.** *Let M be a smooth compact manifold of dimension d and let $\Phi : M \to M$ be a smooth diffeomorphism whose periodic orbits are isolated and transverse. Let $h : M \to \mathbb{R}$ be a generic smooth observable. Then for $D \geq 2d+1$ the map $\Psi_h : M \to \mathbb{R}^D$ defined by*

$$\Psi_h(x) \;=\; \big(\, h(x), h(\Phi(x)), h(\Phi^2(x)), \ldots, h(\Phi^{D-1}(x)) \,\big)$$

is an embedding of M into $\mathbb{R}^D$. The image $\Psi_h(M) \subset \mathbb{R}^D$ is therefore diffeomorphic to M itself.

The theorem says, in terms a practitioner can use, that the topology of the attractor of a dynamical system can be recovered from a sufficiently long lag embedding of any one observable of the system, even if the observable is the only window onto the system that is empirically available. The dynamical system itself need not be known; only the observable is required. The number $D \geq 2d + 1$ is generic — Whitney's embedding theorem is the classical analogue in differential topology — but the empirical content of Takens is sharper: a single one-dimensional time series contains, modulo the technical conditions on the system and the observable, the full topological content of the underlying attractor.

The reception of the theorem in the nonlinear-dynamics community was rapid and substantial. Packard et al. (1980) had introduced the lag embedding empirically a year earlier without a formal theorem behind it, and Takens supplied the formal warrant. Subsequent decades saw the technique deployed across physics, biology, neuroscience and chemical engineering, with extensive methodological literature on the choice of the embedding lag and dimension (Fraser & Swinney, 1986; Kennel et al., 1992), on the estimation of attractor dimension (Grassberger & Procaccia, 1983), and on the detection of determinism in noisy data (Theiler et al., 1992). The theorem did not, however, cross over into the empirical-statistical mainstream of econometrics or applied statistics. Two reasons account for the absence. First, the theorem assumes a deterministic dynamical system on a smooth manifold, and the empirical processes of econometrics are fundamentally stochastic — the assumption is too strong for the domain. Second, the methodological literature that grew around Takens emphasised the qualitative recovery of attractor topology (dimension estimation, Lyapunov exponents, surrogate-data tests for determinism), which the spectral-econometric tradition had no operational use for. The mainstream econometrics of the 1980s and 1990s read Takens, when it read him at all, as a tool for chaotic- deterministic modelling outside its domain of interest, rather than as a structural statement about the algebraic content of any time-series representation.

The reading we develop in the present paper takes a different position on Takens' theorem. The theorem's deepest content, on our reading, is not the chaotic-deterministic application but the *algebraic equivalence* between the lag-embedding representation $(y_t, y_{t-\tau}, \ldots, y_{t-(D-1)\tau})$ and the local kinematic state of a system. The lag embedding is, up to a linear change of basis, the kinematic embedding $(y_t, \Delta y_t, \Delta^2 y_t, \ldots)$ that Newton's calculus would produce; the equivalence is exact at finite order, and we make the connection precise in the next subsection. Once the equivalence is recognised, Takens' theorem stops being a specifically chaotic-determinism result and becomes a structural warrant for the kinematic-state representation as such. The theorem applies to deterministic systems; the structural content applies wherever the kinematic embedding does, including discrete stochastic time series, where the topological reconstruction in Takens' sense generalises to a Markov-of-order- D representation of the conditional distribution of the future trajectory given the present kinematic state.

5.2 The Pascal matrix and the equivalence of bases

The technical observation that converts Takens' theorem from a chaotic-determinism result into a structural statement about the algebraic content of the kinematic representation is the formal equivalence between the lag-embedding basis and the kinematic- embedding basis. The equivalence is recorded as a lemma in Paper 07 (Takens) and is sufficiently brief to be reproduced here.

Lemma 5.2 (Pascal equivalence)**.** *For any positive integer d, the lag-embedding map $L_d : (y_t)_{t\in\mathbb{Z}} \to (y_t, y_{t-1}, \ldots, y_{t-d+1}) \in \mathbb{R}^d$ and the kinematic-embedding map $K_d : (y_t)_{t\in\mathbb{Z}} \to (y_t, \Delta y_t, \Delta^2 y_t, \ldots, \Delta^{d-1} y_t) \in \mathbb{R}^d$ are related by the linear isomorphism $K_d = M_d \cdot L_d$, where the matrix $M_d \in \mathbb{R}^{d\times d}$ has entries*

$$(M_d)_{i,j} \;=\; (-1)^{i-j}\binom{i-1}{j-1}, \qquad 1 \le i, j \le d,$$

giving an upper-triangular matrix of binomial coefficients with alternating signs. The matrix M_d is unimodular ($\det M_d = 1$*) and lies in the special linear group* $\mathrm{SL}(d, \mathbb{Z})$*.*

The lemma is elementary but has substantive content.[2] The two embedding bases — the one Takens' theorem operates on and the one Newton's calculus produces — span the same d-dimensional subspace of the sequence space, and the change of basis between them is given by a fixed integer matrix that does not depend on the data. Any algebraic functional of the embedding that is invariant under the linear SL-action — and in particular any geometric-product functional in the Clifford algebra over the embedding — takes the same value in either basis. The lag embedding and the kinematic embedding are *the same algebraic object*; only the choice of presentation differs.

The empirical consequence is the equivalence of two operative practices that the literatures of the 1980s and 1990s carried out without recognising their identity. The *lag-embedding-based* practice of nonlinear dynamics constructs $(y_t, y_{t-\tau}, \ldots)$ and reads attractor topology off the resulting cloud in $\mathbb{R}^D$; the *difference-based* practice of econometrics constructs $(y_t, \Delta y_t, \Delta^2 y_t)$ implicitly through ARIMA modelling and reads stationarity and unit-root content off the fitted coefficients. The two practices operate on the same algebraic object — they differ only in which basis they choose for the same span — and the algebraic content extracted by one is in principle reachable by the other. The non-recognition of this equivalence has been, on our reading, a structural barrier to the cross-fertilisation of the two literatures over the last forty years; recognition of the equivalence makes the cross-fertilisation immediate.

[2] The operator-theoretic relation between finite differences and differentials underlying the Pascal equivalence had been articulated in this form by Hamilton (1837) as the identity $F(D)\, f\psi(o) = f(1 + \Delta')\, F(D)\, (\psi(o))^{o'}$, generalising a theorem of Herschel and recording the correspondence between Δ-operators and D-operators in terms that the unimodular Pascal matrix M_d later cast in basis-canonical form. The operator-equivalence between differencing and differentiation has, on this reading, a Hamiltonian pedigree that pre-dates the modern formalisation by more than a century.

The choice between the two bases is not gratuitous from the perspective of statistical estimation. The lag-embedding basis has the advantage that each component has the same units (those of the underlying observable) and the same scale (under stationarity, equal sample variance); the kinematic basis has the advantage that each component has a clear physical-kinematic interpretation (level, velocity, acceleration) and that the algebraic structure of the geometric product is most legible when expressed in this basis. The standardisation discussed in Section 6.7 (full Mahalanobis whitening as the strong analogue of arc-length parametrisation; diagonal as its weakening) operates on the kinematic basis and restores the unit comparability across components that the lag basis has by default. Once standardisation is in place, the two bases yield the same statistical content, and the choice between them is a presentation choice rather than a substantive one.

5.3 Lyons' signature: the asymptotic functional

The third side of the triple bridge was supplied, much later, by Lyons (1998) and the rough-path analysis programme that grew from his initial paper. Where Takens' theorem operates on deterministic dynamical systems and yields a topological recovery, and where Newton's calculus operates on smooth analytic functions and yields a local algebraic recovery, Lyons' construction operates on continuous paths of bounded variation (or, more generally, of finite p-variation for some $p \geq 1$) and yields an asymptotic functional recovery: a complete identification of the path, modulo a controlled equivalence relation, by an infinite tower of iterated integrals that the path generates against itself.

The construction is most cleanly stated for paths $X : [0, T] \to \mathbb{R}^d$ of bounded variation. The *signature* of such a path is the sequence

$$\mathrm{Sig}(X) \;=\; (\,1,\; X^{(1)},\; X^{(2)},\; X^{(3)},\; \dots\,), \tag{8}$$

where each level is an iterated integral

$$X^{(k)}_{i_1,\dots,i_k} \;=\; \int_{0<t_1<t_2<\cdots<t_k<T} \mathrm{d}X^{i_1}_{t_1}\,\mathrm{d}X^{i_2}_{t_2}\,\cdots\,\mathrm{d}X^{i_k}_{t_k}$$

with indices $i_1, \dots, i_k \in \{1, \dots, d\}$, giving a d^k-dimensional tensor at level k. The signature is an element of the tensor algebra $T((\mathbb{R}^d))$ and is the natural generalisation of the moments of a probability distribution on $\mathbb{R}$ to functions valued in $\mathbb{R}^d$ over a continuous interval. Two structural properties of the signature are decisive for the present argument.

Multiplicativity (Chen). For any decomposition of the interval $[0, T]$ into two consecutive sub-intervals, the signature of the concatenated path is the tensor product of the signatures of the two pieces: $\mathrm{Sig}(X|_{[0,T]}) = \mathrm{Sig}(X|_{[0,s]}) \otimes \mathrm{Sig}(X|_{[s,T]})$. The tensor product is, in general, non-commutative, and the non-commutativity carries the directional content of the path — the same observation lifted into geometric algebra is the non-commutativity of the bivector products of $\mathrm{Cl}(p, q)$.

Uniqueness (Hambly-Lyons). The signature determines the path up to so-called tree-like equivalence: two paths with the same signature differ at most by a path that goes out

and returns along the same trajectory (Hambly & Lyons, 2010). For any path that is not tree-like — that is, for any path that does not fold back on itself — the signature is a complete invariant. Modulo this equivalence, the signature is the canonical observable of a path of bounded variation.

The connection to GSD is the substantive content of Paper 24 (MT, Theorem 5), which we have already cited in Sections 6.3–6.7 and which we restate informally here for completeness. The 16-channel kinematic path $\mathbf{C}(y;\boldsymbol{\tau}) : [0,T] \to \mathbb{R}^{16}$ associated with a univariate time series $\{y_t\}$ — comprising the three vector components q_1, q_2, q_3, the four scalar/bivector components of the lagged geometric product at each of three lags, and the trivector — has, by direct integration, a Lyons signature $\mathrm{Sig}(\mathbf{C})$. The empirical moment tensor $\mathcal{M}_y^{(c,k)}$ of $16 \times 4 = 64$ coordinates is, on the conjecture of Paper 24 (MT, Theorem 5), the level-≤ 4 truncation of $\mathrm{Sig}(\mathbf{C})$ projected onto the orthonormal Clifford basis $\{e_i, e_{ij}, e_{123}\}$. The truncation is the discrete-statistical version of the asymptotic identification, and the asymptotic identification is the continuum-limit version of the discrete construction. The two agree exactly in the limit $\Delta t \to 0$, and the consistency of the two readings is the technical content that the planned Paper 6 of the GSD programme on the continuum limit (Paper 06 , Continuum) would formalise rigorously once written.

The relevance of Lyons' signature for the present synthesis is not the technical machinery of rough paths in itself but the conceptual confirmation it provides. *The signature is, in the asymptotic-functional limit, what the moment tensor is in finite samples*, and the fact that the moment tensor of GSD is the truncation of the signature of the kinematic path is the quantitative warrant for the claim that GSD operates on the right object. The right object is the kinematic path — local- analytically (Newton-Taylor), topologically-globally (Takens), asymptotically-functionally (Lyons). The three readings are the same.

5.4 The triple bridge as architecture

The three readings — Newton-Taylor, Takens, Lyons — admit a joint reading as three projections of a single algebraic object onto three levels of granularity. The reading is implicit in the preceding subsections and we make it explicit here as the architectural content of the act.

Local-analytic level (Taylor). The kinematic state at an instant $(y, \dot{y}, \ddot{y})$ determines, in the smooth-analytic regime, the trajectory in a neighbourhood of the instant by Taylor expansion. The granularity is a finite-order jet at a point: a finite-dimensional algebraic object that captures the local content of the dynamics.

Topological-global level (Takens). The lag embedding $(y_t, y_{t-\tau}, \ldots, y_{t-(D-1)\tau})$ at sufficient $D \geq 2d+1$ recovers the topology of the attractor of a deterministic dynamical system. The granularity is a finite- dimensional embedding of a manifold: a finite-dimensional geometric object that captures the global topology of the dynamics.

Asymptotic-functional level (Lyons). The iterated-integral signature $\mathrm{Sig}(\mathbf{X})$ of a continuous path identifies the path uniquely up to tree-like equivalence. The granularity is an infinite-dimensional functional: an infinite-dimensional algebraic object that captures the

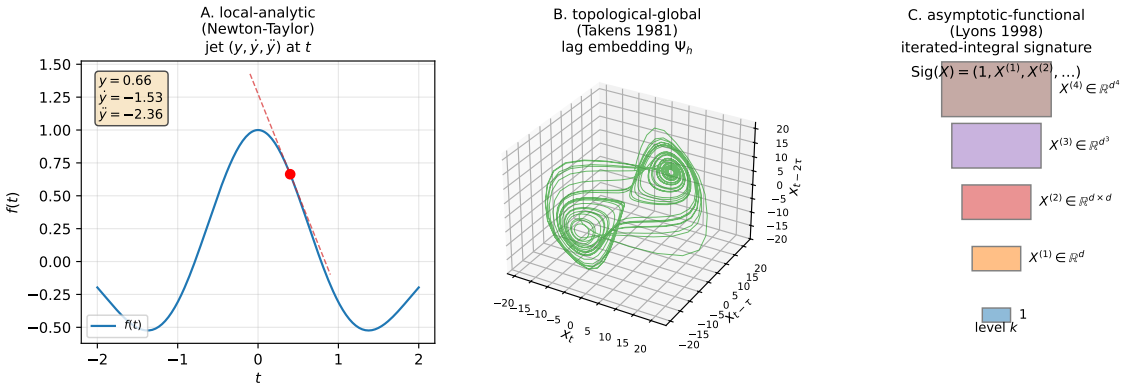

Figure 1: The triple bridge: Newton's kinematic state, Takens' lag-embedded attractor, and Lyons' iterated-integral signature as three levels of granularity of the same underlying algebraic object. The Pascal equivalence (Lemma 5.2) connects the local and the topological levels; the moment-tensor / signature equivalence of Paper 24 (MT, Theorem 5) connects the topological and the asymptotic levels; and Geometric Signal Dynamics is the discrete-statistical realisation that operates on all three simultaneously.

asymptotic functional content of the dynamics.

The Hamiltonian fourth corner: characteristic function. The triple bridge admits a fourth corner that the present paper records as historically significant although not algebraically independent of the three already articulated. Hamilton (1834), in his *General Method in Dynamics*, established that the entire dynamics of a system of attracting or repelling points is reducible to the study of a single function — the *characteristic function* V, identified with the accumulated action of the system from its initial to its final position — that satisfies a pair of partial differential equations whose solution determines the full trajectory of the system. He stated the programme in terms whose ambition is plainly visible:

> *[...] the reduction of the most complex and, probably, of all researches respecting the forces and motions of body, to the study of one characteristic function, the unfolding of one central relation. (Hamilton, 1834, Introductory remarks)*

The structural content of the Hamilton 1834 characteristic function, in the language of the present paper, is that the deterministic-continuum trajectory of a dynamical system admits a *single canonical compression* in the form of V, from which all observable kinematic content can be recovered by differentiation. Hamilton himself returned to this in the *Second Essay* of 1835, proposing the construction as the foundation of a separate branch of analysis: "*a separate branch of algebra, which may be called perhaps the Calculus of Principal Functions; because, in all the chief applications of algebra to physics, and in a very extensive class of purely mathematical questions, it reduces the determination of many mutually*

connected functions to the search and study of one principal or central relation" (Hamilton, 1835, Introductory remarks). The "central relation" that Hamilton 1834-1835 identified for deterministic dynamics is the structural ancestor of the single-object-canonical-compression that GSD performs on empirical-stochastic data. The empirical-stochastic analogue of V is the moment tensor $\mathcal{M}_y^{(c,k)}$ of Paper 24 (MT) that the Hambly-Lyons signature equivalence identifies as the truncated-functional canonical observable of the kinematic path. Hamilton 1834 is, on this reading, the deterministic-continuum fourth corner of the triple bridge: where Newton-Taylor gives the local jet, Takens the topological reconstruction, and Lyons the asymptotic functional, the Hamiltonian characteristic function gives the *central relation* from which all three are derivable in the smooth-deterministic limit. The chain of (5.4) extends, in this reading, to

$$\text{Newton-Taylor} \overset{\text{Taylor}}{\longleftrightarrow} \text{Hamilton 1834} \overset{M_D}{\longleftrightarrow} \text{Takens} \overset{\text{Thm. 5}}{\longleftrightarrow} \text{Lyons},$$

with the kinematic embedding of GSD as the discrete-stochastic realisation that operates on all four levels simultaneously.

The three levels are connected by two formal equivalences. The local and the topological levels are connected by the Pascal equivalence of Lemma 5.2: the lag-embedding basis that Takens operates on and the kinematic-embedding basis that Newton operates on are related by the unimodular matrix M_D, and the algebraic content of the embedding is the same in either basis. The topological and the asymptotic levels are connected by the moment-tensor / signature equivalence of Paper 24 (MT, Theorem 5): the empirical moment tensor on finite samples is the level-≤ 4 truncation of the iterated- integral signature, and the truncation converges to the full signature in the limit of vanishing sampling interval. The two equivalences together produce a single chain

$$\text{Newton-Taylor (local)} \overset{M_D}{\longleftrightarrow} \text{Takens (topological)} \overset{\text{Thm. 5}}{\longleftrightarrow} \text{Lyons (asymptotic)},$$

in which each link is a precise mathematical statement and the chain as a whole is the architectural content of the present act. The kinematic path is the same object at each level; the three readings are the same algebraic content presented at three granularities.

The architectural content has a corollary that we want to state explicitly because it is the rhetorical hinge between Acts II–IV and Act V. The three readings — Newton, Takens, Lyons — were developed in three different mathematical communities at three different historical moments (the seventeenth century for Newton, 1981 for Takens, 1998 for Lyons), and the connections between them were not part of the working repertoire of any of the three communities at the time the readings were established. The Hamiltonian programme of *algebra as the science of Pure Time* (Hamilton, 1837, 1848, 1853) was, on our reading, the one place in which the three readings could have been articulated as the same object — Hamilton's moments-steps-numbers ontology supplies the algebraic substrate, and his identification of the algebra with the science of time supplies the philosophical warrant. The programme was abandoned for the historical reasons traced in Acts II–III before any of the three readings except Newton-Taylor was available, and the three readings developed

in its absence. Geometric Signal Dynamics is, on the reading we have been developing, the recovery of the Hamiltonian articulation on the assembled triple bridge.

5.5 Synthesis of the act

Three propositions consolidate the act and prepare the transition to Act V.

First, the parallel mathematical traditions of nonlinear dynamics (Takens 1981) and rough-path analysis (Lyons 1998) recovered, from outside the spectral-econometric mainstream and from each other, the directional-kinematic content of dynamic data that the spectral-econometric mainstream had structurally truncated. The recoveries were not pieces of a single coordinated programme; they were independent and developed in mutual isolation, and the fact that they converge on the same algebraic object as Newton's kinematic calculus is itself a substantive piece of mathematical synthesis.

Second, the convergence is not metaphorical. The Pascal equivalence (Lemma 5.2) makes the lag-embedding and the kinematic-embedding bases mutually translatable by a fixed integer matrix; the moment-tensor / signature equivalence of Paper 24 (MT, Theorem 5) makes the discrete moment tensor and the asymptotic Lyons signature mutually identifiable up to truncation order. The triple bridge between Newton, Takens and Lyons is therefore a precise architectural statement, not a suggestive analogy.

Third, the convergence is what makes the Hamiltonian programme operationally available. The programme requires an algebra rich enough to host the kinematic state with its grade structure (supplied by Clifford 1878), a probabilistic foundation for the empirical-statistical setting (supplied by Kolmogorov 1933), and a representational substrate that connects the three granularities of the kinematic path (supplied by the triple bridge of the present act). Once all three pieces are in place, the articulation of the kinematic embedding into $\mathrm{Cl}(3,0)$ as the canonical representation of a univariate time series is no longer a parametric choice; it is the unique structurally-supported realisation of the Hamiltonian declaration. The fifth act, which we have already written and to which the present four are the historical preamble, articulates that realisation in the operational detail required for it to function as a research programme.

6 Act V — Geometric Signal Dynamics as instantiation of the *Researches*

The previous four acts traced a fracture and a reunion. Act I recovered the philosophical opposition between *intensio* and *extensio* that Pascal and Port-Royal articulated against the Cartesian reduction of magnitude to scalar coordinate, and showed that Newton's kinematic calculus and Leibniz's monad supplied complementary halves of a direction-bearing representation of dynamic state. Act II recovered the Hamiltonian programme of *moments, steps and numbers* as a declared but only partially executed agenda, and located its abandonment in the middle of the nineteenth century at the precise moment when the mathematical machinery to complete it (Grassmann 1844; Clifford 1878) was becoming available but not yet absorbed into mainstream practice. Act III diagnosed the consolidation of the spectral-econometric tradition (Wold 1938; Cramér 1942; Box-Jenkins 1970; Sims 1980) as the natural inheritance of the abandoned half: a representation that captures only the abelian part of the path module, with all higher-grade information silently dropped or patched back in via successive ad-hoc devices. Act IV showed that the nonlinear dynamics community recovered, under the name of Takens' embedding theorem, exactly the kinematic representation that Newton had outlined and Hamilton had promised, and that the formal equivalence between the kinematic and the lag embeddings via the unimodular Pascal matrix (Paper 07 , Takens) closes one half of a triple bridge whose asymptotic half is given by the Lyons signature (Lyons, 1998; Hambly & Lyons, 2010). The triple bridge is local (Newton-Taylor), global (Takens), and asymptotic (Lyons), all on the same kinematic embedding.

This act argues that Geometric Signal Dynamics (GSD) is the *instantiation* of the Hamiltonian programme on this triple bridge, in the algebraic language Hamilton himself could not yet operationally hold. We do not propose GSD as a successor framework that improves upon Hamilton's: that framing would be both historically imprecise and rhetorically self-defeating. We propose it as a faithful realisation of the agenda Hamilton declared in 1843 and to which he explicitly invited continuation in the closing sentences of the *Researches respecting Quaternions*. The contribution of GSD, on this reading, is not a new foundation but the long-deferred completion of an old one, made possible by 180 years of complementary advances in geometric algebra (Clifford, 1878; Hestenes, 1986; Hestenes & Sobczyk, 1984), in nonlinear dynamics (Takens, 1981), in rough-path theory (Lyons, 1998; Hambly & Lyons, 2010), and in the statistical estimation of multidimensional information functionals (Kraskov et al., 2004; Marčenko & Pastur, 1967).

The argument proceeds in five movements. We first identify in Sections 6.1–6.2 the two basic categories of Hamilton's declared programme — *moments* and *steps* — and verify their literal correspondence with the level y_t and the finite differences Δy_t, $\Delta^2 y_t$ of the GSD kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$. We then argue, in Section 6.3, that the third Hamiltonian category — *numbers* — has been the systematically suppressed half of the inheritance: in his usage a *number* is not an abstract scalar but the algebraic composition of moments and steps, and the canonical number of a series at instant t is, in the GSD formalism, the multivector encoding the geometric product $\mathbf{q}(t)\,\mathbf{q}(t+\tau)$. This is the strongest

philosophical claim of the paper and the place where the rhetorical pivot lands. Appendix A traces the operational trinity of moments, steps and numbers through the public API of the GSD library and shows that the architecture of `gsd.embedding`, `gsd.algebra` and `gsd.signature` is not a coincidence of design but a literal transcription of Hamilton's three categories into executable Python. Section 6.8 situates GSD with respect to the modern programme of geometric algebra in physics, articulated chiefly by Hestenes since 1966, and clarifies that GSD is its sister project on the discrete stochastic empirical-data side, not a competitor or a successor. The act closes with a brief synthesis (Section 6.9) that prepares the ground for Act VI's honest catalogue of limitations.

6.1 Moments

The opening paragraphs of the *Researches respecting Quaternions* are explicit about the ontology Hamilton intends to formalise. In Article 1 of the First Series (Hamilton, 1848, § 1), he writes:

> When we have in any manner been led to form successively the separate conceptions of any number of moments of time, we may afterwards form the new conception of a system, or momental set, to which all these separate moments belong; and may say that this set is of the second, third, fourth, or nth order, according as the number of moments which compose it is 2, 3, 4, or n... A symbol for such a set may be formed by enclosing in parentheses, with commas interposed between them, the separate symbols of the moments which compose the set; thus the symbol of a momental quaternion, or set of the fourth order, will be of the form (a_0, a_1, a_2, a_3).

Two facts about this passage are decisive for the present argument. First, the word *quaternion* is here defined, in its primary sense, as a *set of the fourth order* — a tuple of four moments in time. The algebra of the four-dimensional non-commutative division ring $\mathbb{H}$ that the same paper later derives is, in Hamilton's own ordering, a *secondary* object built upon operations on this primary tuple. The reception history of the *Researches respecting Quaternions* in the twentieth century inverted this ordering: *quaternion* came to mean an element of $\mathbb{H}$, with the temporal-tuple meaning falling out of mathematical practice. We restore the primary meaning in this paper.

Second, the moments are named without further qualification: they are the points of the temporal progression. The level y_t of an observed real-valued time series is, on this reading, exactly a moment in the Hamiltonian sense. The succession of moments $(y_0, y_1, \ldots, y_{T-1})$ is a momental set of order T. No algebraic structure beyond ordering is yet imposed.

We adopt, throughout this paper and consistently with the GSD research programme, the following operational definition.

Definition 6.1 (Moments)**.** Given a real-valued univariate time series $\{y_t\}_{t=0,\ldots,T-1}$, the *moments* of the series are the entries y_t, regarded as ordered points of the temporal progression. The set $\{y_t\}_{t=0,\ldots,T-1}$ is a momental set of order T in the sense of Hamilton (1848, § 1).

The definition is deliberately minimal. It commits to nothing more than what Hamilton committed to in 1843, and in particular it does not prejudge the algebraic operations under which moments may be combined. Those operations are introduced in the next subsection, through the second category of the programme.

6.2 Steps

The second category of the Hamiltonian programme appears in Article 3 of the *Researches respecting Quaternions*, where Hamilton introduces the symbols $a_0, a_1, \ldots$ for what he calls *ordinal relations* between moments — explicitly, the relation of one moment being later or earlier than another. He immediately abridges these to differences, writing $a_0 = a_0' - a_0$, and treats the resulting objects as algebraic quantities in their own right. By Article 6 he is operating on *couples* and *ordinal quaternions* of these differences with characteristics of inversion and transposition. The key passage, which we cite at length because it bears the entire weight of the present argument, reads:

> [The symbol $\sqrt{-1} = (0, 1)$, when used as a multiplier,] transformed one step-couple (a_1, a_2), that is to say, one couple of steps, a_1, a_2, in the progression of time, or one couple of ordinal relations between moments, into another couple of steps or of relations in the same progression of time, according to the law,
>
> $$(0, 1)(a_1, a_2) = (-a_2, a_1).$$
>
> (Hamilton, 1848, § 6)

Two observations are essential. First, Hamilton calls the entries of the couple *steps in the progression of time* — not coordinates on a Cartesian plane and not real or imaginary parts of a complex number, but explicitly differences in the temporal progression. This is the vocabulary of kinematics, not of analytic geometry. Second, the action $(a_1, a_2) \mapsto (-a_2, a_1)$ is a rotation by ninety degrees in the plane spanned by the two steps. Hamilton derives this action algebraically as the result of multiplying by the unit imaginary $\sqrt{-1}$ in his algebra of couples; in the language of modern geometric algebra, the same action is the application of the unit bivector $e_1 \wedge e_2$ to the vector $a_1 e_1 + a_2 e_2$ in $\mathrm{Cl}(2, 0)$. The two operations coincide identically: the unit bivector is the generator of rotation in the plane it spans, and Hamilton's unit imaginary on couples is precisely that generator restricted to two-dimensional couples of temporal differences.

We adopt, accordingly, the following operational definition.

Definition 6.2 (Steps)**.** The *steps* of a univariate time series $\{y_t\}$ are the finite differences

$$\Delta y_t \;=\; y_{t+1} - y_t, \qquad \Delta^2 y_t \;=\; \Delta y_{t+1} - \Delta y_t, \qquad \ldots, \qquad \Delta^k y_t,$$

regarded as algebraic quantities in their own right and not merely as auxiliary devices for stationarisation. The pair $(\Delta y_t, \Delta^2 y_t)$ is a step-couple in the sense of Hamilton (1848, § 6), and the triple $(y_t, \Delta y_t, \Delta^2 y_t)$ is the kinematic embedding of order three.

The first sentence of the definition is the operative one. The econometric tradition deriving from Box & Jenkins (1970) has used finite differences extensively, but as means to an end: the differences are computed in order to render the series stationary so that downstream estimation procedures based on the Wold and Cramér representations remain valid. Under that reading, Δy_t is an artefact of the modelling pipeline and disappears once stationarity has been achieved. Under the Hamiltonian reading we adopt here, Δy_t is a primary object — a step in the progression of time — with its own algebraic structure and its own role in the canonical representation of the series. The two readings are formally compatible (the same numerical quantity is computed in both) but philosophically incompatible: the first treats the step as instrumental, the second as ontological.

The kinematic embedding $\mathbf{q}(t)$ assembles the level y_t and the first two steps Δy_t, $\Delta^2 y_t$ into a single object in $\mathbb{R}^3$. After standardisation by the per-component sample standard deviations σ_y, σ_Δ, $\sigma_{\Delta\Delta}$, this becomes a vector of $\mathrm{Cl}(3,0)$:

$$\mathbf{q}(t) \;=\; \frac{y_t}{\sigma_y}\, e_1 \;+\; \frac{\Delta y_t}{\sigma_\Delta}\, e_2 \;+\; \frac{\Delta^2 y_t}{\sigma_{\Delta\Delta}}\, e_3 \;\in\; \mathbb{R}^3 \subset \mathrm{Cl}(3,0). \tag{9}$$

The choice of dimension three is not conventional: the Geometric Takens Theorem of

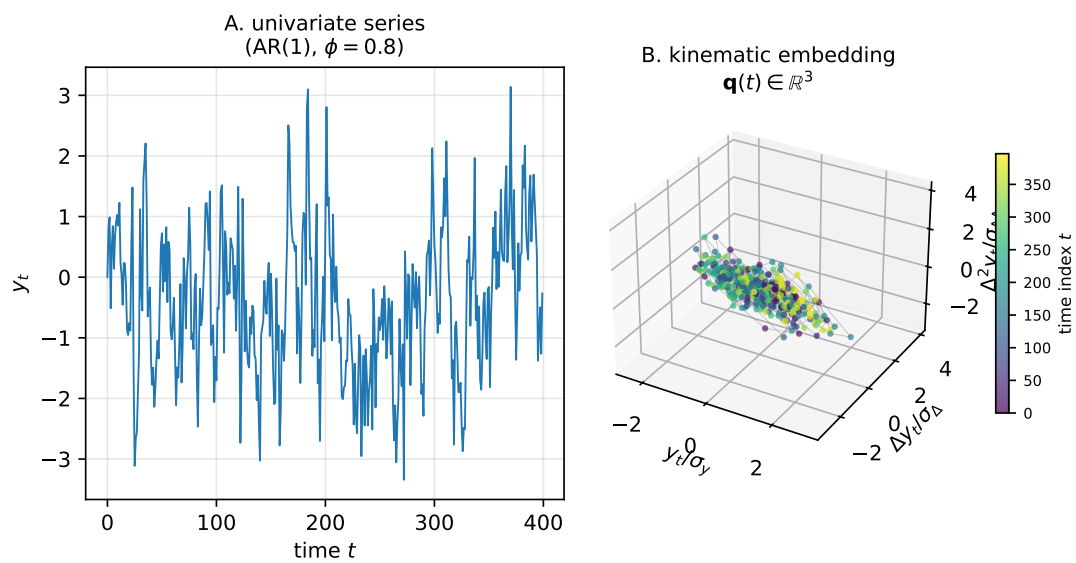

Figure 2: The kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ of a univariate time series, viewed as a discrete trajectory in phase space. The level y_t, the first step Δy_t and the second step $\Delta^2 y_t$ become the three coordinate axes, and the temporal progression of the series becomes a curve. Each point of the curve carries, by construction, what Hamilton's programme would call a *moment* together with its first two *steps* — the algebraic ingredients of a *number* in the sense developed in Section 6.3.

Paper 07 (Takens) establishes that $d^\dagger = \max(2, k^* + 1)$, with k^* the coupling order of the underlying generative process, and that the operationally relevant family $k^* \in \{0, 1, 2\}$ —

level, velocity, acceleration couplings — is exactly covered by $d = 3$. Higher coupling orders require Clifford algebras of higher dimension; the regional optimality of $\mathrm{Cl}(3, 0)$ is discussed honestly in Act VI together with the other limitations of the framework.

The kinematic embedding so defined realises the first two categories of the Hamiltonian programme — moments through the entry y_t/σ_y and steps through the entries $\Delta y_t/\sigma_\Delta$ and $\Delta^2 y_t/\sigma_{\Delta\Delta}$ — within a single algebraic object. We note one historical precedent that bears directly on the autonomy of the second-step component as a representational object in its own right.

Hamilton 1847: the hodograph as autonomous representation of the velocity channel. In a communication to the Royal Irish Academy on 14 December 1846, Hamilton (1847) introduced what he called the *hodograph* of a moving body: the curve traced, from a common origin, by the tip of the velocity vector as the body moves along its orbit. He defined it in terms whose immediacy is worth recording in his own voice:

> *We may always imagine a succession of straight lines, or vectors, to be drawn from some one point, as from a common origin, in such a manner as to represent, by their directions and lengths, the varying directions and degrees (or quantities) of the velocity of the moving point: and the curve which is the locus of the ends of the straight lines so drawn may be called the hodograph of the body, or of its motion [...] (Hamilton, 1847, p. 344)*

The hodograph is, in modern terms, the trajectory of the velocity component of the kinematic state when the velocity is regarded as an autonomous geometric object rather than as a derivative of the position. It is the second component $\Delta y_t/\sigma_\Delta$ of the GSD kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3$, lifted to its own representational space. Hamilton's central result is that under the Newtonian inverse-square law of attraction, the hodograph of any orbit is a circle — what he calls "the Law of the Circular Hodograph" (Hamilton, 1847, p. 349). The result is of substantive geometric content: the dynamical law of the system fixes the geometry of the velocity-channel representation. The present paper's Spectral Identity (6) makes the analogous statement for empirical-stochastic processes — the bivector mean $\mathbb{E}[b_{12}]$ of the kinematic embedding is determined by an integral against the spectral density $f(\lambda)$ with weight $\cos\lambda - 1$, so that the law of the underlying generative process fixes the geometry of the bivector channel. The two readings, separated by a hundred and seventy years, articulate the same structural insight: the dynamics of the system is the geometry of its grade-resolved kinematic representation.

The object lives in a three-dimensional Clifford algebra, which Hamilton did not have, and which is what makes the third and final category of his programme operationally available.

6.3 Numbers

Of the three categories Hamilton names, *number* is the one whose ordinary mathematical meaning has drifted most thoroughly from his. In post-Cantorian usage a number is an

element of one of the standard totally ordered fields $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ — a scalar quantity. In Hamilton's usage a number is, by contrast, an *algebraic composition of moments and steps*: an entity built out of the temporal categories above by means of operations that respect the ordering of time. The unit imaginary $\sqrt{-1}$, in Hamilton's construction, is not a free symbol added to $\mathbb{R}$ to close the polynomial algebra but a specific operation on step-couples (rotation by ninety degrees) reified as an algebraic object. He says this explicitly in Article 6 of the *Researches respecting Quaternions*, deriving $\sqrt{-1} = (0, 1)$ as the multiplier on couples that produces the ninety-degree rotation. The algebra is the algebra of this operation; the imaginary is what the operation looks like when we name it.

The implication for time-series analysis is direct and, as far as we can verify, has not been articulated in this form before in the econometric or signal-processing literatures. The canonical *number* associated with a univariate series at the instant t, in the Hamiltonian sense, is not the scalar value y_t. The scalar y_t is the moment, the first of the three categories. The canonical number is the algebraic composition of the moments and steps that the series exhibits in a temporal neighbourhood of t, realised through an algebra rich enough to host the composition. In $\mathrm{Cl}(3, 0)$, the natural such composition is the geometric product of two consecutive kinematic states. Recall that for vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d \subset \mathrm{Cl}(d, 0)$ the geometric product decomposes uniquely into a scalar and a bivector part:

$$\mathbf{a}\,\mathbf{b} \;=\; \mathbf{a} \cdot \mathbf{b} \;+\; \mathbf{a} \wedge \mathbf{b} \;\in\; \mathrm{Cl}(d, 0).$$

Applied to two states of the kinematic embedding at lag τ, this gives:

$$\mathbf{q}(t)\,\mathbf{q}(t+\tau) \;=\; \underbrace{\mathbf{q}(t) \cdot \mathbf{q}(t+\tau)}_{s(t,\tau)\ \text{(scalar, grade 0)}} \;+\; \underbrace{\mathbf{q}(t) \wedge \mathbf{q}(t+\tau)}_{B(t,\tau)\ \text{(bivector, grade 2)}} \quad . \tag{10}$$

A third grade — the pseudoscalar or trivector — emerges when we compose three states with two distinct lags:

$$\mathcal{T}(t, \tau_1, \tau_2) \;=\; \mathbf{q}(t) \,\wedge\, \mathbf{q}(t+\tau_1) \,\wedge\, \mathbf{q}(t+\tau_2) \;\in\; \Lambda^3\mathbb{R}^3 \;\cong\; \mathbb{R} \cdot e_{123}. \tag{11}$$

The objects s, B and $\mathcal{T}$ — together with the three vector components of $\mathbf{q}(t)$ themselves — exhaust the grade structure of $\mathrm{Cl}(3, 0)$ ($\dim = 1 + 3 + 3 + 1 = 8$). They are the canonical algebraic compositions of moments and steps available in the algebra, and they are the canonical *numbers* of the series at instant t in the Hamiltonian sense.

This is the rhetorical pivot of the paper, and we restate it carefully. The econometric tradition has worked, for nearly a century and a half, with the grade-zero component alone. The level y_t is the moment; the cross-products $\mathbb{E}[y_t y_{t+\tau}]$ and $\mathbb{E}[y_t y_{t+\tau} y_{t+\tau'}]$ are scalar functionals of the moment; the spectral density is a Fourier transform of these functionals. Under the analytic abelianisation entailed by the spectral representation (Cramér, 1942, § 3) the higher-grade information of the geometric product cannot be recovered: the bivector $\mathbf{q}(t) \wedge \mathbf{q}(t+\tau)$ is antisymmetric under exchange of the two arguments, which means it integrates to zero against any symmetric test function in $L^2(\mathbb{R})$ and is therefore invisible to

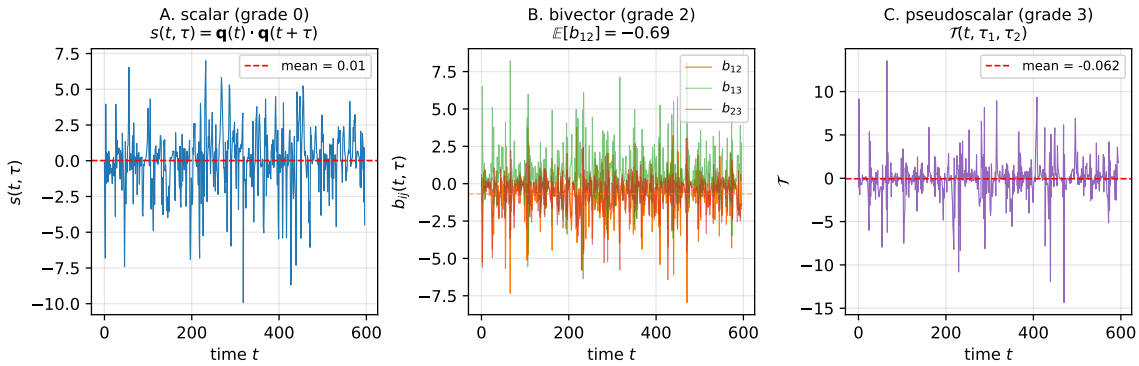

Figure 3: The three grade-wise channels of the multivector composition (10)–(11). Each panel shows the time series of one grade extracted from the same series. The scalar (grade 0) carries the persistence information that classical autocovariance methods exploit; the bivector (grade 2) carries the rotational information of the kinematic state across time, invisible to scalar autocovariance by antisymmetry; the pseudoscalar (grade 3) carries the third-order orientation-asymmetry information. The three channels are algebraically orthogonal and empirically quasi-independent (Section 6.4).

the abelian (commutative) cross-product machinery. The trivector $\mathcal{T}(t,\tau_1,\tau_2)$ is totally antisymmetric in the three arguments and is invisible by an even stronger version of the same mechanism: no scalar functional of the lagged covariances can detect its mean. To recover the bivector and the trivector one must work in an algebra in which their non-commutativity is preserved as structural. That is precisely what $\mathrm{Cl}(3,0)$ is.

The multivector that assembles all available grades of the algebra into a single object,

$$\mathbf{q}(t) \;+\; \mathbf{q}(t)\,\mathbf{q}(t+\tau) \;+\; \mathbf{q}(t)\wedge\mathbf{q}(t+\tau)\wedge\mathbf{q}(t+\tau_2) \;\in\; \mathrm{Cl}(3,0), \tag{12}$$

is what Hamilton would have called a *number* in the sense of his programme: an algebraic composition of moments and steps, built from the temporal categories by operations that respect the ordering of time. We do not propose to install *number* as a technical term of GSD — the post-Cantorian usage of the word for elements of $\mathbb{R}$ is too entrenched, and the gain in fidelity to Hamilton would not justify the friction. The point is conceptual: the object that Hamilton's programme would identify as the canonical algebraic representation of the series at instant t is the multivector (12), with the scalar y_t recognised as its grade-one projection along e_1 rather than as the representation itself. The econometric tradition has worked with the projection alone for nearly a century and a half; what GSD recovers is the rest of the multivector.

The empirical companion of this object is the *moment tensor* of Paper 24 (MT). Where (12) is the algebraic multivector at a single instant, the moment tensor $\mathcal{M}_y^{(c,k)}$ is the empirical first four statistical moments of each of the sixteen Clifford channels of the kinematic path, computed across t. The result is a $16\times 4 = 64$-coordinate empirical signature of the series, which under the Lyons-signature equivalence of Paper 24 (MT, Theorem 5) is precisely the

order-≤ 4 truncation of the Lyons signature of the path $\mathbf{q} : [0, T] \to \mathbb{R}^3 \subset \mathrm{Cl}(3, 0)$. The moment tensor is, in this sense, the discrete-time, finite-truncation version of the infinite-dimensional path-module representation that Hambly-Lyons uniqueness identifies as the canonical observable of a path of bounded variation modulo reparametrisation (Hambly & Lyons, 2010). We have here, on the empirical side, exactly the realisation that the Hamiltonian programme would have aimed for had it had access to sample data and to the rough-path machinery that Lyons constructed in 1998.

6.4 Uncertainty by grade: a category Hamilton could not yet formalise

The reader will have noticed an asymmetry between the present discussion and Hamilton's. The *Researches respecting Quaternions* treat moments, steps and numbers as exact temporal categories; nowhere in them does Hamilton contemplate uncertainty as a structural property of the data. The omission is not a defect of his construction. It is a fact about the state of mathematics in 1843: the modern axiomatisation of probability was eight decades away (Kolmogorov, 1933, 1956), and the spectral representation theory that would absorb stationary stochastic processes into the Wiener-Cramér framework (Wold, 1938; Cramér, 1942; Wiener, 1949) was a century in the future. Hamilton's algebra of Pure Time is, in this respect, the deterministic half of the framework that GSD inherits; the other half — the *stochastic* treatment of moments and steps — was developed between 1933 and 1942 by Kolmogorov, Wold and Cramér, but it was developed at a time when the algebraic representation we now call $\mathrm{Cl}(3, 0)$ was not part of the working toolkit of the statistical community. The two halves were assembled in different decades and under different intellectual constraints, and the joint object — a stochastic algebra of moments and steps in $\mathrm{Cl}(3, 0)$ — has had to wait until now to be articulated.

The shape of the inheritance is worth tracing carefully because it explains where the modern econometric tradition placed uncertainty and where, by that placement, it implicitly excluded the higher grades from probabilistic analysis.

In Kolmogorov's 1933 axiomatisation, uncertainty is encoded by a probability measure P on a measurable space $(\Omega, \mathcal{F})$ of elementary events, and a real-valued random variable is a measurable function $X : \Omega \to \mathbb{R}$. The axiomatisation is neutral with respect to what the random variable represents: nothing in the construction privileges scalars over richer algebraic objects, and indeed measurable functions valued in any Polish space are covered by the standard extension. What privileges scalars in practice is downstream: the stationary-process theory of Wold (1938), the spectral representation of Cramér (1942), and the Lévy decomposition of Lévy (1948) are all formulated for real- or complex-valued processes, and the moments they manipulate ($\mathbb{E}[y_t y_{t+\tau}]$, $\mathbb{E}[y_t y_{t+\tau} y_{t+\tau'}]$, etc.) are scalars by construction. The uncertainty in the data, in this inheritance, lives entirely in the residual ε_t of a scalar model, with $\mathrm{Var}(\varepsilon_t)$ as its first-order summary and the spectral density $f(\lambda)$ as its second-order summary. Higher-order summaries — the polyspectra of Brillinger (1965) and the bispectra of Rosenblatt (1985) — are formal extensions of the same scalar machinery into successive orders of cross-product, each of which is a scalar functional of the scalar process.

GSD inherits Kolmogorov's axiomatisation without modification — the probability space and the measurability requirements are imported verbatim — but it allows the random variable to be valued in $\mathrm{Cl}(3,0)$ rather than in $\mathbb{R}$. The kinematic embedding $\mathbf{q}(t)$ of (9) is, formally, a $\mathrm{Cl}(3,0)$-valued random variable: a measurable function $\Omega \to \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$. The grade-wise products $s(t,\tau)$, $B(t,\tau)$ and $\mathcal{T}(t,\tau_1,\tau_2)$ defined in (10)–(11) are measurable functionals of $\mathbf{q}(t)$, so their distributions inherit well-defined probability laws on the corresponding grade subspaces. The *antisymmetric entropy* decomposition of Paper 04 (HAS) formalises this on the differential-entropy side: under the algebraic orthogonality of grades and the empirical quasi-independence of the corresponding random variables, one has

$$H_{AS}(\tau) \;=\; H_0(\tau) \;+\; H_2(\tau),$$

with H_0 the differential entropy of the scalar component and H_2 the joint differential entropy of the three bivector components. The *three-grade entropy* extension of Paper 18 (JC) closes the decomposition at the available algebraic grade,

$$H_{\mathrm{geom}}(\tau,\tau_2) \;=\; H_0(\tau) \;+\; H_2(\tau) \;+\; H_3(\tau,\tau_2) \;-\; \mathrm{TC}, \tag{13}$$

with H_3 the differential entropy of the pseudoscalar component and TC the joint total correlation across grades, which is empirically negligible ($\mathrm{TC}/H_{\mathrm{geom}} < 0.5\%$ on canonical generative processes; Paper 18 , JC, Theorem 1).

The conceptual content of (13) is the recovery of an asymmetry that the Wold-Cramér inheritance had silently collapsed. In that inheritance, "the uncertainty of the series" is one quantity — the scalar variance, or equivalently the spectrum of power. In the GSD inheritance, uncertainty has three structurally distinct components, each carrying its own information and its own operational interpretation: H_0 measures the persistence uncertainty (how much the level drifts unpredictably from one moment to the next), H_2 measures the rotational uncertainty (how much the bivector-encoded direction of the kinematic state varies), and H_3 measures the volumetric or chiral uncertainty (how much the pseudoscalar-encoded asymmetry between the three lagged states fluctuates). These are not refinements of a single underlying scalar quantity. They are independent random variables — empirically quasi-independent in the sense of (13) and the cited total-correlation bound — with distinct estimators, distinct asymptotic regimes, and distinct interpretations in application.

This expansion of the notion of uncertainty has a precise historical reading. Kolmogorov's axiomatisation of 1933 left the co-domain of the random variable open; by mathematical convenience the subsequent stochastic-process literature fixed it to $\mathbb{R}$ or $\mathbb{C}$, because that is what the analytic machinery of the period (Fourier transforms, L^2 Hilbert space, Wiener integrals) admitted. GSD takes the open co-domain seriously and lets the random variable live where Hamilton's programme would have placed it had Hamilton had Kolmogorov in hand, namely in the algebra $\mathrm{Cl}(3,0)$ where the canonical composition of moments and steps is expressible. The probabilistic content of the kinematic state is then distributed across the three available grades, and the residual ε_t of a scalar model — which the Wold-Cramér inheritance treated as the carrier of all the unmodelled uncertainty —

is recognised as the carrier of the grade-zero uncertainty alone. What the scalar residual cannot detect is whatever uncertainty resides in the rotation and the volumetric channel; and that, on the empirical evidence accumulated by the GSD programme to date, is not a small leftover. The Jarzynski-Crooks proxy of Paper 18 (JC) picks crisis years on financial series at $z = 4.30$ where scalar volatility is unremarkable; the multivariate fragility causality of Paper 17 (FC) detects spillover channels that copula-DCC-GARCH frameworks do not separate from variance spillover; the geometric ergodicity exponent of Paper 25 (GE) discriminates regimes that classical ergodicity tests cannot. Each of these is, in our reading, a piece of uncertainty that the scalar inheritance silently collapsed and that the grade-resolved inheritance recovers.

The point is not that Wold and Cramér were wrong. They were exact within the algebra they had. The point is that they had only the abelian fragment of what the data contained, and that what we are proposing here is the recovery of the rest.

The Heisenberg principle on $\mathrm{Cl}(3,0)$**.** The recovery is not metaphorical. Hitzer & Mawardi (2013) construct a multivector-valued Fourier transform on $\mathrm{Cl}(3,0)$ — the *Clifford Fourier transform*, hereafter CFT — by replacing the scalar imaginary unit $i = \sqrt{-1}$ in the kernel with the pseudoscalar $i_3 = e_{123}$ of the algebra:

$$F\{f\}(\omega) \;=\; \int_{\mathbb{R}^3} f(x)\, e^{-i_3\,\omega\cdot x}\, d^3x, \qquad f : \mathbb{R}^3 \to \mathrm{Cl}(3,0). \tag{14}$$

The construction works because i_3 is central in $\mathrm{Cl}(3,0)$ (i_3 commutes with every multivector and satisfies $i_3^2 = -1$), so the kernel $e^{-i_3\,\omega\cdot x}$ commutes with the multivector- valued integrand and Plancherel-type identities pass through verbatim. On this transform Hitzer and Mawardi prove a Heisenberg- style uncertainty principle: for any $f \in L^2(\mathbb{R}^3, \mathrm{Cl}(3,0))$ with $\int_{\mathbb{R}^3} \|f(x)\|^2 d^3x = F < \infty$ and any constant vectors $a, b \in \mathbb{R}^3$,

$$\left(\int_{\mathbb{R}^3} (a\cdot x)^2\, \|f(x)\|^2\, d^3x\right)\!\left(\int_{\mathbb{R}^3} (b\cdot\omega)^2\, \|F\{f\}(\omega)\|^2\, d^3\omega\right) \;\geq\; (a\cdot b)^2\, \frac{(2\pi)^3}{4}\, F^2, \tag{15}$$

with equality saturated by multivector Gaussians $f(x) = C_0\, e^{-k\,x^2}$, where C_0 is a constant multivector and $k > 0$ is a real scale.

The relevance to the present argument is twofold. First, (14) shows constructively that the spectral machinery *does* extend beyond the abelian fragment: a non-commutative generalisation of the Fourier transform exists on the same $\mathrm{Cl}(3,0)$ in which the GSD kinematic embedding lives, and it admits a Heisenberg principle as direct as the classical one. Wold-Cramér was not wrong; it was the abelian truncation of (14), in which the multivector kernel collapses to its grade-zero projection. Second, the saturator of (15) is a multivector Gaussian, not a scalar one — exactly the regime in which the closed-form theorems of Paper 04 (HAS, Theorem 1) on antisymmetric entropy and Paper 18 (JC, Theorem 2) on the Jarzynski-Crooks correspondence are demonstrated. The grade-resolved uncertainty decomposition of Section 6.4 and the multivector Heisenberg principle of Hitzer & Mawardi (2013) are two faces of the same fact: when the algebra used to host the random variable is

$\mathrm{Cl}(3,0)$, the structural content of "uncertainty" is irreducibly grade-decomposed, and the spectral machinery accompanies that decomposition into all available grades rather than collapsing it onto the scalar.

6.5 The kinematic embedding in classical tests: Dickey-Fuller, Engle-Granger, Johansen

A reader who has followed the argument up to this point may reasonably object that the kinematic embedding cannot be entirely absent from the modern econometric tradition: differencing has been the bread and butter of time-series analysis since Box & Jenkins (1970), and at least the first step Δy_t is ubiquitous. The objection is well-founded and important to address directly. The kinematic embedding does appear in classical tests — not as an articulated geometric object, but as an operational device deployed without the algebraic interpretation that would make it canonical. Three examples are illustrative because they cover the most-used family of unit-root and cointegration tests; in each case the kinematic structure is visible to a reader looking for it, and silent to a reader who is not.

The Dickey-Fuller test. Dickey & Fuller (1979) proposed testing the null hypothesis of a unit root in the autoregression $y_t = \rho y_{t-1} + \varepsilon_t$ by reparametrising the regression as

$$\Delta y_t \;=\; (\rho - 1)\, y_{t-1} \;+\; \varepsilon_t, \tag{16}$$

testing $H_0 : \rho - 1 = 0$ against $H_1 : \rho - 1 < 0$. The reparametrisation is an algebraic identity, but the choice to formulate the test on $(\Delta y_t, y_{t-1})$ rather than on (y_t, y_{t-1}) is not gratuitous: the latter pair has a unit-root covariance that diverges asymptotically, while the former pair has a finite limiting distribution and supports a non-degenerate test statistic. *The pair* $(y_{t-1}, \Delta y_t)$ *is the two-dimensional kinematic embedding* $\mathbf{q}^{(2)}(t-1) = (y_{t-1}, \Delta y_t)$ *in the lag-and-difference basis.* What Dickey and Fuller discovered, in our reading, is that the unit-root test must be formulated on the kinematic embedding because the moment alone is asymptotically degenerate. The Geometric Dickey-Fuller test of Paper 01 (GDF) formalises and extends this discovery: lifting the same test to the full $\mathrm{Cl}(3,0)$ by adding the second step $\Delta^2 y_t$ and reading the result through the bivector mean $\mathbb{E}[B(t,\tau)]$ produces a four-channel diagnostic that recovers the original Dickey-Fuller statistic as one channel and adds three more with independent informational content.

Engle-Granger cointegration. Engle & Granger (1987) proposed testing for cointegration between two integrated series X and Y in two stages: estimate $\hat{\beta}$ from the cointegrating regression $Y_t = \beta X_t + u_t$, then test for a unit root in the residual $\hat{u}_t = Y_t - \hat{\beta} X_t$. The unit-root test on $\hat{u}_t$ is a Dickey-Fuller test, which means it operates on the kinematic embedding of the residual: the pair $(\hat{u}_{t-1}, \Delta\hat{u}_t)$ is the two-dimensional kinematic embedding of the cointegrating residual. The cointegration test is therefore implicitly a test on a *cross-grade* object: it asks whether the residual of a scalar regression between two integrated series

exhibits the geometric signature of stationarity in $\mathrm{Cl}(3,0)$. The Geometric Cointegration framework of Paper 03 (GC) formalises this by moving directly to the cross-level bivector $\mathbb{E}[\mathbf{q}^{(X)}(t) \wedge \mathbf{q}^{(Y)}(t)]$, which detects cointegrating relationships without the intermediate regression step and without requiring the cointegrating direction to be estimated separately.

Johansen's VECM. Johansen (1988, 1991) extended the bivariate cointegration framework to multivariate vector error-correction models (VECMs) of the form

$$\Delta \mathbf{y}_t \;=\; \alpha\,\beta'\,\mathbf{y}_{t-1} \;+\; \sum_{j=1}^{p-1} \Gamma_j\,\Delta \mathbf{y}_{t-j} \;+\; \varepsilon_t, \tag{17}$$

where the matrices α and β encode, respectively, the adjustment speeds and the cointegrating directions. The VECM is, in the language of this paper, a regression of the first step $\Delta \mathbf{y}_t$ on the moment $\mathbf{y}_{t-1}$ and on the first steps $\Delta \mathbf{y}_{t-j}$ at past lags. *It is already a kinematic-embedding regression.* What it lacks is the algebraic interpretation: the matrix β is treated as a parameter to be estimated rather than as a bivector field $\mathbb{E}[\mathbf{q}^{(X_i)}(t) \wedge \mathbf{q}^{(X_j)}(t)]$ to be read directly off the data. The reduced-rank machinery that Johansen developed for estimating β — eigendecomposition of a generalised eigenvalue problem — is, when re-read through the lens of Paper 08 (MPRT), a precursor of the random-matrix-theoretic rank estimator on the multivariate bivector field, which avoids the asymptotic distributional gymnastics of the Johansen procedure by working directly with the Marchenko-Pastur null.

The pattern across the three examples is consistent. The kinematic embedding is present in every case, but it enters as an operational device justified on local technical grounds (asymptotic distributions, residual stationarity, reduced-rank inference) rather than as a canonical algebraic object with intrinsic interpretation. The geometric reading we propose here does not invalidate any of the three tests: Dickey-Fuller, Engle-Granger and Johansen remain correct within their domains. What the geometric reading adds is the articulation: it identifies the embedding as the same canonical object across the three settings, identifies the test statistics themselves as projections onto specific grade components of that object, and opens the path to extending the tests to grade components that the original formulations did not consider — chiefly the bivector and pseudoscalar means, which carry independent information about the rotational and volumetric structure of the data. The Geometric Dickey-Fuller of Paper 01 (GDF), the Geometric Cointegration of Paper 03 (GC), and the multivariate geometric scaling of Paper 08 (MPRT) are the GSD-side companions of these classical procedures, and the relation between the classical and the geometric forms is not one of replacement but of extension by algebraic recognition.

The historical observation we want to highlight is the following. The kinematic embedding has been operating, unrecognised, inside the most cited tests of unit-root and cointegration analysis for forty-five years. The success of those tests — they remain among the most widely deployed inferential tools in applied econometrics — is in no small part *because* they tacitly respect the kinematic structure of the data. What GSD proposes is

not to discard them but to name what they have been doing all along and to extend the recognition to the grades they did not have the algebra to address.

The point generalises beyond unit-root and cointegration testing. Sudjianto (2025) articulates a unified perspective on the modern statistical-learning toolkit under which polynomial expansions, basis-function methods, kernels, gradient-boosted decision trees, and feedforward and self-attention networks are all instances of the same paradigm: an embedding map $\phi : \mathbb{R}^d \to \mathbb{R}^k$ followed by a linear estimator on the image. The transformation is what gives access to non-linear structure of the original inputs; the linear stage is what makes the optimisation tractable. From this vantage point the apparently disparate algorithms of contemporary applied modelling differ in *which* embedding they choose — adaptive trees and sparse one-hot for GBDT, learned continuous representations for neural networks, fixed orthogonal bases for Fourier and wavelet expansions — and in the regime (low-dimensional discrete or high-dimensional continuous) in which the chosen embedding performs.

The kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ of GSD, read in this light, is the canonical embedding of a univariate time series under the Hamiltonian programme: an embedding map that is neither learned nor parametrically chosen but algebraically determined by the requirement that level, velocity and acceleration be jointly representable with comparable variance under standardisation. The moment tensor $\mathcal{M}_y^{(c,k)}$ of Paper 24 (MT) is then the canonical embedding of a series at the level of empirical statistics $\mathbb{R}^T \to \mathbb{R}^{64}$, and the empirical fact that an LDA classifier on this 64-coordinate object matches or surpasses the dedicated estimator batteries of the GSD programme on canonical classification tasks (Paper 24 , MT, Section 8) is precisely the linear stage of Sudjianto (2025)'s paradigm operating on the canonical embedding. Where the modern toolkit chooses among many possible embeddings, the Hamiltonian programme *names one*: the kinematic embedding into $\mathrm{Cl}(3,0)$ followed by its grade decomposition. The cost of this commitment is the regional optimality of $\mathrm{Cl}(3,0)$ for coupling orders $k^* \leq 2$ discussed in Act VIII; the benefit is that the embedding requires no hyperparameter selection, no training data, and no model-selection machinery to be deployed, and that its grade structure carries intrinsic interpretation rather than emergent feature importance. GSD is, in this reading, the discrete time-series instance of the embedding-and-linear-modelling paradigm in which the embedding has been canonicalised by algebraic argument rather than by empirical search.

6.6 The isomorphism $\mathrm{Cl}(3,0) \cong \mathbb{C} \otimes \mathbb{H}$ and Hamilton's geometric biquaternions

Before contrasting GSD with the modern geometric-algebra programme of physics, we record one fact whose historical content is sharper than the textbook-level statement of the isomorphism would suggest. As real algebras of dimension eight,

$$\mathrm{Cl}(3,0) \;\cong\; \mathbb{C} \otimes \mathbb{H} \;=\; \text{biquaternions of Hamilton.} \tag{18}$$

The identification can be read from inside $\mathrm{Cl}(3,0)$ in two movements. The pseudoscalar $i_3 = e_{123}$ satisfies $i_3^2 = -1$ and commutes with every multivector of the algebra; the subalgebra $\{1, i_3\}$ is therefore a copy of $\mathbb{C}$ centrally embedded in $\mathrm{Cl}(3,0)$. The even subalgebra $\mathrm{Cl}(3,0)^+ = \{1, e_{12}, e_{13}, e_{23}\}$ has bivector basis with squares -1 and the quaternion relations $(e_{12})(e_{13}) = e_{23}$, $(e_{13})(e_{23}) = e_{12}$, $(e_{23})(e_{12}) = e_{13}$, and is therefore a copy of $\mathbb{H}$. The whole algebra is the tensor product of these two factors over the centre, recovering precisely Hamilton's biquaternions.

The historical content lies in what Hamilton himself attempted with this object. The system was first announced formally by Hamilton (1844) on 13 November 1843, less than a month after the discovery on the Brougham Bridge: "*Sir Wm. Hamilton proposes to consider some of the consequences which result from the following system of imaginary equations [...]: $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, $ji = -k$, $kj = -i$, $ik = -j$*" (Hamilton, 1844, pp. 424–425). Two decades later he established what is now known as the Cayley-Hamilton theorem in the quaternionic setting, observing that the symbol of any linear operation on quaternions satisfies a biquadratic equation with scalar coefficients determined by the operation (Hamilton, 1862, Article 2) — the canonical statement that any linear operator on a 4-dimensional algebra obeys its own characteristic polynomial, and thereby the algebraic engine that drives the modern matrix- spectral theory underlying the moment-tensor classifier of Paper 24 (MT). In Hamilton (1853), communicated to the Royal Irish Academy in February 1853, he explicitly works with what he calls *bivectors* of the form $\rho + \sqrt{-1}\,\rho_0$, with ρ, ρ_0 "two geometrically real vectors" and $\sqrt{-1}$ "the old and ordinary (or commutative) imaginary of common algebra", and similarly with biquaternions $q + \sqrt{-1}\,q'$. He notes that the conception of such bivectors had occurred to him many years earlier and was "interpretable, geometrically on the author's principles". He then considers, specifically, bivectors with null squares such as $j + hk$, where i, j, k are the non-commutative quaternion units and h is his temporary symbol for $\sqrt{-1}$, so that

$$(j + hk)^2 \;=\; j^2 + h(jk + kj) + h^2k^2 \;=\; -1 + 0{\cdot}h + (-1)(-1) \;=\; 0,$$

and uses these null-squared bivectors to obtain closed-form identities such as $(1+j+hk)^x = 1+x(j+hk)$ in a geometric investigation involving fixed-point iteration of quaternion-valued continued fractions. The investigation produces loci of points $P, Q, R, S, T, U, \ldots$ generated by repeated application of a quaternion-valued transformation, with the alternate points falling on circles in the plane.

Two observations are immediate. First, the role played by Hamilton's h — a commutative imaginary central in the biquaternion algebra — is exactly the role played by the pseudoscalar $i_3 = e_{123}$ in $\mathrm{Cl}(3,0)$ under the isomorphism (18); the quaternion units i, j, k correspond to the bivector basis e_{23}, e_{31}, e_{12} of the even subalgebra. Hamilton was working with the operative content of $\mathrm{Cl}(3,0)$ in 1853, twenty-five years before Clifford gave the basis-canonical presentation of the algebra in Clifford (1878), and he was using its bivectorial content — including null-squared bivectors that have a clear interpretation in the modern theory as light-like elements of an extended signature — to study *discrete*

iterations producing geometric loci. This is, in the structural sense, a problem of discrete dynamical systems, and it is studied with the algebra in which GSD operates today.

Second, Hamilton himself frames the work as the geometric operationalisation of the programme of the *Researches respecting Quaternions*: the bivectors and biquaternions are introduced as "conducting to interesting conclusions" on the same principles, and the application is to a problem of trajectories under iterated transformation. The 1853 paper is, in this reading, an early attempt to do precisely what GSD now does with the full Clifford-algebraic machinery: take the algebra of moments and steps, allow non-trivial geometric content (here: null-squared bivectors), and read the resulting structure as a description of how a system evolves under repeated application of a transformation.

The intermediate phase: *On Symbolical Geometry*, 1846–1849. The 1853 paper on biquaternions (Hamilton, 1853) was not an isolated communication. It was the closing instalment of a sustained programme that Hamilton had developed in the intervening decade in his four-part *On Symbolical Geometry* (Hamilton, 1846–49), published in volumes i-iv of the *Cambridge and Dublin Mathematical Journal* between 1846 and 1849. The series is the operational bridge between the abstract quaternion algebra of Hamilton (1837, 1848) and the geometric biquaternions of 1853, and three of its declarations bear directly on the genealogical reading we are developing here.

First, Hamilton himself, in the introductory remarks of 1846–49, links the new symbolic geometry explicitly to the *Algebra as the Science of Pure Time* programme of nine years earlier:

> *An opinion has been formerly published [in* Trans. Royal Irish Acad.*, vol. xvii. Dublin, 1835] by the writer of the present paper, that it is possible to regard Algebra as a science [...] and that when so regarded, and so separated, its ultimate subject-matter is found in what a great metaphysician has called the inner intuition of* time*. On which account, the writer ventured to characterise Algebra as being the* Science of Pure Time*. (Hamilton, 1846–49, Introductory remarks)*

The continuity is in his own voice. The symbolic geometry of 1846 is the geometric companion of the algebra of Pure Time, not a new project.

Second, the central technical move of the paper is a decomposition that any reader trained in twentieth-century geometric algebra will recognise immediately. In Article 17 (eqns. 38–39) Hamilton writes, for any two lines a and e,

$$\mathrm{e} \div \mathrm{a} \;=\; \mathrm{V}(\mathrm{e} \div \mathrm{a}) \;+\; \mathrm{S}(\mathrm{e} \div \mathrm{a}), \tag{19}$$

that is, the geometric quotient of two lines decomposes into a *vector part* $\mathrm{V}(\mathrm{e} \div \mathrm{a})$ and a *scalar part* $\mathrm{S}(\mathrm{e} \div \mathrm{a})$. This is *literally* the Clifford decomposition $\mathbf{a}\,\mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ thirty-two years before Clifford published it in arbitrary signature $\mathrm{Cl}(p, q)$. Hamilton's "vector" satisfies $\alpha^2 = -1$ (eq. 90 of 1846–49: "*the square of a vector is always a negative scalar*"),

which under the isomorphism $\mathrm{Cl}(3,0)^+ \cong \mathbb{H}$ identifies his vectors with the bivector basis $\{e_{12}, e_{13}, e_{23}\}$ of the even subalgebra of $\mathrm{Cl}(3,0)$.[3]

Third, the geometry he constructs is explicitly coordinate-free. In a footnote to Article 39 he records, in passing and without emphasis:

> *In the method employed in the present essay, no use whatever has hitherto been made of any formula of spherical or even plane trigonometry, any more than of the doctrine of coordinates. (Hamilton, 1846–49, Article 39, footnote)*

The phrase "*the doctrine of coordinates*" is Hamilton's direct designation of the Cartesian programme that Section 2.1 of Act I traced as the post-seventeenth-century mainstream. *On Symbolical Geometry* is, on this reading, the first sustained articulation in modern mathematics of an explicit coordinate-free programme — published contemporaneously with Grassmann (1844) and independently of him.

The intermediate phase clarifies the genealogical line. The coordinate-free programme that the present paper traces from Pascal-Port-Royal through Clifford to Hestenes and GSD is not a retrospective reconstruction; it is a tradition that Hamilton himself articulated explicitly in 1846, in continuity with his 1837 *Algebra as the Science of Pure Time* programme and as preparation for his 1853 biquaternion synthesis. The chain

Pascal-Port-Royal (XVII) $\rightarrow$ Hamilton 1837 abstract quaternions $\rightarrow$ Hamilton 1846 *Symbolical Geometry* $\rightarrow$

is one continuous line, with Symbolical Geometry as the operational bridge that converts the abstract algebra of 1837 into geometric machinery on lines and planes, and the biquaternion extension of 1853 as the final step that promotes the even subalgebra $\mathrm{Cl}(3,0)^+$ to the full $\mathrm{Cl}(3,0)$ via the central imaginary h.

What separated Hamilton 1853 from GSD is the absence of two pieces that a hundred and seventy years would supply. Clifford's basis- canonical presentation (Clifford, 1878) was twenty-five years away when Hamilton wrote, and the grade decomposition $1 \oplus 3 \oplus 3 \oplus 1$ that makes the algebraic structure of $\mathrm{Cl}(3,0)$ legible as "scalar plus vector plus bivector plus pseudoscalar" was not yet articulated. And the statistical machinery for inference on empirical sample paths — the axiomatisation of probability of Kolmogorov (1933, 1956), the spectral representation theorems, the rough-path theory of Lyons (1998) — was three quarters of a century away. The programme Hamilton sketched in 1853 was, in our reading, a research programme awaiting the completion of two parallel mathematical traditions that had not yet crystallised. GSD is, in the same reading, the first realisation of that programme in which both completions are finally available simultaneously.

[3]The terminological inversion deserves explicit acknowledgement. What Hamilton calls a "*vector*" is, under the modern presentation, an element of $\mathrm{Cl}_2(3,0)$ — a bivector. What modern Clifford algebra calls a *vector*, in the strict $\mathrm{Cl}_1(3,0)$ sense, is the *index* of Hamilton's vector (Article 17). The mismatch arises because Hamilton operated within the even subalgebra $\mathrm{Cl}(3,0)^+$ before the basis-canonical presentation by grade became standard.

6.7 The kinematic embedding as intrinsic geometry: a Frenet-Serret reading

A standard textbook line in time-series analysis says that a series *begins to exist* when an ordering is imposed on a sequence of observations by attaching timestamps $0, 1, 2, 3, \ldots$ to them. The order, on this account, is external to the data: it is the observer's grid, and its choice is conventional. The kinematic embedding suggests, against this account, that the order so imposed is largely *label*, not *structure*. What constitutes the series as a geometric object is not the indexing of its moments by integers but the algebraic composition of moments and steps that the embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ assembles. The indexing is a parametrisation; the structure is what survives reparametrisation.

The objection of circularity is unavoidable and worth addressing directly. To form $\Delta y_t = y_{t+1} - y_t$ one must already know which observation is y_t and which is y_{t+1}, so the order cannot be *constituted* by the differences in any literal sense. The objection is correct as far as it goes, but its conclusion is softer than it pretends. The integer index t is a label that identifies which observation is to be processed; once fixed, the geometric content extracted by the embedding — the means and distributions of the grade-wise components $s(t, \tau)$, $B(t, \tau)$, $\mathcal{T}(t, \tau_1, \tau_2)$ — is invariant to monotone reparametrisations of the index after a suitable rectification of the embedding's component scales. A series sampled equispaced and a series sampled non-equispaced under the same generative process yield, after rectification, the same distributional content of the kinematic embedding. The label $0, 1, 2, \ldots$ is conventional; the rectified embedding is not.

Three rectifications, one ontology. The rectification can be performed in three distinct modes, none of which subsumes the others, and the distinction is operationally relevant although ontologically secondary. The first mode is *none*, in which the embedding $\mathbf{q}(t)$ is used in raw units; the covariance of the embedding is the unmodified $\Sigma_{\mathbf{q}}$. The second is *marginal standardisation* (also known as the per-column z-score), dividing each entry of $\mathbf{q}(t)$ by its per-component sample standard deviation $(\sigma_y, \sigma_\Delta, \sigma_{\Delta\Delta})$; this brings the three grades onto comparable scales and yields a covariance with unit diagonal but with non-zero cross-correlations $\tilde{\rho}_{ij} \neq 0$ between the three components. The third is *Mahalanobis whitening* (also known as ZCA whitening, and adopted in the GSD programme as the *Bivector Whitening* of Paper 08 , MPRT, Section 2), $\mathbf{q}'(t) = \Sigma_{\mathbf{q}}^{-1/2}\,\mathbf{q}(t)$, which yields a covariance that is *exactly* the identity, $\Sigma_{\mathbf{q}'} = I$. Under Bivector Whitening the algebraic cross-coupling between $(y, \Delta y, \Delta^2 y)$ induced by the finite-difference stencil is removed at the population level, the Marchenko-Pastur null on the multivariate bivector field becomes exact at $\tau \geq 3$ (Paper 08 , MPRT, Proposition 1), and empirically the cross-correlation between standardised components drops to numerical zero (e.g. $\mathrm{corr}(q_1, q_2) \approx 10^{-7}$ under full whitening versus $\approx 10^{-1}$ under marginal standardisation on canonical synthetic generative processes). Crucially, the three modes are *disjoint*: marginal standardisation does not subsume full whitening (it is unit-diagonal but not decorrelated); full whitening does subsume the unit-variance property of marginal standardisation but adds the orthogonal-

ity property the latter lacks; and the choice of *none* simply leaves the algebraic coupling in place. The GSD library (`gsd.embedding.kinematic_embedding`, version 0.2.0) takes `whitening='full'` as the default for this reason, with `'diagonal'` and `'none'` retained as explicit modes for research and pedagogical use.

For the present argument the distinction is technical: all three modes eliminate, to varying degrees of fidelity, the timestamp parametrisation and preserve the ontological invariance under reparametrisation that the Frenet-Serret reading articulates. Full whitening is the discrete analogue of arc-length parametrisation in the *strong* sense, eliminating both scale and angular distortion; marginal standardisation is its diagonal weakening, eliminating scale only; the raw embedding is its limit case. Where the present paper writes simply "standardisation" below, the reader should understand any of the three modes as compatible with the ontological reading developed here, and should consult Paper 08 (MPRT) for the operational rationale that recommends the full mode in inferential applications.

The analogy with classical differential geometry is precise rather than metaphorical. For a smooth curve $\gamma : I \to \mathbb{R}^3$ in Euclidean three-space, the Frenet-Serret apparatus (do Carmo, 1976) constructs from successive derivatives the moving trihedron $\{T, N, B\}$ — tangent, normal, binormal — and the two scalar invariants curvature $\kappa(s)$ and torsion $\tau(s)$, where s denotes arc-length. The construction starts from a chosen parametrisation of γ, but *after rectification by arc-length* the trihedron and its invariants depend only on the image of γ as a subset of $\mathbb{R}^3$, not on the parametrisation originally chosen. The fundamental theorem of curves makes the statement quantitative: a curve is determined, up to rigid motion, by the pair $(\kappa(s), \tau(s))$. The parametrisation is a choice; the curve as a geometric object is not.

The kinematic embedding plays, for a discrete time series, the same role that the Frenet-Serret triple plays for a smooth curve. The discrete jet $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ is the finite-difference analogue of the continuous jet $(\gamma, \gamma', \gamma'')$; the standardisation by $(\sigma_y, \sigma_\Delta, \sigma_{\Delta\Delta})$ is the discrete analogue of arc-length rectification, eliminating the dependence on the timestamp parametrisation; and the grade-wise components of the lagged geometric product $\mathbf{q}(t)\,\mathbf{q}(t+\tau)$ — scalar, bivector, pseudoscalar — are the algebraic counterparts of curvature, torsion, and the higher orientational invariants that the geometric algebra makes legible by grade rather than by index gymnastics.

The reading projects to three levels of granularity, in exact parallel with the Taylor-Takens-Lyons triple bridge of Act IV. At the discrete-finite level, standardisation eliminates timestamp dependence and the grade-resolved means are reparametrisation invariants. At the smooth-continuum level, in the limit $\Delta t \to 0$ that the planned Paper 6 of the GSD programme (Paper 06 , Continuum) addresses formally, the kinematic embedding becomes the analytic jet $J^2(\mathbb{R})$ and Frenet-Serret applies literally. At the asymptotic-rough-path level, the Lyons signature of the $\mathrm{Cl}(3,0)$-valued path is invariant to reparametrisation modulo the Hambly-Lyons uniqueness theorem (Hambly & Lyons, 2010), which establishes that two paths with the same signature differ at most by a tree-like equivalence — the infinite-dimensional functional analogue of "differ by a rigid motion" in Frenet-Serret. The three levels are, once again, the same proposition projected to different granularities.

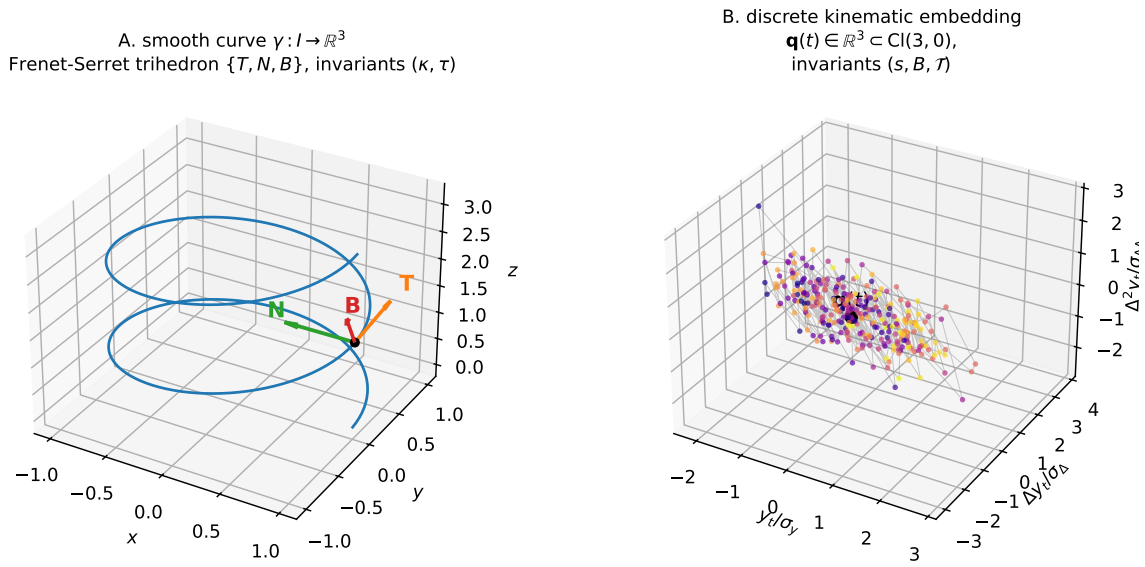

Figure 4: The Frenet-Serret correspondence. A point on a smooth curve carries a tangent T, a normal N, and a binormal B, together with two scalar invariants (κ, τ) that determine the curve up to rigid motion. A point on a discrete time series, lifted via the kinematic embedding, carries a level, a first step, and a second step, together with grade-wise invariants $(s, B, \mathcal{T})$ that determine the series up to Hambly-Lyons equivalence (Hambly & Lyons, 2010). The two are the same construction at two levels of granularity.

The conceptual content of the projection is what underwrites Hamilton's heuristic phrase *algebra as the science of Pure Time. Pure Time*, in his 1837 *Algebra as the Science of Pure Time* essay and in the prefatory pages of the *Researches respecting Quaternions*, is not a real-valued parameter $t \in \mathbb{R}$ to be chosen by the observer; it is the primordial category of progression, ontologically prior to any numerical parametrisation. The Frenet-Serret reading of the kinematic embedding gives this heuristic phrase a technical content: the geometric structure of a time series in $\mathrm{Cl}(3,0)$ — its moments, its bivectorial rotations, its pseudoscalar volumes — is constituted by the algebraic composition of moments and steps and *exists independently of the timestamp the observer attaches to it.* The algebra is the science of Pure Time precisely because its grade-resolved invariants survive reparametrisation, and the external indexing turns out to be parametric scaffolding rather than ontological substance. Hamilton, in 1837, did not have the geometric-differential apparatus to articulate the invariance in this form; we have it now, and the reading we have just given is what his phrase, on our reading, was always pointing at.

Egregium reading: the lagged scalar field as intrinsic surface. A natural extension of the Frenet-Serret reading, prefigured by Hamilton (1867) when he applied Gauss's *Disquisitiones Generales circa Superficies Curvas* of 1827 — the source of the *Theorema Egregium* establishing the curvature of a surface as an intrinsic invariant — to the *Sec-*

ond Method of two-parameter coordinate description, is to read the *lagged scalar field* $s(t,\tau) = \mathbf{q}(t) \cdot \mathbf{q}(t+\tau)$ as a two-parameter surface $(t,\tau) \mapsto s(t,\tau) \in \mathbb{R}$. Its Gaussian curvature $K(t,\tau) = (s_{tt}\, s_{\tau\tau} - s_{t\tau}^2)/(1 + s_t^2 + s_\tau^2)^2$ is, by the Theorema Egregium, an intrinsic invariant of the surface — independent of any embedding into ambient space. A preliminary smoke test on canonical generative processes ($T = 1500$, 50 replicates per class) shows that the proportion of points at which $K(t,\tau)$ is positive separates white noise from AR(1) processes with strong significance (t-statistic of order 8 between $\phi = 0$ and $\phi = 0.5$ or $\phi = 0.9$), suggesting that the sign distribution of K is a candidate intrinsic invariant complementary to the grade-resolved means $(\bar{s}, \bar{b}_{12}, \bar{\mathcal{T}})$. The smoke test does not yet establish a closed-form relationship between K and the spectral or grade-resolved content of the process; that connection is registered as an open conjecture of the GSD programme (GSD Book, 2026).

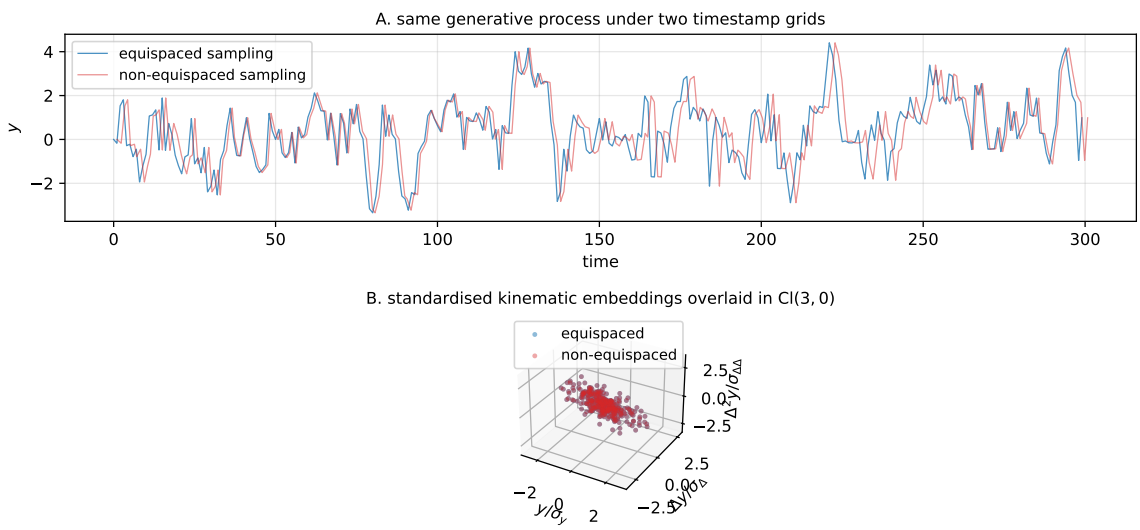

Figure 5: Invariance of the kinematic embedding under reparametrisation. The same generative process under two different timestamp grids (equispaced and non-equispaced) yields two superficially distinct series; after rectification, the standardised kinematic embeddings occupy the same region of phase space, illustrating empirically that the timestamp is parametric scaffolding while the grade-wise structure is the invariant content.

6.8 GSD and the Hestenes programme of geometric algebra

Geometric Signal Dynamics is not the first modern programme to recover the Hamiltonian thread through Clifford algebra. The recovery has been ongoing for sixty years, principally through the work of David Hestenes, beginning with *Spacetime Algebra* (Hestenes, 1966), continuing through *New Foundations for Classical Mechanics* (Hestenes, 1986, 1999) and the *Geometric Calculus* treatise written with Sobczyk (Hestenes & Sobczyk, 1984), and extending into a research community organised around the *International Workshop on Applications*

of Geometric Algebra that he founded. The Hestenes programme has demonstrated, with substantial empirical success, that the standard formal apparatus of theoretical physics — vector calculus, tensor calculus, complex numbers, quaternions, spinors, differential forms — admits a unifying re-expression in $\mathrm{Cl}(p, q)$ for the appropriate signature, and that the re-expression is not merely notationally more compact but conceptually more transparent: the geometrical content of physical equations becomes legible at the level of the algebra itself rather than being encoded in coordinate-dependent index gymnastics. The reformulations of relativistic electrodynamics, of Pauli and Dirac mechanics, and of classical mechanics in terms of multivector kinematics are achievements of the Hestenes programme and are the immediate intellectual ancestor of the present work.

GSD differs from the Hestenes programme in two respects, neither of which is a foundational disagreement. First, the domain. The Hestenes programme operates on smooth analytic paths in physical space and spacetime — the natural setting of classical and quantum mechanics — and its main objects are deterministic field equations and their geometric solutions. GSD operates on stochastic discrete empirical paths in arbitrary observational settings — the natural setting of statistical time-series analysis — and its main objects are sample functionals of the lagged geometric product and their asymptotic and finite-sample properties. The two domains share the underlying algebra (Hestenes uses $\mathrm{Cl}(1, 3)$ for spacetime applications and $\mathrm{Cl}(3, 0)$ for non-relativistic three-dimensional physics; GSD uses $\mathrm{Cl}(3, 0)$ for univariate kinematic embeddings and $\mathrm{Cl}(2d, 0)$ or $\mathrm{Cl}(3d, 0)$ for bivariate and multivariate extensions) but address disjoint application territories.

Second, the role of estimation. Because GSD operates on empirical data, its central objects are not field equations but estimators: the bivector mean $\mathbb{E}[B(t, \tau)]$ as a unit-root diagnostic (Paper 01 , GDF); the antisymmetric entropy $H_{AS} = H_0 + H_2$ as an information-theoretic decomposition (Paper 04 , HAS; Paper 18 , JC); the grade-resolved transfer entropy $\Phi^{(g)}_{X\to Y}$ as a causal-flow decomposition (Paper 05 , GTE; Paper 17 , FC); the moment tensor $\mathcal{M}^{(c,k)}_y$ as a Lyons-signature truncation (Paper 24 , MT); the geometric ergodicity exponent $\hat{\eta}_y$ as a regime classifier (Paper 25 , GE). None of these has a direct counterpart in the Hestenes literature, because the questions they answer (does this empirical series have a unit root; how much information of grade g flows from X to Y; is this panel ergodic; what regime does this series belong to) are not the questions of theoretical physics. They are the questions of statistical inference under stochastic observation, and they require the coupling of geometric algebra with the empirical-process theory of Kraskov et al. (2004); Marčenko & Pastur (1967); Lyons (1998) that the Hestenes programme did not need.

We therefore situate GSD as the sister programme of the Hestenes school on the data-analytic side. The two are nieces of Hamilton, of Grassmann and of Clifford in the same sense; their division of labour is by application domain, not by foundational commitment. The Hestenes programme has demonstrated the unifying power of geometric algebra for theoretical physics; GSD aims to demonstrate the analogous unifying power for empirical signal analysis. Where one ends and the other begins is, ultimately, the boundary between the deterministic continuum and the stochastic sample.

6.9 Synthesis

The argument of this act can be stated very compactly. Hamilton, in 1843, declared a programme to extend the algebraic treatment of *Pure Time* from couples to sets of arbitrary order, with three named categories: *moments*, *steps* and *numbers*. The first two categories are realised, with literal correspondence, by the level and the finite differences of the kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ (Definitions 6.1 and 6.2). The third — *numbers* as algebraic compositions of moments and steps — is realised by the multivector (12) that assembles the vector, scalar, bivector and pseudoscalar grades of the lagged geometric product into a single object, and, on the empirical side, by the moment tensor $\mathcal{M}_y^{(c,k)}$ that Paper 24 (MT, Theorem 5) identifies with the order-≤ 4 truncation of the Lyons signature of the kinematic path.

To this trinity GSD adds four further articulations that Hamilton could not yet have made and that the spectral-econometric tradition made under constraints we now have the algebra to relax. The first (Section 6.4) is the recovery of *uncertainty by grade*: the Wold-Cramér inheritance placed all the probabilistic content of a stationary time series in the scalar variance of the residual, because Kolmogorov's measurable random variable was, by analytic convenience of the period, scalar- valued. Allowing the random variable to live in $\mathrm{Cl}(3,0)$ recovers the three-grade entropy decomposition $H_{\mathrm{geom}} = H_0 + H_2 + H_3 - \mathrm{TC}$ (Paper 04 , HAS; Paper 18 , JC) and reveals that the uncertainty of a series is structurally three independent quantities, not one. The Clifford Fourier transform of Hitzer & Mawardi (2013) provides a multivector Heisenberg principle on the same $\mathrm{Cl}(3,0)$, saturated by multivector Gaussians, which shows constructively that the spectral machinery extends beyond the abelian fragment when one is willing to use a central pseudoscalar kernel rather than a scalar one. The second (Section 6.5) is the recognition that the kinematic embedding has been operating, unrecognised, inside the most cited tests of unit-root and cointegration analysis for forty-five years: the Dickey-Fuller regression on $(\Delta y_t, y_{t-1})$, the Engle-Granger residual unit-root test, and the Johansen VECM are all kinematic-embedding regressions whose test statistics are projections onto specific grade components of the same canonical algebraic object; in the wider modern toolkit unified by Sudjianto (2025) under the embedding-and-linear-modelling paradigm, GSD is the time-series instance whose embedding has been canonicalised by algebraic argument rather than by empirical search. The third (Section 6.6) is the formal identification $\mathrm{Cl}(3,0) \cong \mathbb{C} \otimes \mathbb{H}$ between the algebra in which GSD operates and Hamilton's own biquaternions: the 1853 paper of Hamilton (1853) on the geometric interpretation of biquaternion calculation already worked with bivectors of the form $j+hk$ — null-squared elements that in the modern presentation are light-like vectors of an extended signature — and used them to study fixed-point iterations producing geometric loci, that is, discrete dynamical systems in the sense of the present-day theory. Hamilton was, in 1853, attempting the geometric operationalisation of his own programme on the same algebra GSD now uses, and lacking only the basis-canonical presentation that Clifford was to give twenty-five years later. The fourth (Section 6.7) is the recognition that the embedding is intrinsic geometry in the sense of classical differential geometry: the timestamp

$0, 1, 2, \ldots$ that an observer attaches to a series is parametric scaffolding, not ontological substance, and the standardised kinematic embedding is the Frenet-Serret-style invariant of the series as a curve in $\mathrm{Cl}(3, 0)$. The reading projects to three levels of granularity in exact parallel with the Taylor-Takens-Lyons bridge — discrete-finite via standardisation, smooth-continuum via Frenet-Serret proper, and asymptotic-rough-path via Hambly-Lyons signature uniqueness — and gives Hamilton's phrase *algebra as the science of Pure Time* its technical content: the algebra is the science of Pure Time because its grade-resolved invariants survive reparametrisation.

Together the recoveries — moments, steps, and the recognition of *numbers* as multivectors composed of them; uncertainty by grade rather than aggregated in a scalar residual; the kinematic embedding made explicit where the classical tests had it implicit; the formal identification of $\mathrm{Cl}(3, 0)$ with the biquaternion algebra Hamilton himself was operationalising; and the parametrisation-invariant intrinsic geometry that turns the external timestamp into label rather than structure — constitute the substance of GSD as instantiation of the Hamiltonian programme. The framework operates as the discrete empirical-data sister of the Hestenes geometric-algebra programme for theoretical physics; the operational transcription of moments, steps, and multivector compositions into the public API of the `gsd` library is documented in Appendix A for readers wishing to follow the correspondence into executable code.

The recovery is not novel mathematics: every individual ingredient has been available since 1878 (Clifford), since 1933 (Kolmogorov), since 1981 (Takens), since 1998 (Lyons) or since 1966 (Hestenes). The novelty is that they have not been previously assembled in this configuration with explicit fidelity to the Hamiltonian declaration of 1843. What this paper claims, and what the GSD research programme as a whole aims to substantiate, is that the assembly is faithful and that it is operationally fruitful in time-series analysis, in econometrics, in nonlinear dynamics and statistical mechanics, in causal inference, and in the applied domains where these meet. The claim of fruitfulness is the substance of Act VII (the scope of the programme), and the honest enumeration of what the programme does not yet deliver, or where its delivery is conditional on unverified assumptions, is the substance of Act VIII (limitations and open directions). The present act prepares the ground for both by establishing what the programme *is*, in its own self-understanding, in continuity with the agenda Hamilton declared and to whose continuation he explicitly invited mathematicians yet to come.

7 Scope: ten domains in which the kinematic embedding unblocks structural information

The five acts have argued that Geometric Signal Dynamics is the instantiation of Hamilton's programme of *algebra as the science of Pure Time* on the algebraic substrate of $\mathrm{Cl}(3,0)$ and the empirical-statistical substrate of Kolmogorov's measure- theoretic probability. The argument so far has been historical and structural; the present section turns to the operational scope. We articulate, in ten short subsections corresponding to ten disciplinary domains, what the kinematic embedding unblocks in each domain — what specific problem becomes solvable, or identifiable, or quantifiable that had been opaque, ad-hoc, or patched under the scalar inheritance — and state honestly what the framework does not yet deliver in that domain. The articulation cites the specific paper of the GSD programme in which the unblocking is operationally demonstrated, with the working papers (Paper 01 , GDF; Paper 04 , HAS; Paper 24 , MT; Paper 25 , GE) and the integrating paper on layered detection and classification (Paper 23 , Layered) flagged as the formal-publication channel and the remainder as the research-note channel of the *GSD book repository* (GSD Book, 2026).

The ten domains are organised by thematic affinity rather than by historical priority. The intent is to make it possible for a reader of any one of them to enter the GSD programme through the door most legible to her own discipline.

7.1 Time-series analysis and econometrics

The first natural domain is the one most closely identified with the spectral-econometric inheritance traced in Acts II and III. Under that inheritance, non-stationarity is a violation of modelling assumptions to be removed by transformation (differencing, filtering, cointegration projection) before the analytical machinery can be applied; the transformation produces a stationary residual on which the abelian Fourier-dual analysis proceeds. Under the geometric reading, non-stationarity is a *rotation* in the bivector field of the kinematic embedding that the spectral identity (6) measures directly. The unit-root null becomes the vanishing of the bivector mean rather than a non-degenerate distribution of a $\hat{\rho}$ statistic; the cointegrating direction becomes a cross- level bivector rather than a parameter of an auxiliary regression; the multivariate causal flow decomposes by grade rather than collapsing into impulse-response functions on scalar innovations. The Geometric Dickey-Fuller test (Paper 01 , GDF), the Geometric Cointegration framework (Paper 03 , GC), the Spectral Foundations (Paper 02 , Spectral) and the Geometric Transfer Entropy (Paper 05 , GTE; Paper 17 , FC) jointly rebuild the central inferential tools of empirical econometrics on the kinematic embedding rather than on the scalar process. *Honest limitation:* the rebuilding has been demonstrated on the canonical inferential tasks (unit-root testing, cointegration, transfer entropy) but has not been extended to the full ARFIMA-long-memory or the Markov-switching families, where the spectral-econometric tradition has decades of refinement that the geometric reading does not yet match.

7.2 Nonlinear dynamics and chaos theory

The second domain is the one in which the kinematic embedding was implicitly recovered, under the lag-embedding presentation, in the Takens theorem of 1981. The Pascal equivalence of Lemma 5.2 shows that the methodological apparatus that the nonlinear-dynamics community built on Takens' embedding — attractor reconstruction, dimension estimation, Lyapunov exponents, surrogate-data tests — is operationally compatible with the geometric reading on the kinematic basis, with the algebraic content of the lag-embedding made legible by grade rather than by index gymnastics on the embedded cloud. The Geometric Takens Theorem (Paper 07 , Takens) extends Takens' result to the discrete-stochastic regime in which empirical time series live, replacing the deterministic dynamical system with a linear-AR generative process and the manifold-embedding conclusion with a saturation of the linearised geometric transfer entropy at $d^\dagger = \max(2, k^* + 1)$, where k^* is the coupling order of the underlying generative process. The result extends the territory of Takens-style reasoning from deterministic chaos into the standard regime of empirical econometrics. *Honest limitation:* the extension is rigorous for linear-AR drivers and is empirically robust against non-Gaussian innovations and threshold-AR generative processes, but it does not cover deterministic-chaotic systems with finite attractor dimension where the original Takens theorem applies in its full strength; in that regime our framework offers an algebraically convenient presentation, not a sharper mathematical statement.

7.3 Complex systems and the geometry of complexity

The third domain is the broad area of complex systems, in which a heterogeneous family of complexity measures — Lempel-Ziv complexity, Lyapunov exponents, fractal dimensions, multifractal spectra, permutation and sample entropies — has been developed without an underlying common algebraic substrate. The geometric reading proposes that the structural content of complexity, in each of the cases where it is empirically meaningful, is a specific grade-resolved feature of the kinematic embedding: persistence complexity at grade zero, rotational complexity at grade two, chiral or volumetric complexity at grade three. The antisymmetric entropy decomposition of Paper 04 (HAS) and the three-grade extension of Paper 18 (JC) provide the information-theoretic articulation of this position; the moment- tensor classification of Paper 24 (MT) provides the empirical demonstration that linear classifiers on the moment tensor of $\mathrm{Cl}(3,0)$ outperform dedicated complexity-measure batteries on canonical generative-process classification tasks. The heterogeneous family of complexity measures is, on this reading, the multiple scalar-projection shadow of a single algebraic object that GSD makes legible. *Honest limitation:* the moment-tensor classification has been demonstrated on the discrete generative-process zoo of stationary AR/RW/ARIMA/GARCH/ fBm; it has not yet been demonstrated on chaotic-deterministic systems whose complexity content has historically motivated the field, and the relation between the bivector and trivector energies of GSD and the fractal-dimension family of measures remains conjectural pending the planned Paper 6 on the continuum limit (Paper 06 , Continuum).

7.4 Information theory and entropy

The fourth domain is the information-theoretic core of dynamic data analysis. Shannon entropy and its conditional and mutual extensions are the basic vocabulary of the field, but the empirical-statistical practice has worked almost exclusively with real-valued random variables, for the historical reasons traced in Section 3.2. The geometric reading recovers the multivector-valued extension implicit in Kolmogorov's open codomain: the random variable $\mathbf{q}(t)$ takes values in $\mathrm{Cl}(3,0)$, and its entropy decomposes into a grade-zero scalar entropy, a grade-two bivector entropy, and a grade-three pseudoscalar entropy, with the total-correlation remainder empirically negligible (Paper 18 , JC, Theorem 1). The Hitzer-Mawardi multivector Heisenberg principle (Hitzer & Mawardi, 2013) provides the spectral-frequency dual of the same algebra. The transfer-entropy machinery that has dominated the empirical-information-theoretic literature on causality (Schreiber, 2000) is recovered, in the geometric reading, as the grade-zero projection of a grade-resolved transfer-entropy decomposition (Paper 05 , GTE; Paper 17 , FC) that adds the bivector and trivector channels as independent direction-bearing components of the same underlying flow. *Honest limitation:* the grade-resolved transfer-entropy machinery has been demonstrated on bivariate generative processes with simulated coupling structure; the extension to high-dimensional panels (e.g. FRED-MD with $N \sim 100$) is open and depends on the random-matrix-theoretic scaling of Paper 08 (MPRT) which is itself empirically validated but not yet integrated with the grade-resolved transfer-entropy machinery in a single multivariate pipeline.

7.5 Causal inference and the do-calculus

The fifth domain is the rapidly growing field of causal inference under the do-calculus framework of Pearl (2009). The do-calculus has been classically applied to non-temporal structural causal models, with the extensions to time series (Runge, 2019) typically operating at the level of scalar-valued processes and parametric impulse- response decompositions. The geometric reading provides a graded extension of the do-operator (Paper 15 , Graded): an intervention $\mathrm{do}^{(g)}(\Delta^g X = x_0)$ that fixes the g-th finite difference of X at x_0 rather than the level itself. The Graded Pearl-Takens Theorem of Paper 15 (Graded, Theorem A) establishes that the intervention grade k^* is identifiable from the observational distribution via the grade-resolved transfer entropy if and only if the embedding dimension $d \geq \max(2, k^* + 1)$. The do- calculus is therefore not abandoned but enriched: the identification conditions are graded, the intervention modes are graded, and the test statistics are projections onto specific grade components of the same algebraic object. *Honest limitation:* the graded extension has been demonstrated for linear-Gaussian SCMs with a single intervention grade per edge of the causal graph; multi-grade interventions and the extension to non-Gaussian or non-linear SCMs are open. The do-calculus literature has substantial machinery for handling latent confounders that the geometric reading has not yet absorbed.

7.6 Statistical mechanics and the thermodynamics of trajectories

The sixth domain is the statistical-mechanical analysis of processes, with particular emphasis on the Jarzynski-Crooks fluctuation theorems (Jarzynski, 1997; Crooks, 1999) and the Peters-Adamou ergodicity-economics literature (Peters & Adamou, 2018). The classical apparatus of these literatures requires either explicit control protocols (in the Jarzynski case) or explicit ensemble- versus-time-average comparisons (in the Peters-Adamou case), neither of which is operationally feasible on most empirical data. The geometric reading provides observational proxies for both. The Jarzynski-Crooks correspondence of Paper 18 (JC, Theorem 2) expresses the discrete entropy production rate as $\langle \Sigma \rangle = \kappa(\phi)\, \mathbb{E}[\mathcal{T}(1,2)]^2 + O(\gamma_3^4)$ in the AR(1) skew-normal regime, providing a direct estimable proxy from the kinematic trivector mean. The geometric ergodicity exponent of Paper 25 (GE, Definition 2) provides a single dimensionless number per series, classifying the process into ergodic, frozen non-ergodic, or explosive non- ergodic regimes via the log-log scaling of windowed moment-tensor dispersion. Both proxies operate on observational data without simulation or explicit ensemble. *Honest limitation:* the Jarzynski-Crooks correspondence is established for AR(1) skew-normal innovations and validated empirically on a wider class but is not proved for general non-Markovian processes; the geometric ergodicity exponent has been validated on canonical generative-process families but its formal asymptotic distribution under finite samples is open.

7.7 Risk management and quantitative finance

The seventh domain is the quantitative-finance application of fragility, leverage, and tail-risk analysis. The classical apparatus rests on the Taleb-Douady fragility theorem (Taleb & Douady, 2013), the Adrian-Brunnermeier conditional- value-at-risk framework (Adrian & Brunnermeier, 2016), and the copula-DCC dependence machinery of Engle (2002). Each of these depends on either an explicit option-pricing model (Taleb-Douady), a parametric quantile regression (CoVaR), or a parametric joint-distribution assumption (copula-DCC). The Geometric Fragility paper (Paper 16 , GF) establishes that the trivector mean $\mathbb{E}[\mathcal{T}^Y]$ is a non-parametric Taleb-Douady fragility estimand without option-pricing; the Geometric Fragility Causality paper (Paper 17 , FC) extends this to a non-parametric measure of fragility spillover between series; and the empirical validation of Paper 19 (Validation) demonstrates that the Jarzynski-Crooks proxy of Paper 18 (JC), in combination with VIX, achieves out-of-sample PR-AUC 0.66 on NBER recession-vicinity prediction (vs. VIX alone at 0.62), with Pearson correlation 0.005 vs. VIX (orthogonal information). The geometric apparatus is therefore not merely interpretable but empirically competitive with the established quantitative- finance toolkit. *Honest limitation:* the empirical demonstrations have been on US equity and FX data over 1971-2026; extension to fixed-income, commodities, and crypto-assets is open, and the comparison with the non-parametric quantile literature (Koenker, 2005) is not yet systematic.

7.8 Engineering: condition monitoring and fault detection

The eighth domain is the engineering practice of condition monitoring and predictive maintenance. The classical apparatus rests on time-series of sensor readings combined with feature-extraction batteries (root-mean-square, kurtosis, spectral features, wavelet coefficients) and threshold-based remaining- useful-life estimators. The geometric reading provides a non-parametric grade-resolved feature battery: persistence energy (grade 0) for the slow drift of the operating point, rotational energy (grade 2) for the kinematic dynamics of the sensor signal, and pseudoscalar energy (grade 3) for the chiral asymmetry that often precedes failure. The antisymmetric-entropy work of Paper 04 (HAS) demonstrates on the CMAPSS turbofan- engine benchmark that the divergence between bivector energy and bivector entropy is a degradation indicator that fires 35–45 cycles before classical RUL thresholds; the three-grade entropy work of Paper 18 (JC) extends this with a chiral fragility indicator that achieves CMAPSS lead-time 0.45 of life, matching or surpassing the dedicated trivector-based estimator of Paper 17 (FC). The detection is non-parametric and does not require domain-specific tuning. *Honest limitation:* the demonstrations have been on the CMAPSS benchmark; extension to rotating machinery, structural-health monitoring, and electrical- grid signals is open. The integration with physics-of-failure domain knowledge (specific to each engineering application) has not yet been systematic.

7.9 Dynamic inequality and the geometry of trajectories

The ninth domain is the analysis of inequality and mobility in longitudinal income and wealth panels. The classical apparatus rests on cross-sectional inequality measures (Gini, Atkinson) and on rank-rank intergenerational mobility coefficients (Chetty et al., 2014). Both are static or quasi-static measures; neither captures the geometry of the individual trajectories that constitute the panel. The Geometric Ergodicity paper (Paper 25 , GE, Section 4) proposes the panel ergodicity exponent $\hat{\eta}^{\text{panel}}$ as a non-parametric classifier of the panel into ergodic, frozen, or explosive regimes, with the canonical-DGP demonstration showing that quintile-locked frozen panels have $\hat{\eta}^{\text{panel}} = +0.01 \pm 0.04$ (statistically indistinguishable from zero) while explosive heterogeneous-compounding panels have $\hat{\eta}^{\text{panel}} > 4$. The signal-to-noise of the classification is sufficient to distinguish economies with identical Gini coefficients and identical rank-rank slopes. The geometric reading makes available, on the same algebraic substrate, a per-trajectory fragility score $\mathbb{E}[\mathcal{T}_i]$ at each individual unit of the panel (Paper 16 , GF), providing both an aggregate measure ($\hat{\eta}^{\text{panel}}$) and an individual measure ($\mathbb{E}[\mathcal{T}_i]$ for each i) of the dynamic structure of inequality. *Honest limitation:* the application has been simulated on canonical panel-DGPs; the extension to real longitudinal income panel data (PSID, LIS cross-country, administrative registries) is the explicit subject of a forthcoming applied-extension paper (Paper 26 , DI) and remains empirical work in progress.

7.10 Machine learning and regime classification

The tenth domain is contemporary machine learning, in particular the regime-classification problems that arise in time-series analysis under the embedding-and-linear-modelling paradigm of Sudjianto (2025). The classical machine-learning toolkit chooses an embedding $\phi : \mathbb{R}^d \to \mathbb{R}^k$ — adaptive trees, sparse one-hot encodings, learned continuous representations, fixed orthogonal bases — and applies a linear or weakly-nonlinear classifier to its image. The choice of embedding is the central methodological lever and is typically made by hyperparameter search or empirical comparison. The moment-tensor representation of Paper 24 (MT) provides an *algebraically determined* embedding for univariate time series — the 64-coordinate moment tensor on the Clifford-channel representation of the kinematic path — that is neither learned nor selected by hyperparameter, and that is empirically demonstrated to match or surpass dedicated estimator batteries on canonical generative-process classification tasks (Paper 24 , MT, Section 8). The classification is not a proposal of a new machine-learning algorithm; it is a proposal of a new embedding that the standard machine-learning toolkit can operate on. The Hambly-Lyons uniqueness theorem (Hambly & Lyons, 2010) provides the asymptotic warrant: the moment tensor is the truncation of the canonical observable of the kinematic path modulo tree-like equivalence. *Honest limitation:* the demonstrations have been on synthetic generative processes; extension to large-scale empirical benchmarks (UCR time-series classification archive, M-competition forecasting datasets) is open and is the explicit subject of the integrating paper (Paper 23 , Layered).

7.11 Synthesis and further domains

The ten domains share a common structural pattern. In each, the classical apparatus has accumulated, over decades of methodological refinement, a heterogeneous family of devices for handling the specific empirical recalcitrance of its problem space; in each, the underlying object that the devices are recovering admits a single algebraic articulation in $\mathrm{Cl}(3,0)$ that the kinematic embedding makes operationally legible; and in each, the GSD programme has demonstrated, on a specific paper of the research note or working-paper line, that the geometric articulation is empirically competitive with or superior to the classical apparatus on canonical tasks. The articulation is not a proposal to discard the classical apparatus, which has earned its place by sustained empirical success; it is a proposal to recognise the algebraic structure that the apparatus has been silently respecting and to make that structure operationally explicit, with the consequent simplification of the methodological landscape and the consequent recovery of interpretability that the grade-resolved decomposition provides.

The ten are not exhaustive. Three further domains, in which the geometric apparatus is in principle applicable but has not yet been substantively developed, are worth registering as natural next directions. The first is *neuroscience and connectivity analysis* (fMRI BOLD signals, EEG/MEG dynamics, cellular-level calcium imaging), where the multivariate graded transfer entropy of Paper 17 (FC) combined with the random-matrix-theoretic scaling of Paper 08 (MPRT) would provide a non-parametric graded-causal-graph discovery

framework. The second is *climate science*, where the same machinery applied to multivariate sensor arrays would provide a graded analogue of the PCMCI-style causal discovery on benchmark climate datasets such as CauseMe. The third is *audio and music signal processing*, where the kinematic embedding of an acoustic signal combined with the grade-resolved decomposition would provide an algebraic substrate for rhythm, harmonic structure, and timbral asymmetry that the spectral-only apparatus does not natively articulate. Each of these is a substantial research direction in its own right and is registered here as part of the operational scope of the programme rather than as a delivered application.

8 Limitations and open directions

A research programme that proposes itself as the long-deferred realisation of an unfinished agenda is well-served by stating explicitly what it does *not* yet deliver. We have articulated, throughout the body of the paper, the operational scope of GSD across ten disciplinary domains; the present section collects the structural limitations, the conjectures that remain unproven, and the methodological gaps that the programme has identified but not closed. The intent is honest enumeration, not self-criticism. Each limitation is, in our reading, a research direction rather than a defect, and the paper would betray its own genre — synthesis rather than manifesto — if it did not state them plainly.

8.1 Core papers planned but not yet written

Three of the originally articulated core papers of the GSD programme (GSD Book, 2026) are at the time of writing planned but not yet drafted. Each addresses a structural question whose resolution would tighten the foundations on which the present synthesis rests.

Continuum limit and rough-path connection (Paper 06). The discrete kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ becomes, in the limit $\Delta t \to 0$, the analytic jet $J^2(\mathbb{R})$ of a smooth path; the moment-tensor / signature correspondence of Paper 24 (MT, Theorem 5) becomes, in the same limit, an exact identification with the Lyons signature on the $\mathrm{Cl}(3,0)$-valued kinematic path. The convergence is widely expected on the basis of standard limit theorems for stochastic calculus and rough-path theory (Lyons, 1998; Hambly & Lyons, 2010), but it has not been formally proved in the GSD context. The formal proof is the central content of the planned Paper 06 (Paper 06 , Continuum). While the proof is open, every asymptotic claim of the present paper that invokes the continuum limit — the Itô-Stratonovich identification of the bivector, the convergence of the moment tensor to the signature, the Frenet-Serret reading at the smooth-continuum level — should be read as conjectural with the operational discrete content secured.

Gauge invariance of the embedding (Paper 09). The kinematic embedding depends on the choice of standardisation $(\sigma_y, \sigma_\Delta, \sigma_{\Delta\Delta})$ in the diagonal mode and on $\Sigma_{\mathbf{q}}^{-1/2}$ in the full Mahalanobis mode. The two modes yield different numerical values of the grade-wise statistics, and the question of which observables are invariant under the choice — and which are gauge-dependent — has been raised but not formally resolved. Conjecturally, *ratios* of grade-wise quantities ($\Phi^{(2)}/\Phi^{(0)}$, the mechanism angle $\theta = \arctan(\Phi^{(2)}/\Phi^{(0)})$, the asymmetry coefficient $\mathcal{A}$) are gauge-invariant, while *magnitudes* are not. A formal characterisation of the gauge structure as a fibre bundle over the space of standardisations, with explicit identification of gauge-invariant versus gauge-dependent observables, is the content of the planned Paper 09. Until that paper is written, every quantitative claim of the present synthesis should be read as conditional on a specified standardisation mode, and the choice of mode should be declared explicitly in any operational deployment.

Wasserstein metric on signatures (Paper 10). The windowed geometric ergodicity exponent of Paper 25 (GE, Definition 2) detects regime change as a deviation from the linear

log-log scaling of the moment-tensor dispersion across windows. A more refined diagnostic — the Wasserstein distance $d_W(\mu_t, \mu_{t'})$ between distributions of the GTE signature $(\Phi^{(0)}, \Phi^{(2)})$ at distinct epochs — is the content of the planned Paper 10 of the programme. The Wasserstein distance provides a continuous quantification of structural change that is insensitive to the discrete event-detection of the regime classifier; it would unify the windowed-stationarity diagnostic of Paper 05, the ergodicity exponent of Paper 25, and the fragility-spillover detection of Paper 17 under a single metric on the space of GSD signatures. The construction is implicit in the existing machinery but the formal development is open.

8.2 Conjectures declared in completed papers

Several results in the working-paper line of the programme are declared with empirical support but partial formal proof. We register them here so that the synthesis does not present them with greater authority than their current status warrants.

The *Lyons-signature equivalence* of Paper 24 (MT, Theorem 5) — that the empirical moment tensor of order ≤ 4 is the truncation of the Lyons signature on the $\mathrm{Cl}(3,0)$-valued kinematic path — is empirically supported by the smoke-test results reported in Section 8 of that paper but the explicit term-by-term identification between moment-tensor entries and iterated-integral signature components is not yet written. The Hambly-Lyons uniqueness of the truncated tensor as a canonical observable on the path module is, accordingly, conjectural pending the explicit identification.

The *Jarzynski-Crooks correspondence* of Paper 18 (JC, Theorem 2) is proved for AR(1) skew-normal innovations and validated empirically across a wider class of generative processes; the extension to general non-Markovian processes is open. The proxy $|\mathbb{E}[\mathcal{T}]|^2/\mathrm{Var}(\mathcal{T})$ is operationally useful in empirical applications, but its theoretical interpretation as discrete entropy production rests on the AR(1) skew-normal proof and on numerical agreement under more general regimes rather than on a complete theorem.

The *spectral conjecture* of Paper 05 (GTE, Conjecture 1) — that $\Phi^{(2)}_{XY} \geq c \cdot |\mathrm{Im}\, f_{XY}(\lambda^*)|$ for some universal constant c — is declared with motivation but without proof. It is the multivariate-causal analogue of the Spectral Identity (6) and would close the bridge between the abelian cross-spectrum and the grade-resolved transfer entropy.

8.3 Limitations of the empirical estimators

The non-parametric estimators on which the GSD programme rests — the KSG mutual-information estimator (Kraskov et al., 2004), the Marchenko-Pastur rank estimator of Paper 08 (MPRT), the moment-tensor classifier of Paper 24 (MT) — inherit the finite-sample limitations of the underlying machinery.

The KSG estimator has known bias that grows with the dimension of the conditioning argument, and the high-grade content of the GSD signature involves multi-dimensional conditioning that pushes this bias into the regime where it can affect quantitative conclusions. Reported in Paper 07 (Takens, Table 3), the bias on the linearised geometric transfer

entropy at $T = 3000$, $d = 5$ reaches -8.8%, which is operationally significant. The empirical countermeasures (block bootstrap on the residuals, calibrated null distributions on permuted innovations) work but are computationally expensive and do not eliminate the bias at the level of the sufficient statistic.

The Marchenko-Pastur null on the bivector covariance requires the full Mahalanobis whitening of Section 6.7 and breaks empirically under the diagonal whitening. The break is not a defect of the theorem but a consequence of the algebraic cross-coupling that the diagonal mode does not remove; the operational consequence is that any inferential application of Paper 08 (MPRT) requires the full whitening mode, which has higher computational cost and is sensitive to the conditioning of the Gram matrix $\Sigma_{\mathbf{q}}$ at small samples.

The moment-tensor classifier of Paper 24 (MT) achieves strong empirical performance on canonical generative-process classification tasks, but its asymptotic distribution under the null of an arbitrary stationary process is not known in closed form. Calibration of significance thresholds for hypothesis- testing applications relies on permutation null distributions that are computationally expensive to estimate and that may not respect the underlying coupling structure of the data.

8.4 Out-of-scope of the Geometric Takens Theorem

The Geometric Takens Theorem of Paper 07 (Takens, Theorem 2) covers linear AR(p) and AR(p)-$I(p_{\text{int}})$ drivers with linear coupling via $\Delta^{k^*}X$, VECM couplings (effective $k^* = 0$ relative to cointegrating error), non-Gaussian innovations (Student-t with $\nu > 2$, χ^2-skewed), TAR- threshold and GARCH-in-level dynamics. Three families of generative process are explicitly out of scope. *Pure triangular cointegration* $Y_t = \beta X_t + u_t$ with no temporal lag between the two series violates the lag-structure assumption of the proof and requires the multivariate $\mathrm{Cl}(2d, 0)$ extension of Paper 03 (GC); the unified treatment of triangular and lagged cointegration in a single algebraic framework is open. *Quadratic and higher-order coupling* $Y_t = \phi Y_{t-1} + c\sigma^2_{X,t-1} + \varepsilon$ couples through a non-linear functional of the driver and is invisible to the linear kinematic embedding; recovery of this content would require an algebraic extension to tensor-product or higher-grade-coupled embeddings. *Long-memory and chaotic-deterministic processes* (ARFIMA, strange-attractor systems with finite Lyapunov dimension) violate the finite-order state-space assumption of the proof and are covered, in part, by the planned Paper 06 on the continuum limit.

8.5 The regional optimality of $\mathrm{Cl}(3, 0)$

The choice of $\mathrm{Cl}(3, 0)$ as the algebraic substrate is justified by the Geometric Takens Theorem as the optimal embedding dimension for coupling orders $k^* \in \{0, 1, 2\}$ — level, velocity and acceleration coupling, the operationally relevant family in physics, economics and engineering. Higher-order coupling regimes ($k^* \geq 3$, that is, jerk, snap, and beyond) require Clifford algebras of higher dimension, $\mathrm{Cl}(d, 0)$ with $d > 3$, and the GSD programme has not yet developed the parallel machinery for these regimes systematically. The multivariate cointegration extension to $\mathrm{Cl}(2d, 0)$ is in the working-paper line of Paper 03 (GC), but

the trivariate and higher-order graded transfer entropy frameworks are pending. The honest reading is that GSD is a regionally optimal framework for the operationally dominant family of empirical generative processes, not a universally optimal framework, and that the extension to the full Clifford ladder is a research direction rather than a delivered theorem.

8.6 Three tensions worth declaring as such

Beyond the technical limitations enumerated above, three conceptual tensions are best addressed by declaring them as ongoing rather than as resolved.

The independence of the grades. The grade decomposition $H_{\text{geom}} = H_0 + H_2 + H_3 - \text{TC}$ is empirically quasi-additive, with the total-correlation remainder $\text{TC}/H_{\text{geom}} < 0.5\%$ on canonical generative processes (Paper 18 , JC, Theorem 1). The exact algebraic independence at the level of the Clifford grades is structural, but the statistical near-independence at the level of the empirical estimators is a contingent fact about the canonical generative-process zoo and is not a theorem. Pathological processes for which the grades are highly entangled statistically are constructible in principle, and the operational scope of the additive decomposition is therefore conditional on the relevant generative-process class.

The status of the kinematic embedding as canonical. The choice of $(y_t, \Delta y_t, \Delta^2 y_t)$ as the canonical embedding is supported by three convergent arguments — algebraic completeness, the Pascal equivalence with the lag embedding, the Frenet-Serret reading as intrinsic geometry — but none of these uniquely identifies the kinematic basis among linear-algebraically equivalent alternatives (e.g., orthogonal polynomial bases on the same span). The kinematic basis has operative virtues (each component carries a clear physical interpretation; the components have comparable scales after standardisation), but the claim of *canonical* status is philosophical rather than mathematical and depends on the Hamiltonian programme's identification of moments and steps as primitive temporal categories. We do not know how to prove canonicity in a stronger sense.

The synthesis as historical reading. The reconstruction that places GSD as the realisation of an unfinished seventeenth-to-nineteenth-century programme is, in the philosophical-historical sense, a reading rather than a historical reconstruction. Hamilton, Grassmann, Clifford, Kolmogorov, Takens, Lyons and Hestenes did not assemble the pieces into the programme we have just articulated; the assembly is retrospective, and other plausible historiographical readings of the same materials are available. The synthesis we propose has the operational virtue of articulating the GSD programme's identity in a single coherent narrative, but it should be read as such — as identity- articulation rather than as historical reconstruction proper — and other historians and philosophers of mathematics would reasonably find different organising threads through the same texts.

9 Closing

The five acts and the scope chapter have argued that Geometric Signal Dynamics is the discrete-statistical realisation of a research programme declared by William Rowan Hamilton in 1837 and 1843, abandoned in its full generality during the second half of the nineteenth century, and made operationally accessible in the early twenty-first by the gradual completion of three mathematical traditions that Hamilton's programme depended on but that the programme's own century could not supply. The limitations chapter has registered, with deliberate straightforwardness, the gaps and open directions that the present synthesis does not close. We close the paper by returning to the Hamiltonian text from which it takes its title and its motivation.

Hamilton's 1853 paper *On the Geometrical Interpretation of Some Results Obtained by Calculation with Biquaternions* (Hamilton, 1853), communicated to the Royal Irish Academy in February of that year, is three pages long. In the opening paragraph Hamilton describes the conception of bivectors in $\mathbb{C} \otimes \mathbb{H}$ as having occurred to him "many years ago" and as "interpretable, geometrically on the author's principles"; he develops the calculus to a specific problem of fixed-point iteration producing geometric loci; and he closes by remarking, in a tone characteristic of the nineteenth-century Royal Society communications, that the geometric proofs of his algebraic results "will not be found difficult to confirm" by readers willing to follow the construction. The paper is, on a casual reading, a brief technical communication on a specific application of the biquaternion calculus.

On the reading we have developed here, it is more than that. It is the last sustained statement, in Hamilton's own voice, of the *programme* that his 1837 *Algebra as the Science of Pure Time* and his 1843–1848 *Researches respecting Quaternions* had declared and that the academy of his time would not pick up. The bivectors of 1853, written in his notation as $j+hk$ with h the central commutative imaginary, are exactly the elements of the algebra that we now write as $\mathrm{Cl}(3,0)$ under the isomorphism $\mathrm{Cl}(3,0) \cong \mathbb{C} \otimes \mathbb{H}$ (Section 6.6). The fixed-point iteration he studies is, in the structural sense, a problem of discrete dynamical systems on this algebra. And the geometric interpretation he proposes is what Section 6.3 of Act V articulates as the reading of the multivector (12) as the canonical algebraic representation of the kinematic state at each instant. Hamilton in 1853 was writing about the algebra in which Geometric Signal Dynamics now operates.

The earlier paper of 1843 closes with a sentence that has served, throughout the present synthesis, as both motivation and rhetorical anchor. We reproduce it here in full:

> *. . . the author indulges a hope that the papers containing an account of those subsequent investigations respecting Quaternions, which he has made, and (in part) communicated to the Academy, since the date prefixed to this First Series of Researches, will tend to place the subject in a still clearer point of view: and, by exhibiting more fully to mathematicians its interest and its importance, increase the likelihood of their contributing their aid to its development. (Hamilton, 1848, opening pages)*

The Geometric Signal Dynamics programme is, modestly, the development to which Hamilton's hope was addressed. The contributing aid has been long in coming — the algebra

of Clifford was twenty-five years away when Hamilton wrote, the axiomatic probability of Kolmogorov ninety years away, the embedding theorem of Takens a hundred and thirty-eight years away, the rough-path theory of Lyons a hundred and fifty-five years away — but the pieces have arrived, and the programme of *algebra as the science of Pure Time* is now operationally articulable on empirical-statistical data.

The articulation we have just performed is, by the standards of nineteenth-century communications to the Royal Irish Academy, a brief and late-arriving note. It does not introduce a new theorem; it assembles results already available in Papers 01–25 of the GSD programme (GSD Book, 2026), in the broader literatures of geometric algebra, rough paths, information theory, and causal inference, and in the philosophical traditions that ran from Pascal and Port-Royal through Newton and Leibniz to Hamilton himself. The contribution is architectural, not technical. It states what the architecture *is*, in continuity with the agenda Hamilton declared and to whose continuation he explicitly invited mathematicians yet to come.

We close, accordingly, in the same understated key in which the title is set. The present paper is some brief, late-arriving notes on the connection between discrete-time stochastic processes and the Hamiltonian programme of *algebra as the science of Pure Time*. The notes are an addendum, with a peculiar remark, to a programme older than Geometric Signal Dynamics itself. Whether the addendum proves useful — whether the synthesis we have articulated finds operative purchase beyond the GSD research programme that motivated it, and contributes to the development that Hamilton hoped for — is something the readers of this note will collectively decide, as mathematicians and applied scientists working on dynamic data already do every day, by what they choose to take seriously and to build on.

Postscript: Two Foundational Inquiries

The synthesis above presents Geometric Signal Dynamics as a continuation of Hamilton's programme. The continuation is also a foundation. We close, in Hamilton's manner of declaring a programme before all of its applications have been written down, with two inquiries that we believe sit beneath the rest of the development.

First inquiry: what is the geometry of a discrete time series? A real-valued time series $y\colon \{1,\dots,T\} \to \mathbb{R}$ is, classically, a function on a one-dimensional Cartesian timeline. The Cartesian timeline carries no intrinsic geometry: its only invariant is order. Whatever geometric content time series possess must come from elsewhere — from the kinematic structure of the values themselves, not from the timestamps that index them.

The kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ supplies that elsewhere. It assigns to the series a three-dimensional kinematic profile at each instant, and so admits intrinsic geometric invariants of two natural kinds: the Frenet–Serret curvature of the bivector trajectory $B(t) = \mathbf{q}(t-1) \wedge \mathbf{q}(t)$ as a 1-submanifold of $\mathbb{R}^3$, and the Gaussian curvature of the lagged scalar surface $s(t,\tau) = \mathbf{q}(t)\cdot\mathbf{q}(t+\tau)$ as a 2-submanifold of $\mathbb{R}^3$. Both invariants are intrinsic in the sense of Gauss's *Theorema Egregium*: they depend only on the metric of the constructed manifolds, not on the rigid relocations of the latter. Neither construction requires the timestamp axis for anything beyond the original lookup table $t \mapsto y_t$. The geometry is in $\mathbf{q}$, not in t.

We therefore submit the inquiry: *what is the geometry of a time series?* — and propose the kinematic embedding as the 3-dimensional ground from which the question becomes formulable. Under this view, Takens's delay-embedding theorem appears as a particular reparametrisation of the same kinematic ground; Lyons's path signature as the abelian projection of the moment tensor that the same ground supports.

Second inquiry: what is the non-abelian extension of Cramér–Kolmogorov? A probability measure μ on $\mathbb{R}^d$ is, classically, characterised by its Fourier transform $\phi_\mu(\xi) = \mathbb{E}[e^{i\xi^\top X}]$, a complex-valued function on $\mathbb{R}^d$. The kernel $e^{i\xi^\top X}$ is abelian by virtue of the central imaginary unit $i \in \mathbb{C}$. Cramér–Kolmogorov probability theory is, in this precise sense, an *abelian* theory: its underlying group structure is the additive group $\mathbb{R}^d$, and the characteristic function takes values in the abelian group $U(1)$.

Hamilton's 1843 *Principle of Fluctuation* did not prescribe an abelian kernel. The abelianisation was Cramér's choice in 1942, not Hamilton's in 1843. Within $\mathrm{Cl}(3,0)$ the kernel admits non-abelian generalisations of two natural kinds: a kernel valued in the centre $\langle 1, i_3\rangle \cong \mathbb{C}$, recovering Cramér; and a kernel valued in the bivector subspace $\mathrm{Cl}(3,0)^{(2)} \cong \mathfrak{spin}(3)$, valued in the non-abelian group $\mathrm{Spin}(3) \cong SU(2)$ once exponentiated. The latter produces a hierarchy

$$\{\hat{\Phi}_\mu^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$$

of characteristic functions indexed by the irreducible representations of $SU(2)$, related by the

Plancherel formula $\|\mu\|_{L^2}^2 = \sum_j (2j+1)\, \|\hat{\Phi}_\mu^{(j)}\|_{HS}^2$. The classical Cramér–Wold–Kolmogorov theory recovers as the $j = 0$ projection. The cases $j = 1/2$ and $j = 1$ alone suffice to expose, respectively, central and rotational structure of the joint distribution that the abelian theory subsumes silently in its choice of representation.

We therefore submit the second inquiry: *what is the non-abelian extension of Cramér–Kolmogorov?* — and propose Spin(3), the simply-connected double cover of $SO(3)$, as the natural group; the Plancherel-graded family $\{\hat{\Phi}_\mu^{(j)}\}$ as the natural object; and the abelian case as the projection one chooses, not the theory one is forced into.

Closing. The two inquiries are independent in their objects and parallel in their motivation. The geometric inquiry asks after the structure of the carrier (*where does the signal live?*); the probabilistic inquiry asks after the structure of the law (*what does the signal sample?*). The synthesis above argues that both inquiries admit canonical answers in $\mathrm{Cl}(3,0)$, and that the further papers of the programme treat them in detail. The reader is invited to consider that two such canonical extensions, sitting at the foundations and not at the periphery, are unlikely to be coincidences — and that the present synthesis, far from being the end of the programme, may be its preface.

A The operational trinity in the GSD library

The three categories of the Hamiltonian programme — moments, steps, numbers — are realised in the public API of the GSD library (`gsd==0.2.0`) with one-to-one correspondence. We sketch the correspondence here without entering into implementation detail; the library documentation (GSD Book, 2026) contains the full reference. The point of this appendix is not to advertise software but to make the philosophical claim of the body checkable: the architecture of the library is not a contingent design choice but a literal transcription of Hamilton's three categories into executable Python.

Moments. The level y_t enters the library as the input array passed to any top-level analysis function. The standardised moment y_t/σ_y is computed inside `gsd.embedding.kinematic_embedding` as the first column of the returned embedding matrix $Q \in \mathbb{R}^{T\times 3}$. The unstandardised level is preserved, when needed, by passing `standardize=False` to the same function (this is the choice mandated by the geometric ergodicity exponent of Paper 25 (GE), where per-window restandardisation would erase the scale information that distinguishes ergodic from non-ergodic series).

Steps. The first and second steps $\Delta y_t/\sigma_\Delta$ and $\Delta^2 y_t/\sigma_{\Delta\Delta}$ are the second and third columns of the embedding matrix. The library function `gsd.embedding.apply_diff` performs the finite-difference operation that produces them; `gsd.embedding.finite_diff_coefs` returns the binomial coefficients of Δ^k for arbitrary k, so that higher-order steps are available without re-implementation when the regional optimality of $d = 3$ is exceeded.

Numbers. The geometric product $\mathbf{q}(t)\,\mathbf{q}(t+\tau)$ and its scalar/bivector decomposition are computed by `gsd.algebra.auto_scalar_bivector`, which returns the time series $s(t;\tau)$ and $B(t;\tau)$ as separate arrays. The trivector $\mathcal{T}(t;\tau_1,\tau_2)$ is computed by `gsd.algebra.trivector_pseudoscala` The cross-product version $\mathbf{q}^{(A)}(t)\,\mathbf{q}^{(B)}(t+\tau)$ for two distinct series is computed by `gsd.algebra.cross_scalar_` The full multivector (12), together with bootstrap confidence intervals on the grade-wise means and the conditional fragility estimators of Paper 16 (GF), is assembled into the container class `gsd.GSDSignature` by the one-liner `gsd.analyze(y, tau, tau2)`. The class exposes the grade-wise time series as attributes (`.scalar`, `.bivector`, `.pseudoscalar`), the grade-wise means as properties, and the empirical moment tensor of Paper 24 (MT) via `gsd.moment_tensor.compute`.

Architecture. The architecture of the library transcribes the Hamiltonian categories directly: a *moments* layer (`gsd.embedding`), a *steps* layer (also `gsd.embedding` through the difference operators), and a *numbers* layer (`gsd.algebra` for the per-instant multivector, `gsd.signature` for the empirical signature container, `gsd.moment_tensor` for the Lyons-truncation moment tensor). A reader familiar with the *Researches respecting Quaternions* can read the API diagram of GSD as a transcription of Article 1 of the First Series (moments), Article 6 (steps and the unit imaginary on couples), and Articles 6–8 of the Second

Series (numbers as compositions in the quaternion algebra) into the executable Python of 2026. We do not claim that this transcription is the only possible one or that alternative library designs would necessarily fail; we claim only that it is faithful, and that its faithfulness is part of what the paper asks the reader to take seriously.

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